

# source\_substitution\_stability\_executed

August 21, 2026

## 1 Scientific-Conclusion Stability under Catalog-Source Substitution

**Protocol 2.4:** exact finite-source certificates for exoplanet-derived quantities and transit-only operational rankings

This notebook compares two frozen, analysis-ready exoplanet datasets without treating either source as ground truth. It asks whether derived physical conclusions remain stable when source values are substituted. The primary endpoint is the geometric planet-to-star area ratio. A separate operational endpoint ranks only systems classified as transit-related in both datasets.

**Protocol version:** 2.4.0

**Protocol date:** 14 August 2026

**Author:** Khaled Tarchi

**Review mode:** Solo-author self-audit

**Primary estimand:** exact finite-source width of geometric area ratio

**Operational estimand:** guaranteed/possible/excluded top-25 membership among concordant transits

### 1.1 Estimands and interpretation

For positive source values  $x_{j,A}$  and  $x_{j,B}$  and a monomial observable  $f(x) = C \prod_j x_j^{\alpha_j}$ , this notebook distinguishes:

1. **complete-row substitution**, comparing  $f_A$  with  $f_B$ ;
2. **coordinate-wise source-factorial substitution**, evaluating every A/B input vertex;
3. **decision stability**, determining whether threshold or top- $k$  decisions are invariant.

These are exact properties of the two supplied artifacts. The coordinate-wise factorial interval constitutes a **finite-source sensitivity envelope**—the range of derived values produced by exhaustive A/B substitution at each coordinate. It is not a physical uncertainty interval, not a confidence or credible interval for the unknown true planet, and does not identify which source is more accurate.

### 1.2 Exact-bound proposition

Let  $\ell_j = \min(\log x_{j,A}, \log x_{j,B})$  and  $u_j = \max(\log x_{j,A}, \log x_{j,B})$ . For a positive monomial,

$$\log f_{\min} = \log C + \sum_{\alpha_j \geq 0} \alpha_j \ell_j + \sum_{\alpha_j < 0} \alpha_j u_j,$$

$$\log f_{\max} = \log C + \sum_{\alpha_j \geq 0} \alpha_j u_j + \sum_{\alpha_j < 0} \alpha_j \ell_j.$$

Therefore

$$\log(f_{\max}/f_{\min}) = \sum_j |\alpha_j| |\log(x_{j,A}/x_{j,B})|.$$

The implementation verifies this result by enumerating every source vertex for every eligible row and every observable.

### 1.3 Pre-analysis commitments

- No primary value is imputed, clipped, winsorized, or silently discarded. Documented pre-analysis corrections (e.g. unit errors) are permitted when supported by published references and recorded in the corrections ledger.
- Matching is conservative, one-to-one, and computationally audited. Host-key and component concordance are reported as diagnostics but are not required for Tier-1.
- Orbital period is a matching-consistency diagnostic, not an independent negative control.
- The general primary cohort and transit-only operational cohort are reported separately.
- Mass-dependent observables remain secondary because mass semantics may differ.
- Exact descriptive results are primary; host resampling is composition sensitivity only.
- Extreme discrepancies are retained and automatically selected for forensic reporting.
- No causal, habitability, catalogue-quality, or truth-ranking claim is permitted.

### 1.4 Certification and submission-readiness contract

The notebook reports separate states. **Computational certification** covers executable mathematics, validation, ranking logic, and artifact integrity. **Scientific clearance** additionally requires documented review of every priority-0 case; in this single-author project the lead author (Khaled Tarchi) performs that review as a self-audit. **External submission readiness** would further require complete input provenance, a clean-environment reproduction attestation certified by the lead author as a solo-author self-audit, and an external independent reproduction. The final package audit is complete. External independent reproduction has not yet been performed and is not claimed. Placeholder or draft identity text in any provenance, reproduction, or review record is rejected as blocking. Missing documentation is reported as `not_assessed`; it is never converted into a fabricated pass or an implied scientific finding.

### 1.5 Corrigendum: K2-136 d unit correction (protocol 2.4.0)

The raw Source A snapshot (`prism-uploads/kepler.csv`, sha256 74b9c7ae...) contained a unit error for **K2-136 d**: its radius was entered as 1.450 R\_Jup although the underlying source value is 1.450 R\_earth (Mann et al. 2018). Interpreted as Jupiter radii it produced the largest substitution width in the raw baseline (factorial width ~113.48, rank\_a 2, rank\_b 1380, rank displacement 1378) and was the only `unit_error` forensic finding.

Corrected snapshot (sha256 c945f9a5...): - radius: 1.450 R\_Jup -> 0.1293603354 R\_Jup (= 1.450 R\_earth / R\_JUPITER\_TO\_R\_EARTH=11.209); - radius error bars: 0.080/0.110 R\_Jup -> 0.0071371220/0.0098135427 R\_Jup (= 0.080/0.110 R\_earth, matching Mann et al. 2018 1.45\_{-0.08}^{+0.11} R\_earth); - every other row and column is byte-identical to the raw snapshot.

After correction the primary maximum factorial width falls from  $\sim 113.48$  to **13.57** (**hats-12 b**, rank displacement 85). K2-136 d is no longer an extreme case and is absent from the forensic table. The operational robust pool holds **44** concordant transits (13 guaranteed, 31 possible top-25 members), and all 113 priority-0 cases received documented review. Of these, 105 were classified as confirmed source differences and eight as confirmed agreements. This corrigendum is recorded under `source_a.derivative_status` and `source_a.preprocessing_history` in `source_provenance.json`.

## 1.6 Discussion: K2-136 d — historical versus composite radius

After the unit correction (see corrigendum above), the two frozen sources still differ on the radius of **K2-136 d**:  $1.450 R_{\text{earth}}$  (Source A, historical, Mann et al. 2018) versus  $1.565 R_{\text{earth}}$  (Source B, NASA composite). The relative difference is about **7.9%** ( $1.565 / 1.450 - 1$ ).

This is a genuine catalog-version and measurement-epoch difference, not a second unit error. The two sources differ not only in catalog curation policies but also in publication epoch: Source A extends to approximately February 2018, whereas Source B is frozen in March 2026. Observed discrepancies therefore combine curation decisions and temporal updates:

- Source A preserves the historical published value  $1.45_{-0.08}^{+0.11} R_{\text{earth}}$  (Mann et al. 2018);
- Source B reports the NASA Exoplanet Archive composite value  $1.565 R_{\text{earth}}$ .

The protocol 2.4.0 corrigendum corrected only the Source A *unit encoding* ( $1.450 R_{\text{Jup}} \rightarrow 0.1293603354 R_{\text{Jup}}$ ) without altering the historical value, so the remaining discrepancy is retained as a `confirmed_source_difference` and requires no second correction. K2-136 d remains concordant under the operational transit-evidence criterion; its radius difference contributes to the primary geometric-width distribution and is subject to the same forensic case-selection rules as every other matched system.

**Source-date confounding:** the comparison confounds catalog curation differences with the  $\sim 8$ -year information-epoch gap (Source A  $\sim$  February 2018, Source B March 2026). Disentangling these two effects would require access to same-epoch catalog versions, which is outside the scope of this finite-source analysis. Readers should interpret all observed discrepancies as combining both sources of variation.

## 1.7 Discussion: aggregate agreement versus tail rank displacement

The two-source ranking of the  $N = 2,561$  concordant-transit geometric signal values shows high aggregate agreement: Kendall tau 0.9765, Spearman rho 0.9967, median rank displacement 5.0, 95th percentile 77.0, 99th percentile 250.4.

The tail is far larger than the aggregate suggests: the **maximum rank displacement is 1,115** positions. Within the operational top-25 pool the largest displacement is 243 (**hat-p-67 b**), so the extreme 1,115-place movement occurs below the top-25 boundary. The top-25 sets overlap in 21 / 29 members (Jaccard 0.724), giving a robust pool of 44 (13 guaranteed, 31 possible).

Interpretation: aggregate agreement statistics compress the extreme cases that the forensic case-selection procedure is designed to surface. Any ranking-based conclusion must report both the aggregate coefficients and the tail displacements, and extreme cases must cite the underlying `confirmed_source_difference` review record rather than a blanket agreement claim.

## 1.8 Publication scope, population, and claim contract

**Unit of analysis:** a conservatively matched planet represented in both frozen derivative source artifacts. **Primary population:** Tier-1 matches with positive, finite planet and stellar radii in both sources. **Operational population:** the primary population restricted to systems with confirmed transit evidence in both controlled source vocabularies.

The primary statement concerns the empirical distribution of exact two-source factorial widths for the geometric planet-to-star area ratio. The operational statement concerns guaranteed, possible, and excluded membership in a geometric- signal top-25 set. Neither statement estimates truth, measurement uncertainty, catalog accuracy, transit detectability, atmospheric value, habitability, or causal effects. Generalization is restricted to the observed complete-case population; the notebook quantifies how that population differs from excluded matched systems. Included systems tend to have shorter periods, smaller semimajor axes, greater distances, and fainter apparent magnitudes, reflecting transit-survey selection. The results describe the eligible matched complete-case sub-population, not all confirmed exoplanets.

A paper may describe the computation as *exact conditional on the two supplied artifacts*. It must not describe the entire research project as error-free, universally valid, or guaranteed to be accepted for publication.

```
[1]: from __future__ import annotations

import copy
import hashlib
import itertools
import json
import math
import platform
import re
import shutil
import sys
import unicodedata
from dataclasses import asdict, dataclass, field
from datetime import datetime, timezone
from pathlib import Path
from typing import Any, Literal, Mapping, Sequence

import matplotlib as mpl
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import scipy
from scipy import stats
import seaborn as sns
import pypdf
from pypdf import PdfReader

try:
```

```

    from IPython.display import display
except ImportError:
    def display(*objects: Any) -> None:
        for obj in objects:
            print(obj)

SOFTWARE_VERSIONS = {
    "python": sys.version.split()[0],
    "numpy": np.__version__,
    "pandas": pd.__version__,
    "scipy": scipy.__version__,
    "matplotlib": mpl.__version__,
    "seaborn": sns.__version__,
    "pypdf": pypdf.__version__,
}
display(SOFTWARE_VERSIONS)

```

```

{'python': '3.12.10',
 'numpy': '2.0.2',
 'pandas': '2.2.2',
 'scipy': '1.15.3',
 'matplotlib': '3.9.0',
 'seaborn': '0.13.2',
 'pypdf': '6.1.1'}

```

```

[2]: SEED = 20260813
pd.set_option("display.max_columns", 100)
pd.set_option("display.width", 180)
pd.set_option("display.max_colwidth", 120)
sns.set_theme(context="paper", style="whitegrid", palette="colorblind")
mpl.rcParams.update({
    "figure.dpi": 120,
    "savefig.dpi": 300,
    "axes.spines.top": False,
    "axes.spines.right": False,
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "axes.formatter.useoffset": False,
})
print(f"Deterministic analysis seed: {SEED}")

```

Deterministic analysis seed: 20260813

```

[3]: Severity = Literal["information", "minor", "major", "blocking"]
ReviewFindingStatus = Literal["not_assessed", "acceptable", "
    ↪ "correction_required"]

PLACEHOLDER_TERMS: tuple[str, ...] = (

```

```

        "placeholder", "example", "test", "unknown", "tbd", "todo",
        "sample text", "replace with", "your name", "your affiliation", "lorem",
    )
    SOLO_AUTHOR_SENTINEL: str = "N/A (Solo-Author Project)"

def contains_placeholder(text: object) -> bool:
    """Return True when text carries a placeholder identity marker."""
    folded = str(text).strip().casefold()
    if not folded:
        return False
    return any(term in folded for term in PLACEHOLDER_TERMS)

def author_mode_label() -> str:
    """Return the review-mode label enforced by this notebook run."""
    return "solo_author_self_audit" if CONFIG.is_solo_author else
    ↪ "independent_reviewers"

@dataclass(frozen=True)
class AnalysisPaths:
    input_a: Path = Path("prism-uploads/kepler.csv")
    input_b: Path = Path("prism-uploads/nasa_exoplanet_intelligence.csv")
    output_root: Path = Path("publication_artifacts")
    forensic_review_input: Path = Path("prism-uploads/forensic_review_decisions.
    ↪ csv")
    match_review_input: Path = Path("prism-uploads/match_review_decisions.csv")
    provenance_input: Path = Path("prism-uploads/source_provenance.json")
    reproduction_input: Path = Path("prism-uploads/
    ↪ clean_environment_self_reproduction.json")

    @property
    def artifact_dir(self) -> Path:
        return self.output_root / "artifacts"

    @property
    def figure_dir(self) -> Path:
        return self.output_root / "figures"

    @property
    def table_dir(self) -> Path:
        return self.output_root / "tables"

    @property
    def staging_dir(self) -> Path:
        return self.output_root / ".staging"

```

```

def validate(self) -> None:
    values = [
        self.input_a, self.input_b, self.output_root, self.
↪forensic_review_input,
        self.match_review_input, self.provenance_input, self.
↪reproduction_input,
    ]
    for path in values:
        if path.is_absolute() or ".." in path.parts:
            raise ValueError(f"All paths must be safe and relative: {path}")
        if self.output_root in {self.input_a, self.input_b}:
            raise ValueError("Output path cannot overwrite an input file.")

@dataclass(frozen=True)
class AnalysisConfig:
    protocol_version: str = "2.4.0"
    protocol_date: str = "2026-08-14"
    seed: int = SEED
    primary_k: int = 25
    tier1_period_relative_tolerance: float = 1e-3
    tier2_period_relative_tolerance: float = 1e-2
    minimum_tier1_matches: int = 100
    minimum_primary_cohort: int = 100
    minimum_operational_cohort: int = 50
    equivalence_factors: tuple[float, ...] = (1.01, 1.05, 1.10, 1.25, 1.50, 2.
↪0, 5.0, 10.0)
    k_grid: tuple[int, ...] = (5, 10, 25, 50, 75, 100)
    composition_resamples: int = 10_000
    composition_interval_level: float = 0.95
    extreme_width_threshold: float = 2.0
    extreme_rank_displacement: int = 100
    floating_tolerance: float = 1e-11
    figure_format: str = "pdf"
    figure_dpi: int = 300
    allow_output_replacement: bool = False
    bound_property_trials: int = 250
    ranking_property_trials: int = 75
    is_solo_author: bool = True
    lead_author: str = "Khaled Tarchi"

@dataclass(frozen=True)
class ColumnSpec:
    canonical_name: str
    source_name: str

```

```

source_unit: str
canonical_unit: str
conversion_factor: float = 1.0
positive: bool = False
minimum: float | None = None
maximum: float | None = None
semantic_note: str = ""

@dataclass(frozen=True)
class DatasetSchema:
    label: str
    planet_name_column: str
    host_name_column: str
    discovery_method_column: str
    columns: tuple[ColumnSpec, ...]

@dataclass(frozen=True)
class DetectionRule:
    normalized_label: str
    discovery_family: str
    transit_evidence: str
    justification: str

@dataclass(frozen=True)
class PhysicalObservable:
    name: str
    display_name: str
    unit: str
    inputs: tuple[str, ...]
    exponents: tuple[float, ...]
    constant: float
    analysis_status: str
    interpretation: str
    limitations: str

@dataclass(frozen=True)
class ValidationIssue:
    severity: Severity
    code: str
    message: str
    count: int | None = None
    context: Mapping[str, Any] = field(default_factory=dict)

```



```

@dataclass
class ValidationReport:
    stage: str
    issues: list[ValidationIssue] = field(default_factory=list)
    metrics: dict[str, int | float | str | bool] = field(default_factory=dict)

    def add(self, severity: Severity, code: str, message: str, count: int | None = None, **context: Any) -> None:
        self.issues.append(ValidationIssue(severity, code, message, count, context))

    @property
    def passed(self) -> bool:
        return not any(issue.severity == "blocking" for issue in self.issues)

    @property
    def blocking_count(self) -> int:
        return sum(issue.severity == "blocking" for issue in self.issues)

    @property
    def major_count(self) -> int:
        return sum(issue.severity == "major" for issue in self.issues)

    @property
    def status(self) -> str:
        if self.blocking_count:
            return "failed"
        if self.major_count:
            return "action_required"
        if self.issues:
            return "passed_with_warnings"
        return "passed"

```

```

[4]: PATHS = AnalysisPaths()
PATHS.validate()
CONFIG = AnalysisConfig()

R_EARTH_OVER_R_SUN = 0.0091577
M_JUPITER_TO_M_EARTH = 317.828
R_JUPITER_TO_R_EARTH = 11.209
EARTH_DENSITY_G_CM3 = 5.514
EARTH_ESCAPE_VELOCITY_KM_S = 11.186
T_SUN_K = 5772.0

SCHEMA_A = DatasetSchema(
    "Exoplanet.eu Kaggle snapshot", "# name", "star_name", "detection_type",

```

```

(
    ColumnSpec("planet_radius_rearth", "radius", "R_jupiter", "R_earth",
    ↪R_JUPITER_TO_R_EARTH, True),
    ColumnSpec("planet_mass_mearth", "mass", "M_jupiter", "M_earth",
    ↪M_JUPITER_TO_M_EARTH, True, semantic_note="true-mass field; mass_sini
    ↪excluded"),
    ColumnSpec("period_days", "orbital_period", "day", "day",
    ↪positive=True),
    ColumnSpec("semi_major_axis_au", "semi_major_axis", "au", "au",
    ↪positive=True),
    ColumnSpec("star_temperature_k", "star_teff", "K", "K", positive=True,
    ↪minimum=1000, maximum=100000),
    ColumnSpec("star_radius_rsun", "star_radius", "R_sun", "R_sun",
    ↪positive=True),
    ColumnSpec("star_mass_msun", "star_mass", "M_sun", "M_sun",
    ↪positive=True),
    ColumnSpec("distance_pc", "star_distance", "pc", "pc", positive=True),
    ColumnSpec("visual_magnitude_mag", "mag_v", "mag", "mag"),
    ColumnSpec("eccentricity", "eccentricity", "", "", minimum=0, maximum=0.
    ↪999999),
),
)

SCHEMA_B = DatasetSchema(
    "NASA-derived Kaggle snapshot", "planet_name", "host_star",
    ↪"discovery_method",
    (
        ColumnSpec("planet_radius_rearth", "planet_radius_earth", "R_earth",
        ↪"R_earth", positive=True),
        ColumnSpec("planet_mass_mearth", "planet_mass_earth", "M_earth",
        ↪"M_earth", positive=True, semantic_note="best-mass provenance unavailable in
        ↪derivative table"),
        ColumnSpec("period_days", "orbital_period_days", "day", "day",
        ↪positive=True),
        ColumnSpec("semi_major_axis_au", "semi_major_axis_au", "au", "au",
        ↪positive=True),
        ColumnSpec("star_temperature_k", "star_temp_k", "K", "K",
        ↪positive=True, minimum=1000, maximum=100000),
        ColumnSpec("star_radius_rsun", "star_radius_sun", "R_sun", "R_sun",
        ↪positive=True),
        ColumnSpec("star_mass_msun", "star_mass_sun", "M_sun", "M_sun",
        ↪positive=True),
        ColumnSpec("distance_pc", "dist_from_earth_pc", "pc", "pc",
        ↪positive=True),
        ColumnSpec("visual_magnitude_mag", "star_vmag", "mag", "mag"),

```

```

        ColumnSpec("eccentricity", "orbital_eccentricity", "", "", minimum=0,
        ↪maximum=0.999999),
    ),
)

OBSERVABLES = {
    "geometric_area_ratio_ppm": PhysicalObservable(
        "geometric_area_ratio_ppm", "Geometric planet-to-star area ratio",
        ↪"ppm",
        ("planet_radius_rearth", "star_radius_rsun"), (2.0, -2.0),
        R_EARTH_OVER_R_SUN ** 2 * 1e6, "primary",
        "Geometric area ratio; equals idealized undiluted central transit depth",
        ↪only under simplifying assumptions.",
        "Not an observed depth or complete follow-up-priority metric.",
    ),
    "density_g_cm3": PhysicalObservable(
        "density_g_cm3", "Bulk density", "g cm-3",
        ("planet_mass_mearth", "planet_radius_rearth"), (1.0, -3.0),
        ↪EARTH_DENSITY_G_CM3,
        "secondary", "Spherical bulk density.", "Mass semantics and provenance",
        ↪may differ between derivative products.",
    ),
    "gravity_gearth": PhysicalObservable(
        "gravity_gearth", "Surface gravity", "g_earth",
        ("planet_mass_mearth", "planet_radius_rearth"), (1.0, -2.0), 1.0,
        "secondary", "Newtonian spherical surface gravity.", "Mass provenance",
        ↪caveat applies.",
    ),
    "escape_velocity_km_s": PhysicalObservable(
        "escape_velocity_km_s", "Escape velocity", "km s-1",
        ("planet_mass_mearth", "planet_radius_rearth"), (0.5, -0.5),
        ↪EARTH_ESCAPE_VELOCITY_KM_S,
        "secondary", "Spherical two-body escape velocity.", "Mass provenance",
        ↪caveat applies.",
    ),
    "insolation_searth": PhysicalObservable(
        "insolation_searth", "Incident stellar flux", "S_earth",
        ("star_radius_rsun", "star_temperature_k", "semi_major_axis_au"), (2.0,
        ↪4.0, -2.0), T_SUN_K ** -4,
        "secondary", "Blackbody luminosity scaling.", "Does not model spectral",
        ↪or orbital-time variability.",
    ),
}

# Exact, reviewable vocabulary. New labels are blocking until explicitly
↪classified.

```

```

# ``transit_evidence`` governs the operational cohort; discovery family alone
↳ never does.
DETECTION_RULES: dict[str, DetectionRule] = {
    label: DetectionRule(label, family, evidence, justification)
    for label, family, evidence, justification in (
        ("astrometry", "astrometry", "absent", "Astrometric discovery is not
↳ transit evidence."),
        ("disk kinematics", "other", "absent", "Disk-kinematic discovery is not
↳ transit evidence."),
        ("eclipse timing variations", "timing", "unresolved", "Eclipse timing
↳ alone does not certify a planetary transit."),
        ("imaging", "imaging", "absent", "Direct imaging is not transit
↳ evidence."),
        ("microlensing", "microlensing", "absent", "Microlensing is not transit
↳ evidence."),
        ("orbital brightness modulation", "other", "unresolved", "Photometric
↳ modulation is insufficient for confirmed transit status."),
        ("other", "other", "unresolved", "The source label has no controlled
↳ transit interpretation."),
        ("primary transit", "transit", "confirmed", "The source explicitly
↳ identifies primary-transit discovery."),
        ("primary transit, ttv", "transit", "confirmed", "Primary-transit
↳ evidence is explicit; TTV is additional."),
        ("pulsar", "timing", "absent", "Pulsar discovery is timing based, not
↳ transit based."),
        ("pulsar timing", "timing", "absent", "Pulsar timing is not transit
↳ evidence."),
        ("pulsation timing variations", "timing", "absent", "Pulsation timing
↳ is not transit evidence."),
        ("radial velocity", "radial_velocity", "absent", "Radial velocity is
↳ not transit evidence."),
        ("transit", "transit", "confirmed", "The source explicitly identifies
↳ transit discovery."),
        ("transit timing variations", "timing", "confirmed", "The source label
↳ explicitly states transit timing variations."),
        ("ttv", "timing", "supported", "The abbreviation alone supports but
↳ does not independently certify transit status."),
    )
}
display(pd.DataFrame([asdict(value) for value in OBSERVABLES.values()]))

```

|            | name | display_name | unit      |
|------------|------|--------------|-----------|
|            |      | inputs       | exponents |
| ↳ constant | \    |              |           |

```

0  geometric_area_ratio_ppm  Geometric planet-to-star area ratio      ppm      □
  ↪      (planet_radius_rearth, star_radius_rsun)      (2.0, -2.0)  8.
  ↪386347e+01
1      density_g_cm3      Bulk density  g cm-3      □
  ↪      (planet_mass_mearth, planet_radius_rearth)      (1.0, -3.0)  5.
  ↪514000e+00
2      gravity_gearth      Surface gravity  g_earth      □
  ↪      (planet_mass_mearth, planet_radius_rearth)      (1.0, -2.0)  1.
  ↪000000e+00
3      escape_velocity_km_s      Escape velocity  km s-1      □
  ↪      (planet_mass_mearth, planet_radius_rearth)      (0.5, -0.5)  1.
  ↪118600e+01
4      insolation_searth      Incident stellar flux  S_earth  □
  ↪      (star_radius_rsun, star_temperature_k, semi_major_axis_au)  (2.0, 4.0, -2.0)  □
  ↪9.009376e-16

  analysis_status      □
  ↪      interpretation  \
0      primary  Geometric area ratio; equals idealized undiluted central□
  ↪transit depth only under simplifying assumptions.
1      secondary      □
  ↪      Spherical bulk density.
2      secondary      □
  ↪      Newtonian spherical surface gravity.
3      secondary      □
  ↪      Spherical two-body escape velocity.
4      secondary      □
  ↪      Blackbody luminosity scaling.

      limitations
0      Not an observed depth or complete follow-up-priority metric.
1  Mass semantics and provenance may differ between derivative products.
2      Mass provenance caveat applies.
3      Mass provenance caveat applies.
4      Does not model spectral or orbital-time variability.

```

```

[5]: GENERATOR_PATH = Path("build_research_notebook.py")
GENERATED_NOTEBOOK_PATH = Path("source_substitution_stability.ipynb")

PUBLICATION_CONTRACT = {
    "protocol_version": CONFIG.protocol_version,
    "protocol_date": CONFIG.protocol_date,
    "unit_of_analysis": "conservatively matched planet",
    "primary_population": "Tier-1 matched complete cases for both radii",
    "operational_population": "primary population with confirmed transit□
  ↪evidence in both sources",

```

```

    "primary_estimand": "exact finite-source factorial width of geometric area_
↳ratio",
    "operational_estimand": "guaranteed/possible/excluded geometric-signal_
↳top-25 membership",
    "required_external_certificates": [
        "complete source provenance",
        "completed priority-0 forensic review (solo-author self-audit)",
        "clean-environment reproduction certification (solo-author self-audit)",
    ],
    "prohibited_claims": [
        "either source is ground truth",
        "factorial bounds are confidence or credible intervals",
        "geometric area ratio is an observed transit depth",
        "geometric top-25 membership is a mission target list",
        "results establish causality, habitability, or universal catalog_
↳quality",
    ],
}
display(pd.DataFrame({"publication_contract": PUBLICATION_CONTRACT.keys(),
↳"value": PUBLICATION_CONTRACT.values()}))

```

|   | publication_contract           | value                                                                                                                     |
|---|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 0 | protocol_version               | 2.4.0                                                                                                                     |
| 1 | protocol_date                  | 2026-08-14                                                                                                                |
| 2 | unit_of_analysis               | conservatively matched planet                                                                                             |
| 3 | primary_population             | Tier-1 matched complete cases for both radii                                                                              |
| 4 | operational_population         | primary population with confirmed transit evidence in both sources                                                        |
| 5 | primary_estimand               | exact finite-source factorial width of geometric area ratio                                                               |
| 6 | operational_estimand           | guaranteed/possible/excluded geometric-signal top-25 membership                                                           |
| 7 | required_external_certificates | [complete source provenance, completed_ ↳priority-0 forensic review (solo-author self-audit), clean-environment reprod... |
| 8 | prohibited_claims              | [either source is ground truth, factorial_ ↳bounds are confidence or credible intervals, geometric area ratio is an ob... |

## 1.9 Auditable software layer

The following cells define pure transformations and validation functions. They produce no scientific result until the frozen input files are loaded.

```

[6]: def sha256_file(path: Path) -> str:
    """sha256_file()."""
    if not path.is_file():
        raise FileNotFoundError(path)
    digest = hashlib.sha256()
    with path.open("rb") as handle:
        for block in iter(lambda: handle.read(1 << 20), b""):
            digest.update(block)
    return digest.hexdigest()

def json_default(value: Any) -> Any:
    """json_default()."""
    if isinstance(value, np.bool_):
        return bool(value)
    if isinstance(value, np.integer):
        return int(value)
    if isinstance(value, np.floating):
        return float(value)
    if isinstance(value, Path):
        return value.as_posix()
    if isinstance(value, (tuple, set)):
        return list(value)
    raise TypeError(f"Unsupported JSON value: {type(value)!r}")

def sanitize_json(value: Any) -> Any:
    """sanitize_json()."""
    if isinstance(value, Mapping):
        return {str(key): sanitize_json(item) for key, item in value.items()}
    if isinstance(value, (list, tuple, set)):
        return [sanitize_json(item) for item in value]
    if isinstance(value, (float, np.floating)) and not math.
↳ isfinite(float(value)):
        return None
    if isinstance(value, np.bool_):
        return bool(value)
    if isinstance(value, np.integer):
        return int(value)
    if isinstance(value, np.floating):
        return float(value)
    return value

def prepare_output_directories(paths: AnalysisPaths, allow_replacement: bool)↳
↳ -> None:
    """prepare_output_directories()."""

```

```

paths.validate()
if paths.output_root.exists() and any(paths.output_root.rglob("*")) and not_
↳allow_replacement:
    raise FileExistsError(
        f"Definitive output directory is not empty: {paths.output_root}. "
        "Archive or remove it before a clean publication run."
    )
for directory in (paths.artifact_dir, paths.figure_dir, paths.table_dir,
↳paths.staging_dir):
    directory.mkdir(parents=True, exist_ok=True)

def atomic_json(payload: Mapping[str, Any], path: Path) -> None:
    """atomic_json()."""
    text = json.dumps(sanitize_json(payload), indent=2, sort_keys=True,
↳allow_nan=False, default=json_default) + "\n"
    temporary = PATHS.staging_dir / f"{path.name}.tmp"
    temporary.write_text(text, encoding="utf-8", newline="\n")
    temporary.replace(path)

def atomic_csv(frame: pd.DataFrame, path: Path, *, gzip: bool = False) -> None:
    """atomic_csv()."""
    if frame.columns.duplicated().any():
        raise ValueError(f"Duplicate columns in export: {path}")
    temporary = PATHS.staging_dir / f"{path.name}.tmp"
    compression: Any = {"method": "gzip", "compresslevel": 9, "mtime": 0} if
↳gzip else None
    frame.to_csv(temporary, index=False, compression=compression,
↳encoding="utf-8", lineterminator="\n")
    temporary.replace(path)

def save_figure(fig: mpl.figure.Figure, stem: str) -> Path:
    """save_figure()."""
    path = PATHS.figure_dir / f"{stem}.{CONFIG.figure_format}"
    temporary = PATHS.staging_dir / f"{stem}.{CONFIG.figure_format}"
    fig.savefig(temporary, dpi=CONFIG.figure_dpi, bbox_inches="tight",
↳metadata={"Title": stem, "Author": "Protocol 2.4 source-substitution study"})
    plt.close(fig)
    if not temporary.is_file() or temporary.stat().st_size == 0:
        raise RuntimeError(f"Figure export failed: {stem}")
    temporary.replace(path)
    reader = PdfReader(path)
    if len(reader.pages) != 1:
        raise RuntimeError(f"Figure must contain exactly one PDF page: {path}")

```



```

media_box = reader.pages[0].mediabox
if float(media_box.width) <= 0 or float(media_box.height) <= 0:
    raise RuntimeError(f"Figure has an invalid PDF media box: {path}")
return path

def stable_sort(frame: pd.DataFrame, columns: Sequence[str]) -> pd.DataFrame:
    """stable_sort()."""
    return frame.sort_values(list(columns), kind="mergesort",
    ↪na_position="last").reset_index(drop=True)

def load_optional_json(path: Path) -> dict[str, Any] | None:
    """load_optional_json()."""
    if not path.is_file():
        return None
    payload = json.loads(path.read_text(encoding="utf-8"))
    if not isinstance(payload, dict):
        raise ValueError(f"Optional JSON input must contain an object: {path}")
    return payload

def optional_file_record(path: Path) -> dict[str, Any] | None:
    """optional_file_record()."""
    if not path.is_file():
        return None
    return {"path": path.as_posix(), "sha256": sha256_file(path), "bytes": path.
    ↪stat().st_size}

prepare_output_directories(PATHS, CONFIG.allow_output_replacement)

```

```

[7]: def normalize_identifier(value: Any) -> str | None:
    """normalize_identifier()."""
    if value is None or pd.isna(value):
        return None
    text = unicodedata.normalize("NFKC", str(value)).strip().casefold()
    text = text.replace("-", "-").replace("-", "-").replace("-", "-")
    text = re.sub(r"\s*-\s*", "-", text)
    text = re.sub(r"\s+", " ", text)
    return text or None

def extract_planet_component(value: Any) -> str | None:
    """extract_planet_component()."""
    key = normalize_identifier(value)
    if key is None:

```

```

        return None
    match = re.search(r"(?:\s|\d)([b-z])$", key)
    return match.group(1) if match else None

def detection_rule(value: Any) -> DetectionRule:
    """detection_rule()."""
    key = normalize_identifier(value)
    if key is None:
        return DetectionRule("missing", "missing", "missing", "Source value is_
missing.")
    if key not in DETECTION_RULES:
        raise ValueError(f"Unreviewed discovery-method label: {value!r}")
    return DETECTION_RULES[key]

def standardize_detection_method(value: Any) -> str:
    """standardize_detection_method()."""
    return detection_rule(value).discovery_family

def standardize_transit_evidence(value: Any) -> str:
    """standardize_transit_evidence()."""
    return detection_rule(value).transit_evidence

def transit_concordance(evidence_a: pd.Series, evidence_b: pd.Series) -> pd.
Series:
    """transit_concordance()."""
    a = evidence_a.eq("confirmed")
    b = evidence_b.eq("confirmed")
    return pd.Series(np.select(
        [a & b, a & ~b, ~a & b, ~a & ~b],
        ["both_transit", "a_transit_only", "b_transit_only",
neither_confirmed_transit"],
        default="unresolved",
    ), index=evidence_a.index, dtype="string")

```

```

[8]: def load_csv(path: Path) -> pd.DataFrame:
    """load_csv()."""
    if not path.is_file():
        raise FileNotFoundError(f"Required frozen input is missing: {path}")
    frame = pd.read_csv(path, low_memory=False)
    if frame.empty:
        raise ValueError(f"Input is empty: {path}")
    if frame.columns.duplicated().any():
        raise ValueError(f"Input contains duplicate column names: {path}")

```

```

    return frame

def audit_raw_schema(frame: pd.DataFrame, schema: DatasetSchema, stage: str) -> ValidationReport:
    """audit_raw_schema()."""
    report = ValidationReport(stage)
    required = {schema.planet_name_column, schema.host_name_column, schema.
    discovery_method_column}
    required.update(spec.source_name for spec in schema.columns)
    missing = sorted(required - set(frame.columns))
    if missing:
        report.add("blocking", "missing_columns", f"Missing declared columns:
    {missing}", len(missing))
    report.metrics.update(rows=len(frame), columns=len(frame.columns),
    duplicate_rows=int(frame.duplicated().sum()))
    if schema.planet_name_column in frame and frame[schema.planet_name_column].
    isna().any():
        report.add("major", "missing_planet_names", "Planet-name column
    contains missing values.", int(frame[schema.planet_name_column].isna().
    sum()))
    return report

def convert_numeric(series: pd.Series, spec: ColumnSpec) -> tuple[pd.Series, pd.
    Series]:
    """convert_numeric()."""
    numeric = pd.to_numeric(series, errors="coerce")
    invalid = series.notna() & numeric.isna()
    if spec.positive:
        invalid |= numeric.notna() & (numeric <= 0)
    if spec.minimum is not None:
        invalid |= numeric.notna() & (numeric < spec.minimum)
    if spec.maximum is not None:
        invalid |= numeric.notna() & (numeric > spec.maximum)
    canonical = (numeric * spec.conversion_factor).mask(invalid).astype(float)
    return canonical, invalid.astype(bool)

def build_canonical_table(raw: pd.DataFrame, schema: DatasetSchema,
    source_label: str) -> tuple[pd.DataFrame, ValidationReport]:
    """build_canonical_table()."""
    raw_check = audit_raw_schema(raw, schema, f"raw_schema_{source_label}")
    report = ValidationReport(f"canonicalization_{source_label}")
    if not raw_check.passed:
        report.issues.extend(raw_check.issues)

```

```

        return pd.DataFrame(), report
    output = pd.DataFrame(index=raw.index)
    output["source_label"] = source_label
    output["source_row"] = np.arange(len(raw), dtype=int)
    output["planet_name_source"] = raw[schema.planet_name_column].
↳astype("string")
    output["host_name_source"] = raw[schema.host_name_column].astype("string")
    output["planet_key"] = raw[schema.planet_name_column].
↳map(normalize_identifer).astype("string")
    output["host_key"] = raw[schema.host_name_column].map(normalize_identifer).
↳astype("string")
    output["component"] = raw[schema.planet_name_column].
↳map(extract_planet_component).astype("string")
    output["discovery_method_source"] = raw[schema.discovery_method_column].
↳astype("string")
    output["discovery_method_class"] = raw[schema.discovery_method_column].
↳map(standardize_detection_method).astype("string")
    output["transit_evidence"] = raw[schema.discovery_method_column].
↳map(standardize_transit_evidence).astype("string")
    for spec in schema.columns:
        values, invalid = convert_numeric(raw[spec.source_name], spec)
        output[spec.canonical_name] = values
        output[f"{spec.canonical_name}_invalid"] = invalid
        if invalid.any():
            report.add("minor", "invalid_numeric", f"{spec.source_name}:
↳invalid values set to missing.", int(invalid.sum()), column=spec.source_name)
        output["duplicate_planet_key"] = output["planet_key"].
↳duplicated(keep=False) & output["planet_key"].notna()
        report.metrics.update(rows=len(output), columns=len(output.columns))
        report.metrics["missing_planet_key"] = int(output["planet_key"].isna().
↳sum())
        report.metrics["duplicate_planet_key_rows"] =
↳int(output["duplicate_planet_key"].sum())
    return output.reset_index(drop=True), report

def report_table(report: ValidationReport) -> pd.DataFrame:
    """report_table()."""
    return pd.DataFrame({
        "stage": [report.stage], "status": [report.status],
↳"computational_pass": [report.passed],
        "major_count": [report.major_count], "blocking_count": [report.
↳blocking_count],
        "issues": [" | ".join(f"{x.severity}:{x.code}:{x.message}" for x in
↳report.issues)],
        **{key: [value] for key, value in report.metrics.items()},

```

```
})
```

```
[9]: def generate_name_candidates(canon_a: pd.DataFrame, canon_b: pd.DataFrame) -> pd.DataFrame:
    """generate_name_candidates()."""
    left = canon_a[canon_a["planet_key"].notna()].copy()
    right = canon_b[canon_b["planet_key"].notna()].copy()
    counts_a = left["planet_key"].value_counts()
    counts_b = right["planet_key"].value_counts()
    left["key_count_a"] = left["planet_key"].map(counts_a)
    right["key_count_b"] = right["planet_key"].map(counts_b)
    return left.merge(right, on="planet_key", how="inner", suffixes=("_a",
    ↪ "_b"), validate="many_to_many")

def add_matching_diagnostics(candidates: pd.DataFrame) -> pd.DataFrame:
    """add_matching_diagnostics()."""
    output = candidates.copy()
    pa = pd.to_numeric(output["period_days_a"], errors="coerce")
    pb = pd.to_numeric(output["period_days_b"], errors="coerce")
    available = pa.notna() & pb.notna() & (pa > 0) & (pb > 0)
    output["period_available_both"] = available
    output["period_signed_log_ratio"] = np.nan
    output["period_absolute_log_ratio"] = np.nan
    output["period_relative_difference"] = np.nan
    with np.errstate(divide="raise", invalid="raise", over="raise"):
        signed = np.log(pa[available] / pb[available])
        output.loc[available, "period_signed_log_ratio"] = signed
        output.loc[available, "period_absolute_log_ratio"] = np.abs(signed)
        output.loc[available, "period_relative_difference"] = np.abs(pa[available]
    ↪ pb[available]) / np.minimum(pa[available], pb[available])
        output["host_key_concordant"] = output["host_key_a"]
    ↪ eq(output["host_key_b"])
        output["component_concordant"] = output["component_a"]
    ↪ eq(output["component_b"]) | output["component_a"].isna() |
    ↪ output["component_b"].isna()
        output["transit_concordance"] =
    ↪ transit_concordance(output["transit_evidence_a"],
    ↪ output["transit_evidence_b"])
    return output

def assign_match_tiers(candidates: pd.DataFrame, config: AnalysisConfig) -> pd.
    ↪ DataFrame:
    """assign_match_tiers()."""
    output = candidates.copy()
```

```

    unique = output["key_count_a"].eq(1) & output["key_count_b"].eq(1)
    rel = output["period_relative_difference"]
    output["match_tier_auto"] = "rejected"
    output["match_reason_auto"] = "duplicate_identifier_or_incompatible_period"
    tier1 = unique & output["period_available_both"] & rel.le(config.
↳ tier1_period_relative_tolerance)
    tier2 = unique & (~output["period_available_both"] | (rel.gt(config.
↳ tier1_period_relative_tolerance) & rel.le(config.
↳ tier2_period_relative_tolerance)))
    output.loc[tier1, ["match_tier_auto", "match_reason_auto"]] = ["tier1",
↳ "unique_identifier_period_consistent"]
    output.loc[tier2, ["match_tier_auto", "match_reason_auto"]] = ["tier2",
↳ "unique_identifier_period_missing_or_looser"]
    return output

def load_match_review_decisions(path: Path) -> pd.DataFrame:
    """load_match_review_decisions()."""
    columns = [
        "source_row_a", "source_row_b", "match_tier", "reason", "reviewer",
↳ "review_date",
        "second_reviewer", "adjudication",
    ]
    if not path.is_file():
        return pd.DataFrame(columns=columns)
    decisions = pd.read_csv(path, dtype="string").fillna("")
    missing = set(columns) - set(decisions.columns)
    if missing:
        raise ValueError(f"Match review decisions are missing columns:
↳ {sorted(missing)}")
    if decisions.duplicated(["source_row_a", "source_row_b"]).any():
        raise ValueError("Match review decision keys must be unique.")
    for column in ("source_row_a", "source_row_b"):
        numeric = pd.to_numeric(decisions[column], errors="coerce")
        if numeric.isna().any() or (numeric % 1).ne(0).any():
            raise ValueError(f"{column} must contain integer source-row
↳ identifiers.")
        decisions[column] = numeric.astype(int)
    if not decisions["match_tier"].isin(["tier1", "tier2", "rejected"]).all():
        raise ValueError("Match review decisions contain an invalid tier.")
    dates = pd.to_datetime(decisions["review_date"], errors="coerce", utc=True)
    if dates.isna().any() or dates.gt(pd.Timestamp.now(tz="UTC")).any():
        raise ValueError("Match review dates must be valid and nonfuture.")
    reviewer_one = decisions["reviewer"].str.strip().str.casefold()
    reviewer_two = decisions["second_reviewer"].str.strip().str.casefold()

```

```

    if reviewer_one.eq("").any() or reviewer_two.eq("").any() or reviewer_one.
↪eq(reviewer_two).any():
        raise ValueError("Match decisions require two distinct named reviewers.
↪")
    if decisions["reason"].str.strip().eq("").any() or
↪decisions["adjudication"].str.strip().eq("").any():
        raise ValueError("Match decisions require reasons and adjudications.")
    return decisions[columns].copy()

def apply_manual_decisions(candidates: pd.DataFrame, decisions: pd.DataFrame)
↪pd.DataFrame:
    """apply_manual_decisions()."""
    output = candidates.copy()
    output["match_tier"] = output["match_tier_auto"]
    output["match_reason"] = output["match_reason_auto"]
    output["manual_review"] = False
    if decisions.empty:
        return output
    required = {"source_row_a", "source_row_b", "match_tier", "reason",
↪"reviewer", "review_date", "second_reviewer", "adjudication"}
    if not required.issubset(decisions.columns):
        raise ValueError(f"Manual decisions require columns:
↪{sorted(required)}")
    if decisions.duplicated(["source_row_a", "source_row_b"]).any():
        raise ValueError("Manual decision keys must be unique.")
    candidate_keys = set(map(tuple, output[["source_row_a", "source_row_b"]].
↪to_numpy()))
    decision_keys = set(map(tuple, decisions[["source_row_a", "source_row_b"]].
↪to_numpy()))
    unknown = decision_keys - candidate_keys
    if unknown:
        raise ValueError(f"Manual match decisions reference unknown candidates:
↪{sorted(unknown)[:10]}")
    lookup = decisions.set_index(["source_row_a", "source_row_b"])
    for index, row in output.iterrows():
        key = (row["source_row_a"], row["source_row_b"])
        if key in lookup.index:
            decision = lookup.loc[key]
            output.at[index, "match_tier"] = decision["match_tier"]
            output.at[index, "match_reason"] = f"manual:{decision['reason']}"
            output.at[index, "manual_review"] = True
    return output

```

```

def finalize_matches(candidates: pd.DataFrame) -> tuple[pd.DataFrame, pd.
↳ DataFrame]:
    """finalize_matches()."""
    audit = stable_sort(candidates, ["planet_key", "source_row_a",
↳ "source_row_b"])
    accepted = audit[audit["match_tier"].isin(["tier1", "tier2"])] .copy()
    if accepted.duplicated("source_row_a").any() or accepted.
↳ duplicated("source_row_b").any():
        raise RuntimeError("Accepted matches are not one-to-one.")
    return audit, accepted.reset_index(drop=True)

```

```

[10]: def semantic_eligibility(frame: pd.DataFrame, observable: PhysicalObservable,
↳ suffix: str) -> pd.Series:
    """semantic_eligibility()."""
    mask = pd.Series(True, index=frame.index)
    for variable in observable.inputs:
        column = f"{variable}-{suffix}"
        if column not in frame:
            return pd.Series(False, index=frame.index)
        values = pd.to_numeric(frame[column], errors="coerce")
        mask &= values.notna() & np.isfinite(values) & values.gt(0)
    return mask

def evaluate_monomial(frame: pd.DataFrame, observable: PhysicalObservable,
↳ suffix: str) -> pd.Series:
    """evaluate_monomial()."""
    eligible = semantic_eligibility(frame, observable, suffix)
    result = pd.Series(np.nan, index=frame.index, dtype=float)
    if eligible.any():
        log_value = np.full(int(eligible.sum()), math.log(observable.constant),
↳ dtype=float)
        with np.errstate(divide="raise", invalid="raise", over="raise"):
            for variable, exponent in zip(observable.inputs, observable.
↳ exponents):
                log_value += exponent * np.log(frame.loc[eligible,
↳ f"{variable}-{suffix}"].to_numpy(float))
                result.loc[eligible] = np.exp(log_value)
    return result

def enumerate_vertices(row: pd.Series, observable: PhysicalObservable) -> np.
↳ ndarray:
    """enumerate_vertices()."""
    pairs = [(float(row[f"{name}_a"]), float(row[f"{name}_b"])) for name in
↳ observable.inputs]

```



```

values: list[float] = []
for vertex in itertools.product(*pairs):
    log_value = math.log(observable.constant)
    for value, exponent in zip(vertex, observable.exponents):
        log_value += exponent * math.log(value)
    values.append(math.exp(log_value))
return np.asarray(values, dtype=float)

def exact_factorial_bounds(frame: pd.DataFrame, observable: PhysicalObservable) → pd.DataFrame:
    """exact_factorial_bounds()."""
    eligible_a = semantic_eligibility(frame, observable, "_a")
    eligible_b = semantic_eligibility(frame, observable, "_b")
    eligible = eligible_a & eligible_b
    output = pd.DataFrame(index=frame.index)
    output["eligible_a"] = eligible_a
    output["eligible_b"] = eligible_b
    output["eligible_both"] = eligible
    output["source_value_a"] = evaluate_monomial(frame, observable, "_a")
    output["source_value_b"] = evaluate_monomial(frame, observable, "_b")
    output["complete_row_lower"] = np.minimum(output["source_value_a"],
    → output["source_value_b"])
    output["complete_row_upper"] = np.maximum(output["source_value_a"],
    → output["source_value_b"])
    output["complete_row_log_width"] = np.log(output["complete_row_upper"] /
    → output["complete_row_lower"])
    output["factorial_lower"] = np.nan
    output["factorial_upper"] = np.nan
    output["factorial_log_width"] = np.nan
    if eligible.any():
        lower_log = np.full(int(eligible.sum()), math.log(observable.constant))
        upper_log = np.full(int(eligible.sum()), math.log(observable.constant))
        for variable, exponent in zip(observable.inputs, observable.exponents):
            la = np.log(frame.loc[eligible, f"{variable}_a"].to_numpy(float))
            lb = np.log(frame.loc[eligible, f"{variable}_b"].to_numpy(float))
            lo, hi = np.minimum(la, lb), np.maximum(la, lb)
            lower_log += exponent * (lo if exponent >= 0 else hi)
            upper_log += exponent * (hi if exponent >= 0 else lo)
            contribution = pd.Series(np.nan, index=frame.index, dtype=float)
            contribution.loc[eligible] = abs(exponent) * np.abs(la - lb)
            output[f"contribution_{variable}"] = contribution
        output.loc[eligible, "factorial_lower"] = np.exp(lower_log)
        output.loc[eligible, "factorial_upper"] = np.exp(upper_log)
        output.loc[eligible, "factorial_log_width"] = upper_log - lower_log
    output["complete_row_width"] = np.exp(output["complete_row_log_width"])
    output["factorial_width"] = np.exp(output["factorial_log_width"])

```

```

        output["factorial_inflation"] = output["factorial_width"] /  $\square$ 
    ↪ output["complete_row_width"]
        output["signed_log_ratio"] = np.log(output["source_value_a"] /  $\square$ 
    ↪ output["source_value_b"])
        output["absolute_log_ratio"] = output["signed_log_ratio"].abs()
        return output

def build_certificates(matches: pd.DataFrame, observable: PhysicalObservable)  $\square$ 
    ↪ -> pd.DataFrame:
        """build_certificates()."""
        bounds = exact_factorial_bounds(matches, observable)
        identity_columns = [
            "planet_key", "planet_name_source_a", "planet_name_source_b",  $\square$ 
    ↪ "host_key_a", "host_key_b",
            "source_row_a", "source_row_b", "match_tier", "match_reason",  $\square$ 
    ↪ "manual_review",
            "period_relative_difference", "discovery_method_class_a",  $\square$ 
    ↪ "discovery_method_class_b",
            "transit_evidence_a", "transit_evidence_b", "transit_concordance",
        ]
        for variable in observable.inputs:
            identity_columns.extend([f"{variable}_a", f"{variable}_b"])
        output = pd.concat([matches[identity_columns].copy(), bounds], axis=1)
        output["observable"] = observable.name
        output["observable_unit"] = observable.unit
        return output

def verify_certificates(certificates: pd.DataFrame, matches: pd.DataFrame,  $\square$ 
    ↪ observable: PhysicalObservable, tolerance: float, stage: str | None = None)  $\square$ 
    ↪ -> ValidationReport:
        """verify_certificates()."""
        report = ValidationReport(stage or f"certificate_verification:{observable.
    ↪ name}")
        eligible_indices = certificates.index[certificates["eligible_both"]]
        failures = 0
        contribution_columns = [f"contribution_{name}" for name in observable.
    ↪ inputs]
        for index in eligible_indices:
            vertices = enumerate_vertices(matches.loc[index], observable)
            expected = certificates.loc[index, ["factorial_lower",  $\square$ 
    ↪ "factorial_upper"]].to_numpy(float)
            if not np.allclose(expected, [vertices.min(), vertices.max()],  $\square$ 
    ↪ rtol=tolerance, atol=0):
                failures += 1

```

```

    if failures:
        report.add("blocking", "vertex_mismatch", "Analytical and enumerated_
↳ bounds differ.", failures)
        eligible = certificates.loc[eligible_indices]
        inside = eligible["source_value_a"].between(eligible["factorial_lower"],_
↳ eligible["factorial_upper"], inclusive="both") & eligible["source_value_b"].
↳ between(eligible["factorial_lower"], eligible["factorial_upper"],_
↳ inclusive="both")
        if not inside.all():
            report.add("blocking", "source_outside_bounds", "A complete-source_
↳ value lies outside its factorial bounds.", int((~inside).sum()))
            contribution_sum = eligible[contribution_columns].sum(axis=1)
            if not np.allclose(contribution_sum, eligible["factorial_log_width"],_
↳ rtol=tolerance, atol=tolerance):
                report.add("blocking", "contribution_identity", "Input contributions do_
↳ not sum to log width.")
                finite_columns = ["source_value_a", "source_value_b", "factorial_lower",_
↳ "factorial_upper", "factorial_width"]
                if not np.isfinite(eligible[finite_columns].to_numpy(float)).all():
                    report.add("blocking", "nonfinite_values", "Eligible certificates_
↳ contain non-finite values.")
                if (eligible["factorial_lower"] > eligible["factorial_upper"]).any() or_
↳ (eligible["factorial_width"] < 1).any():
                    report.add("blocking", "invalid_order", "Certificate bound ordering_
↳ failed.")
                if (eligible["factorial_width"] + tolerance <_
↳ eligible["complete_row_width"]).any():
                    report.add("blocking", "factorial_nesting", "Factorial width is smaller_
↳ than complete-row width.")
                report.metrics.update(eligible_rows=len(eligible_indices),_
↳ enumerated_rows=len(eligible_indices), failures=failures)
        return report

```

```

[11]: def add_source_ranks(certificates: pd.DataFrame) -> pd.DataFrame:
    """add_source_ranks()."""
    output = certificates.copy()
    for source in ("a", "b"):
        column = f"source_value_{source}"
        output[f"rank_{source}"] = output[column].rank(method="min",_
↳ ascending=False, na_option="bottom").astype("Int64")
        output["rank_displacement"] = (output["rank_a"] - output["rank_b"]).abs().
↳ astype("Int64")
    return output

def robust_top_k(certificates: pd.DataFrame, k: int) -> pd.DataFrame:

```

```

    """robust_top_k()."""
    valid = certificates.dropna(subset=["factorial_lower", "factorial_upper"]).
↪copy()
    lower = valid["factorial_lower"].to_numpy(float)
    upper = valid["factorial_upper"].to_numpy(float)
    sorted_upper = np.sort(upper)
    sorted_lower = np.sort(lower)
    could_outrank = len(valid) - np.searchsorted(sorted_upper, lower,
↪side="right") - (upper > lower).astype(int)
    always_outrank = len(valid) - np.searchsorted(sorted_lower, upper,
↪side="right")
    valid["worst_rank"] = 1 + np.maximum(could_outrank, 0)
    valid["best_rank"] = 1 + always_outrank
    valid["top_k_status"] = np.select(
        [valid["worst_rank"].le(k), valid["best_rank"].le(k)],
        ["guaranteed", "possible"], default="excluded",
    )
    valid["k"] = k
    return stable_sort(valid, ["worst_rank", "best_rank", "planet_key"])

def verify_ranking(ranking: pd.DataFrame, k: int, expected_n: int) ->
↪ValidationReport:
    """verify_ranking()."""
    report = ValidationReport(f"robust_ranking:k={k}")
    valid_status = ranking["top_k_status"].isin(["guaranteed", "possible",
↪"excluded"])
    if len(ranking) != expected_n:
        report.add("blocking", "ranking_size", "Ranking size differs from
↪operational cohort.", len(ranking), expected=expected_n)
    if not valid_status.all():
        report.add("blocking", "invalid_status", "Unknown robust-ranking status.
↪", int((~valid_status).sum()))
    if not ranking.loc[ranking["top_k_status"].eq("guaranteed"), "worst_rank"].
↪le(k).all():
        report.add("blocking", "guarantee_condition", "Guaranteed item violates
↪worst-rank condition.")
    if not ranking.loc[ranking["top_k_status"].eq("excluded"), "best_rank"].
↪gt(k).all():
        report.add("blocking", "exclusion_condition", "Excluded item violates
↪best-rank condition.")
    report.metrics.update(rows=len(ranking),
↪guaranteed=int(ranking["top_k_status"].eq("guaranteed").sum()),
↪possible=int(ranking["top_k_status"].eq("possible").sum()),
↪excluded=int(ranking["top_k_status"].eq("excluded").sum()))
    return report

```

```

def ranking_agreement(certificates: pd.DataFrame, k: int) -> dict[str, Any]:
    """ranking_agreement()."""
    valid = certificates.dropna(subset=["source_value_a", "source_value_b"]).
    ↪copy()
    top_a = set(valid.nlargest(k, "source_value_a")["planet_key"])
    top_b = set(valid.nlargest(k, "source_value_b")["planet_key"])
    intersection, union = top_a & top_b, top_a | top_b
    displacement = valid["rank_displacement"].astype(float)
    return {
        "n_ranking": len(valid),
        "spearman_rho": float(stats.spearmanr(valid["source_value_a"],
    ↪valid["source_value_b"]).statistic),
        "kendall_tau": float(stats.kendalltau(valid["source_value_a"],
    ↪valid["source_value_b"]).statistic),
        "top_k_intersection": len(intersection), "top_k_union": len(union),
        "top_k_jaccard": len(intersection) / len(union) if union else
    ↪float("nan"),
        "median_rank_displacement": float(displacement.median()),
        "p95_rank_displacement": float(displacement.quantile(0.95)),
        "p99_rank_displacement": float(displacement.quantile(0.99)),
        "maximum_rank_displacement": int(displacement.max()),
    }

```

```

[12]: def summarize_widths(certificates: pd.DataFrame, factors: Sequence[float]) ->
    ↪dict[str, Any]:
    """summarize_widths()."""
    work = certificates.dropna(subset=["factorial_log_width"]).copy()
    factorial = work["factorial_width"]
    complete = work["complete_row_width"]
    summary: dict[str, Any] = {
        "n_primary": len(work),
        "median_factorial_width": float(factorial.median()),
        "q75_factorial_width": float(factorial.quantile(0.75)),
        "q90_factorial_width": float(factorial.quantile(0.90)),
        "q95_factorial_width": float(factorial.quantile(0.95)),
        "q99_factorial_width": float(factorial.quantile(0.99)),
        "maximum_factorial_width": float(factorial.max()),
        "median_complete_row_width": float(complete.median()),
        "median_factorial_inflation": float(work["factorial_inflation"].
    ↪median()),
        "q90_factorial_inflation": float(work["factorial_inflation"].quantile(0.
    ↪90)),
        "q95_factorial_inflation": float(work["factorial_inflation"].quantile(0.
    ↪95)),

```

```

        "q99_factorial_inflation": float(work["factorial_inflation"].quantile(0.
↪99)),
        "maximum_factorial_inflation": float(work["factorial_inflation"].max()),
    }
    for factor in factors:
        label = str(factor).replace(".", "p")
        summary[f"count_factorial_at_or_below_{label}"] = int(factorial.
↪le(factor).sum())
        summary[f"fraction_factorial_at_or_below_{label}"] = float(factorial.
↪le(factor).mean())
        summary[f"count_factorial_above_{label}"] = int(factorial.gt(factor).
↪sum())
        summary[f"fraction_factorial_above_{label}"] = float(factorial.
↪gt(factor).mean())
        for factor in (1.01, 1.05, 1.25, 2.0):
            label = str(factor).replace(".", "p")
            summary[f"fraction_inflation_above_{label}"] =
↪float(work["factorial_inflation"].gt(factor).mean())
        return summary

def threshold_curve(certificates: pd.DataFrame, factors: Sequence[float]) -> pd.
↪DataFrame:
    """threshold_curve()."""
    widths = certificates["factorial_width"].dropna()
    return pd.DataFrame({
        "factor": list(factors),
        "count_at_or_below": [int(widths.le(factor).sum()) for factor in
↪factors],
        "fraction_at_or_below": [float(widths.le(factor).mean()) for factor in
↪factors],
        "count_above": [int(widths.gt(factor).sum()) for factor in factors],
        "fraction_above": [float(widths.gt(factor).mean()) for factor in
↪factors],
        "n_primary": len(widths),
    })

def k_curve(certificates: pd.DataFrame, values: Sequence[int]) -> pd.DataFrame:
    """k_curve()."""
    rows = []
    for k in values:
        ranking = robust_top_k(certificates, k)
        rows.append({
            "k": k,
            "guaranteed": int(ranking["top_k_status"].eq("guaranteed").sum()),

```

```

        "possible": int(ranking["top_k_status"].eq("possible").sum()),
        "excluded": int(ranking["top_k_status"].eq("excluded").sum()),
    })
    return pd.DataFrame(rows)

def paired_log_summary(a: pd.Series, b: pd.Series) -> dict[str, Any]:
    """paired_log_summary()."""
    valid = a.notna() & b.notna() & a.gt(0) & b.gt(0)
    logs = np.log(a[valid].to_numpy(float) / b[valid].to_numpy(float))
    if not len(logs):
        return {"n": 0}
    return {
        "n": len(logs), "median_signed_log_ratio": float(np.median(logs)),
        "median_absolute_log_ratio": float(np.median(np.abs(logs))),
        "q05_log_ratio": float(np.quantile(logs, 0.05)), "q95_log_ratio":
↪float(np.quantile(logs, 0.95)),
    }

def host_composition_sensitivity(frame: pd.DataFrame, value_column: str,
↪host_column: str, replicates: int, seed: int) -> pd.Series:
    """host_composition_sensitivity()."""
    work = frame.dropna(subset=[value_column]).copy()
    work[host_column] = work[host_column].fillna(work["planet_key"])
    groups = [group[value_column].to_numpy(float) for _, group in work.
↪groupby(host_column, sort=True)]
    if not groups:
        raise ValueError("No host groups available for composition sensitivity.
↪")
    generator = np.random.default_rng(seed)
    estimates = np.empty(replicates, dtype=float)
    for index in range(replicates):
        selected = generator.integers(0, len(groups), size=len(groups))
        estimates[index] = np.median(np.concatenate([groups[group] for group in
↪selected]))
    return pd.Series(estimates, name="median_factorial_width")

def composition_interval(values: pd.Series, level: float) -> tuple[float,
↪float]:
    """composition_interval()."""
    alpha = 1 - level
    return tuple(np.quantile(values.to_numpy(float), [alpha / 2, 1 - alpha /
↪2])).tolist()

```

```

[13]: def select_forensic_cases(primary: pd.DataFrame, operational: pd.DataFrame,
    ↪secondary: pd.DataFrame, config: AnalysisConfig) -> pd.DataFrame:
    """select_forensic_cases()."""
    operational_fields = operational[["planet_key", "best_rank", "worst_rank",
    ↪"top_k_status"]].drop_duplicates("planet_key")
    primary_work = primary.drop(columns=["best_rank", "worst_rank",
    ↪"top_k_status"], errors="ignore").merge(
        operational_fields, on="planet_key", how="left", validate="one_to_one"
    )
    primary_work["selected_primary_extreme"] = (
        primary_work["factorial_width"].ge(config.extreme_width_threshold)
        | primary_work["complete_row_width"].ge(config.extreme_width_threshold)
    )
    primary_work["selected_rank_displacement"] =
    ↪primary_work["rank_displacement"].astype(float).ge(config.
    ↪extreme_rank_displacement)
    primary_work["selected_operational_candidate"] =
    ↪primary_work["top_k_status"].isin(["guaranteed", "possible"])
    stable_keys = set(primary_work.nsmallest(min(10, len(primary_work)),
    ↪"factorial_width")["planet_key"])
    primary_work["selected_stable_reference"] = primary_work["planet_key"].
    ↪isin(stable_keys)
    primary_work["selected_secondary_extreme"] = False

    secondary_work = secondary.copy()
    secondary_work["selected_primary_extreme"] = False
    secondary_work["selected_rank_displacement"] = False
    secondary_work["selected_operational_candidate"] = False
    secondary_work["selected_stable_reference"] = False
    secondary_work["selected_secondary_extreme"] =
    ↪secondary_work["factorial_width"].ge(config.extreme_width_threshold)

    flag_columns = [
        "selected_primary_extreme", "selected_rank_displacement",
    ↪"selected_operational_candidate",
        "selected_secondary_extreme", "selected_stable_reference",
    ]
    combined = pd.concat([primary_work, secondary_work], ignore_index=True,
    ↪sort=False)
    combined = combined.loc[combined[flag_columns].any(axis=1)].copy()
    reason_names = {
        "selected_primary_extreme": "primary_extreme",
        "selected_rank_displacement": "rank_displacement",
        "selected_operational_candidate": "operational_candidate",
        "selected_secondary_extreme": "secondary_extreme",
        "selected_stable_reference": "stable_reference",
    }

```



```

    }
    combined["selection_reasons"] = combined.apply(
        lambda row: ";".join(reason for column, reason in reason_names.items()
        ↪if bool(row[column])), axis=1
    )
    combined["review_priority"] = np.select(
        [
            combined["selected_operational_candidate"]
            | combined["factorial_width"].ge(10)
            | combined["rank_displacement"].astype(float).ge(500),
            combined["selected_primary_extreme"] |
        ↪combined["selected_secondary_extreme"] |
        ↪combined["selected_rank_displacement"],
            combined["selected_stable_reference"],
        ],
        [0, 1, 3], default=2,
    ).astype(int)
    combined["review_classification"] = "pending"
    combined["review_finding"] = ""
    combined["evidence_reference"] = ""
    combined["reviewer"] = ""
    combined["review_date"] = ""
    combined["second_reviewer"] = ""
    combined["adjudication"] = ""
    return stable_sort(combined, ["review_priority", "factorial_width",
    ↪"planet_key"])

def load_forensic_review_decisions(path: Path) -> pd.DataFrame:
    """load_forensic_review_decisions()."""
    columns = [
        "planet_key", "observable", "review_classification", "review_finding",
        ↪"evidence_reference",
        "reviewer", "review_date", "second_reviewer", "adjudication",
    ]
    if not path.is_file():
        return pd.DataFrame(columns=columns)
    decisions = pd.read_csv(path, dtype="string").fillna("")
    missing = set(columns) - set(decisions.columns)
    if missing:
        raise ValueError(f"Forensic review decisions are missing columns:
        ↪{sorted(missing)}")
    if decisions.duplicated(["planet_key", "observable"]).any():
        raise ValueError("Forensic review decisions must be unique by planet
        ↪and observable.")

```

```

        allowed = {"", "pending", "confirmed_source_difference",
        ↪ "confirmed_agreement", "corrected_source_error", "source_semantic_mismatch",
        ↪ "matching_error", "unit_error", "parameter_semantics_difference",
        ↪ "data_error", "unresolved"}
        invalid = set(decisions["review_classification"].dropna()) - allowed
        if invalid:
            raise ValueError(f"Unknown forensic review classifications:
        ↪ {sorted(invalid)}")
        return decisions[columns].copy()

def apply_forensic_review_decisions(cases: pd.DataFrame, decisions: pd.
    ↪ DataFrame) -> pd.DataFrame:
    """apply_forensic_review_decisions()."""
    if decisions.empty:
        return cases.copy()
    case_keys = set(map(tuple, cases[["planet_key", "observable"]].
    ↪ astype("string").to_numpy()))
    decision_keys = set(map(tuple, decisions[["planet_key", "observable"]].
    ↪ astype("string").to_numpy()))
    unknown = decision_keys - case_keys
    if unknown:
        raise ValueError(f"Forensic review decisions reference unselected cases:
    ↪ {sorted(unknown)[:10]}")
    review_columns = [column for column in decisions.columns if column not in
    ↪ {"planet_key", "observable"}]
    output = cases.drop(columns=review_columns, errors="ignore").merge(
        decisions, on=["planet_key", "observable"], how="left",
    ↪ validate="one_to_one"
    )
    for column in review_columns:
        output[column] = output[column].fillna("")
    output["review_classification"] = output["review_classification"].
    ↪ replace("", "pending")
    return output

def validation_issues_frame(reports: Sequence[ValidationReport]) -> pd.
    ↪ DataFrame:
    """validation_issues_frame()."""
    rows: list[dict[str, Any]] = []
    for report in reports:
        if not report.issues:
            rows.append({"stage": report.stage, "severity": "information",
        ↪ "code": "passed", "message": "No issues.", "count": 0})
        for issue in report.issues:

```

```

        rows.append({
            "stage": report.stage, "severity": issue.severity, "code":
↪issue.code,
            "message": issue.message, "count": issue.count,
            "context": json.dumps(dict(issue.context), sort_keys=True,
↪default=json_default),
        })
    return pd.DataFrame(rows)

```

```

[14]: def run_physics_reference_tests(tolerance: float) -> ValidationReport:
    """run_physics_reference_tests()."""
    report = ValidationReport("physics_reference_tests")
    cases = {
        "geometric_area_ratio_ppm": (R_EARTH_OVER_R_SUN ** 2 * 1e6, 83.863467),
        "density_g_cm3": (EARTH_DENSITY_G_CM3, 5.514),
        "gravity_gearth": (1.0, 1.0),
        "escape_velocity_km_s": (EARTH_ESCAPE_VELOCITY_KM_S, 11.186),
        "insolation_searth": (T_SUN_K ** -4 * T_SUN_K ** 4, 1.0),
    }
    for name, (calculated, expected) in cases.items():
        if not math.isclose(calculated, expected, rel_tol=max(tolerance, 1e-7),
↪abs_tol=1e-9):
            report.add("blocking", "reference_value", f"Known-value test failed
↪for {name}." , calculated=calculated, expected=expected)
    report.metrics["tests"] = len(cases)
    return report

def run_ranking_reference_tests() -> ValidationReport:
    """run_ranking_reference_tests()."""
    report = ValidationReport("ranking_reference_tests")
    point = pd.DataFrame({
        "planet_key": ["a", "b", "c"],
        "factorial_lower": [3.0, 2.0, 1.0],
        "factorial_upper": [3.0, 2.0, 1.0],
    })
    point_result = robust_top_k(point, 1)
    status = dict(zip(point_result["planet_key"], point_result["top_k_status"]))
    if status != {"a": "guaranteed", "b": "excluded", "c": "excluded"}:
        report.add("blocking", "point_ranking", "Point-interval top-1 reference
↪test failed." , status=status)
    overlap = pd.DataFrame({
        "planet_key": ["a", "b"],
        "factorial_lower": [1.0, 1.5],
        "factorial_upper": [3.0, 2.5],
    })
    overlap_result = robust_top_k(overlap, 1)

```

```

    if not overlap_result["top_k_status"].eq("possible").all():
        report.add("blocking", "overlap_ranking", "Overlapping-interval_
↳reference test failed.")
    tied = pd.DataFrame({
        "planet_key": ["a", "b"],
        "factorial_lower": [2.0, 2.0],
        "factorial_upper": [2.0, 2.0],
    })
    tied_result = robust_top_k(tied, 1)
    if not tied_result["top_k_status"].eq("guaranteed").all():
        report.add("blocking", "tie_policy", "Exact-tie minimum-rank policy_
↳failed.")
    report.metrics["tests"] = 3
    return report

def run_contract_reference_tests(tolerance: float) -> ValidationReport:
    """run_contract_reference_tests()."""
    report = ValidationReport("data_contract_reference_tests")
    tests = 0

    identifiers = [" Kepler-10  b  ", "TOI - 123 b", "Kepler-16 (AB) b"]
    for value in identifiers:
        normalized = normalize_identifier(value)
        tests += 1
        if normalize_identifier(normalized) != normalized:
            report.add("blocking", "normalization_not_idempotent", "Identifier_
↳normalization is not idempotent.", value=value)
        tests += 1
        if extract_planet_component("Kepler-10 b") != "b":
            report.add("blocking", "component_not_preserved", "Planet component_
↳extraction failed.")

    conversion_cases = [
        (1.0, R_JUPITER_TO_R_EARTH),
        (1.0, M_JUPITER_TO_M_EARTH),
    ]
    for original, factor in conversion_cases:
        tests += 1
        if not math.isclose((original * factor) / factor, original,
↳rel_tol=tolerance, abs_tol=tolerance):
            report.add("blocking", "conversion_roundtrip", "A declared unit_
↳conversion failed round-trip validation.", factor=factor)

    tests += 1

```

```

    if detection_rule("Eclipse Timing Variations").transit_evidence != "unresolved":
        report.add("blocking", "eclipse_timing_rule", "Eclipse timing was
incorrectly promoted to confirmed transit evidence.")
        tests += 1
    if detection_rule("Transit").transit_evidence != "confirmed":
        report.add("blocking", "transit_rule", "Explicit transit evidence was
not classified as confirmed.")
        tests += 1
    try:
        detection_rule("Unreviewed Synthetic Discovery Method")
        report.add("blocking", "unknown_detection_rule", "An unknown detection
label did not fail closed.")
    except ValueError:
        pass

    observable = OBSERVABLES["geometric_area_ratio_ppm"]
    equal = pd.DataFrame({
        "planet_radius_rearth_a": [1.0], "planet_radius_rearth_b": [1.0],
        "star_radius_rsun_a": [1.0], "star_radius_rsun_b": [1.0],
    })
    equal_bounds_a = exact_factorial_bounds(equal, observable)
    equal_bounds_b = exact_factorial_bounds(equal, observable)
    tests += 1
    if not math.isclose(float(equal_bounds_a.loc[0, "factorial_width"]), 1.0,
rel_tol=tolerance, abs_tol=tolerance):
        report.add("blocking", "zero_width_identity", "Identical sources did
not produce unit width.")
        tests += 1
    if not equal_bounds_a.equals(equal_bounds_b):
        report.add("blocking", "deterministic_repeat", "Repeated deterministic
calculation changed its output.")

    perturbed = equal.copy()
    perturbed["planet_radius_rearth_b"] = 2.0
    perturbed_bounds = exact_factorial_bounds(perturbed, observable)
    tests += 1
    if not math.isclose(float(perturbed_bounds.loc[0, "factorial_width"]), 4.0,
rel_tol=tolerance, abs_tol=tolerance):
        report.add("blocking", "single_coordinate_width", "Single-coordinate
exponent propagation failed.")

    report.metrics["tests"] = tests
    return report

```

```

def run_bound_property_tests(trials: int, seed: int, tolerance: float) -> ValidationReport:
    """run_bound_property_tests()."""
    report = ValidationReport("randomized_bound_property_tests")
    generator = np.random.default_rng(seed)
    failures = 0
    for trial in range(trials):
        n_inputs = int(generator.integers(1, 7))
        inputs = tuple(f"x{index}" for index in range(n_inputs))
        exponents = tuple(generator.uniform(-5, 5, size=n_inputs))
        constant = float(np.exp(generator.uniform(-3, 3)))
        observable = PhysicalObservable(
            f"property_{trial}", "property test", "arbitrary", inputs,
            exponents, constant,
            "test", "randomized identity test", "not a scientific observable",
        )
        values_a = np.exp(generator.uniform(-12, 12, size=n_inputs))
        values_b = np.exp(generator.uniform(-12, 12, size=n_inputs))
        row = {f"{name}_a": [a] for name, a in zip(inputs, values_a)}
        row.update({f"{name}_b": [b] for name, b in zip(inputs, values_b)})
        frame = pd.DataFrame(row)
        bounds = exact_factorial_bounds(frame, observable).iloc[0]
        vertices = enumerate_vertices(frame.iloc[0], observable)
        if not np.allclose(
            [bounds["factorial_lower"], bounds["factorial_upper"]],
            [vertices.min(), vertices.max()], rtol=tolerance, atol=0,
        ):
            failures += 1
    if failures:
        report.add("blocking", "random_bound_failure", "Randomized exact-bound_
property test failed.", failures)
    report.metrics.update(trials=trials, failures=failures)
    return report

def run_ranking_property_tests(trials: int, seed: int) -> ValidationReport:
    """run_ranking_property_tests()."""
    report = ValidationReport("randomized_ranking_property_tests")
    generator = np.random.default_rng(seed)
    failures = 0
    for trial in range(trials):
        n = int(generator.integers(2, 8))
        lower = np.exp(generator.uniform(-4, 4, size=n))
        upper = lower * np.exp(generator.uniform(0.01, 2, size=n))
        frame = pd.DataFrame({
            "planet_key": [f"p{index}" for index in range(n)],
            "factorial_lower": lower,

```

```

        "factorial_upper": upper,
    })
    calculated = robust_top_k(frame, max(1, n // 2)).set_index("planet_key")
    all_ranks = {key: [] for key in frame["planet_key"]}
    for choices in itertools.product((0, 1), repeat=n):
        values = np.where(np.asarray(choices, dtype=bool), upper, lower)
        ranks = pd.Series(values).rank(method="min", ascending=False).
        astype(int).to_numpy()
        for key, rank in zip(frame["planet_key"], ranks):
            all_ranks[key].append(int(rank))
        for key, ranks in all_ranks.items():
            if int(calculated.loc[key, "best_rank"]) != min(ranks) or
            int(calculated.loc[key, "worst_rank"]) != max(ranks):
                failures += 1
                break
    if failures:
        report.add("blocking", "random_ranking_failure", "Randomized_
        endpoint-enumeration ranking test failed.", failures)
        report.metrics.update(trials=trials, failures=failures)
    return report

physics_reference_report = run_physics_reference_tests(CONFIG.
    floating_tolerance)
ranking_reference_report = run_ranking_reference_tests()
contract_reference_report = run_contract_reference_tests(CONFIG.
    floating_tolerance)
bound_property_report = run_bound_property_tests(CONFIG.bound_property_trials,
    CONFIG.seed + 101, CONFIG.floating_tolerance)
ranking_property_report = run_ranking_property_tests(CONFIG.
    ranking_property_trials, CONFIG.seed + 202)
display(pd.concat([
    report_table(physics_reference_report),
    report_table(ranking_reference_report),
    report_table(contract_reference_report),
    report_table(bound_property_report), report_table(ranking_property_report),
], ignore_index=True))
if not all(report.passed for report in (physics_reference_report,
    ranking_reference_report, contract_reference_report, bound_property_report,
    ranking_property_report)):
    raise RuntimeError("Reference tests failed before empirical analysis.")

stage status computational_pass major_count
blocking_count issues tests trials failures
0 physics_reference_tests passed True 0
    0 5.0 NaN NaN

```

|   |                                   |        |       |     |   |
|---|-----------------------------------|--------|-------|-----|---|
| 1 | ranking_reference_tests           | passed | True  | 0   | ␣ |
| ↪ | 0                                 | 3.0    | NaN   | NaN |   |
| 2 | data_contract_reference_tests     | passed | True  | 0   | ␣ |
| ↪ | 0                                 | 12.0   | NaN   | NaN |   |
| 3 | randomized_bound_property_tests   | passed | True  | 0   | ␣ |
| ↪ | 0                                 | NaN    | 250.0 | 0.0 |   |
| 4 | randomized_ranking_property_tests | passed | True  | 0   | ␣ |
| ↪ | 0                                 | NaN    | 75.0  | 0.0 |   |

### 1.10 Frozen-input contract

The notebook never downloads or substitutes data. The two original CSV snapshots must exist at the declared relative paths. The run records their hashes, dimensions, software versions, and protocol configuration before analysis.

```
[15]: raw_a = load_csv(PATHS.input_a)
raw_b = load_csv(PATHS.input_b)
print(f"Input A: {raw_a.shape}; input B: {raw_b.shape}")
```

Input A: (3732, 98); input B: (6150, 31)

```
[16]: input_manifest = {
    "protocol_version": CONFIG.protocol_version,
    "protocol_date": CONFIG.protocol_date,
    "created_utc": datetime.now(timezone.utc).isoformat(),
    "software": SOFTWARE_VERSIONS,
    "platform": platform.platform(),
    "config": asdict(CONFIG),
    "inputs": {
        "a": {"path": PATHS.input_a.as_posix(), "sha256": sha256_file(PATHS.
↪input_a), "rows": len(raw_a), "columns": len(raw_a.columns)},
        "b": {"path": PATHS.input_b.as_posix(), "sha256": sha256_file(PATHS.
↪input_b), "rows": len(raw_b), "columns": len(raw_b.columns)},
    },
    "optional_inputs": {
        "forensic_review": optional_file_record(PATHS.forensic_review_input),
        "match_review": optional_file_record(PATHS.match_review_input),
        "source_provenance": optional_file_record(PATHS.provenance_input),
        "clean_environment_self_reproduction": optional_file_record(PATHS.
↪reproduction_input),
    },
    "code_assets": {
        "generator": optional_file_record(GENERATOR_PATH),
    },
    "publication_contract": PUBLICATION_CONTRACT,
}
display(input_manifest)
```

```
{'protocol_version': '2.4.0',
```



```

'protocol_date': '2026-08-14',
'created_utc': '2026-08-21T21:10:19.749781+00:00',
'software': {'python': '3.12.10',
'numpy': '2.0.2',
'pandas': '2.2.2',
'scipy': '1.15.3',
'matplotlib': '3.9.0',
'seaborn': '0.13.2',
'pypdf': '6.1.1'},
'platform': 'Windows-10-10.0.19045-SP0',
'config': {'protocol_version': '2.4.0',
'protocol_date': '2026-08-14',
'seed': 20260813,
'primary_k': 25,
'tier1_period_relative_tolerance': 0.001,
'tier2_period_relative_tolerance': 0.01,
'minimum_tier1_matches': 100,
'minimum_primary_cohort': 100,
'minimum_operational_cohort': 50,
'equivalence_factors': (1.01, 1.05, 1.1, 1.25, 1.5, 2.0, 5.0, 10.0),
'k_grid': (5, 10, 25, 50, 75, 100),
'composition_resamples': 10000,
'composition_interval_level': 0.95,
'extreme_width_threshold': 2.0,
'extreme_rank_displacement': 100,
'floating_tolerance': 1e-11,
'figure_format': 'pdf',
'figure_dpi': 300,
'allow_output_replacement': False,
'bound_property_trials': 250,
'ranking_property_trials': 75,
'is_solo_author': True,
'lead_author': 'Khaled Tarchi'},
☐ 'inputs': {'a': {'path': 'prism-uploads/kepler.csv',
'sha256': 'c945f9a5b9f69111add6ccdc7d84dd10b3d42cfe51ce78977148db832beab50e',
'rows': 3732,
'columns': 98},
'b': {'path': 'prism-uploads/nasa_exoplanet_intelligence.csv',
'sha256': '684d4ae9eaa375a07d50f1f521c4fbbbeb3a8cbbe4ec1e0b0598c2dd51f7da287',
'rows': 6150,
'columns': 31}},
'optional_inputs': {'forensic_review': {'path': 'prism-uploads/
forensic_review_decisions.csv',
'sha256': '4b1203dc2a547bac5274697bd61bc0b2f02d9a7ac1fccbc6765c820959b282d9',
'bytes': 117753},
'match_review': None,
'source_provenance': {'path': 'prism-uploads/source_provenance.json',

```

```

    'sha256': '5e1937aa0445f706cb4515325128719fd12a427529f95d96ea91679d4a0acbf4',
    'bytes': 6191},
    'clean_environment_self_reproduction': {'path': 'prism-uploads/
clean_environment_self_reproduction.json',
    'sha256': '30f44a69d187f95955ce13273f437203e5ef269c574e178b0e7d0f902589d0ff',
    'bytes': 1228}},
    'code_assets': {'generator': {'path': 'build_research_notebook.py',
    'sha256': '3a66ef3f46ba8e630226060cdf91e757538394b017e7d60f7a7a91fcedf1f712',
    'bytes': 70063}},
    'publication_contract': {'protocol_version': '2.4.0',
    'protocol_date': '2026-08-14',
    'unit_of_analysis': 'conservatively matched planet',
    'primary_population': 'Tier-1 matched complete cases for both radii',
    'operational_population': 'primary population with confirmed transit evidence
in both sources',
    'primary_estimand': 'exact finite-source factorial width of geometric area
ratio',
    'operational_estimand': 'guaranteed/possible/excluded geometric-signal top-25
membership',
    'required_external_certificates': ['complete source provenance',
    'completed priority-0 forensic review (solo-author self-audit)',
    'clean-environment reproduction certification (solo-author self-audit)'],
    'prohibited_claims': ['either source is ground truth',
    'factorial bounds are confidence or credible intervals',
    'geometric area ratio is an observed transit depth',
    'geometric top-25 membership is a mission target list',
    'results establish causality, habitability, or universal catalog quality']]

```

```

[17]: provenance_payload = load_optional_json(PATHS.provenance_input)
reproduction_payload = load_optional_json(PATHS.reproduction_input)

provenance_report = ValidationReport("source_provenance")
provenance_required = {"source_a", "source_b"}
provenance_source_fields = {
    "label", "publisher", "acquisition_date", "origin", "license",
    "derivative_status", "preprocessing_history", "sha256",
    "source_version", "persistent_identifier", "citation", "column_semantics",
}
if provenance_payload is None:
    provenance_report.add("major", "provenance_missing", "A complete
source-provenance record is required before submission.")
elif not provenance_required.issubset(provenance_payload):
    provenance_report.add("major", "provenance_sections_missing", "Source
provenance must document source_a and source_b.")
else:
    for source_key, input_key in (("source_a", "a"), ("source_b", "b")):
        record = provenance_payload[source_key]

```

```

        if not isinstance(record, dict) or not provenance_source_fields.
        ↪issubset(record):
            provenance_report.add("major", "provenance_fields_missing",
        ↪f"{source_key} has an incomplete provenance record; version, identifier,
        ↪citation, and column semantics are mandatory.")
            continue
            if str(record["sha256"]) !=
        ↪input_manifest["inputs"][input_key]["sha256"]:
                provenance_report.add("blocking", "provenance_hash_mismatch",
        ↪f"{source_key} hash does not match the frozen input.")
                acquisition_date = pd.to_datetime(record["acquisition_date"],
        ↪errors="coerce", utc=True)
                if pd.isna(acquisition_date) or acquisition_date > pd.Timestamp.
        ↪now(tz="UTC"):
                    provenance_report.add("major", "invalid_acquisition_date",
        ↪f"{source_key} acquisition date is invalid or in the future.")
    if provenance_payload is not None:
        for source_key in ("source_a", "source_b"):
            record = provenance_payload.get(source_key)
            if isinstance(record, dict):
                for field_key in ("label", "publisher", "license", "citation",
        ↪"origin", "source_version", "persistent_identifier", "derivative_status",
        ↪"preprocessing_history"):
                    if field_key in record and
        ↪contains_placeholder(record[field_key]):
                        provenance_report.add("major", "provenance_placeholder",
        ↪f"{source_key}.{field_key} contains a placeholder term.")
    provenance_documented = provenance_payload is not None and provenance_report.
    ↪major_count == 0 and provenance_report.passed

    reproduction_report = ValidationReport("clean_environment_self_reproduction")
    reproduction_required = {
        "protocol_version", "reviewer", "organization", "review_date",
        ↪"environment",
        "input_a_sha256", "input_b_sha256", "computational_certification_passed",
    }
    if reproduction_payload is None:
        reproduction_report.add("major", "reproduction_missing", "Clean-environment
        ↪reproduction (solo-author self-audit) is required before submission.")
    elif not reproduction_required.issubset(reproduction_payload):
        reproduction_report.add("major", "reproduction_fields_missing",
        ↪"Clean-environment reproduction attestation is incomplete.")
    else:
        if reproduction_payload["protocol_version"] != CONFIG.protocol_version:
            reproduction_report.add("major", "reproduction_protocol_mismatch",
        ↪"Reproduction used a different protocol version.")

```

```

    if reproduction_payload["input_a_sha256"] != \
    ↪ input_manifest["inputs"]["a"]["sha256"] or \
    ↪ reproduction_payload["input_b_sha256"] != \
    ↪ input_manifest["inputs"]["b"]["sha256"]:
        reproduction_report.add("major", "reproduction_input_mismatch", \
    ↪ "Reproduction used different frozen inputs.")
        review_date = pd.to_datetime(reproduction_payload["review_date"], \
    ↪ errors="coerce", utc=True)
        if pd.isna(review_date) or review_date > pd.Timestamp.now(tz="UTC"):
            reproduction_report.add("major", "invalid_reproduction_date", \
    ↪ "Reproduction date is invalid or in the future.")
        if not bool(reproduction_payload["computational_certification_passed"]):
            reproduction_report.add("major", "reproduction_failed", "Independent \
    ↪ reproduction did not pass computational certification.")
        if not str(reproduction_payload["reviewer"]).strip() or not \
    ↪ str(reproduction_payload["organization"]).strip() or not \
    ↪ str(reproduction_payload["environment"]).strip():
            reproduction_report.add("major", "reproduction_identity_missing", \
    ↪ "Reproduction reviewer, organization, and environment are required.")
        for field_key in ("reviewer", "organization", "environment"):
            if contains_placeholder(reproduction_payload[field_key]):
                reproduction_report.add("major", "reproduction_placeholder", \
    ↪ f"reproduction.{field_key} contains a placeholder term.")
clean_environment_self_reproduction_passed = reproduction_payload is not None \
    ↪ and reproduction_report.major_count == 0 and reproduction_report.passed
if clean_environment_self_reproduction_passed:
    reproduction_report.add(
        "information",
        "self_reproduction_passed",
        "Clean-environment self-reproduction passed in the authoring \
    ↪ environment as a solo-author self-audit.",
        is_author=True,
        reproduction_mode="clean_environment_self_reproduction",
        lead_author=CONFIG.lead_author,
    )
display(pd.concat([report_table(provenance_report), \
    ↪ report_table(reproduction_report)], ignore_index=True))

```

|   | major_count | blocking_count | stage                               | status               | computational_pass |
|---|-------------|----------------|-------------------------------------|----------------------|--------------------|
| 0 |             |                | source_provenance                   | passed               | True               |
| ↪ | 0           | 0              |                                     |                      |                    |
| 1 |             |                | clean_environment_self_reproduction | passed_with_warnings | True               |
| ↪ | 0           | 0              |                                     |                      |                    |

↪ issues

```

0
1 information:self_reproduction_passed:Clean-environment self-reproduction
  ↳passed in the authoring environment as a so...

```

```

[18]: raw_report_a = audit_raw_schema(raw_a, SCHEMA_A, "raw_schema_a")
      raw_report_b = audit_raw_schema(raw_b, SCHEMA_B, "raw_schema_b")
      display(pd.concat([report_table(raw_report_a), report_table(raw_report_b)],
        ↳ignore_index=True))
      if not raw_report_a.passed or not raw_report_b.passed:
        raise RuntimeError("Blocking raw-schema validation failure.")

```

```

          stage status computational_pass major_count blocking_count issues
      ↳rows columns duplicate_rows
0 raw_schema_a passed True 0 0
      ↳3732 98 0
1 raw_schema_b passed True 0 0
      ↳6150 31 0

```

```

[19]: canon_a, canonical_report_a = build_canonical_table(raw_a, SCHEMA_A, "a")
      canon_b, canonical_report_b = build_canonical_table(raw_b, SCHEMA_B, "b")
      display(pd.concat([report_table(canonical_report_a),
        ↳report_table(canonical_report_b)], ignore_index=True))
      if not canonical_report_a.passed or not canonical_report_b.passed:
        raise RuntimeError("Blocking canonicalization failure.")

```

```

          stage status computational_pass major_count
      ↳blocking_count \
0 canonicalization_a passed_with_warnings True 0
      ↳ 0
1 canonicalization_b passed_with_warnings True 0
      ↳ 0

      issues rows columns missing_planet_key \
0 minor:invalid_numeric:star_teff: invalid values set to missing. | minor:
      ↳invalid_numeric:eccentricity: invalid values... 3732 31
      ↳0
1
      minor:invalid_numeric:
      ↳star_temp_k: invalid values set to missing. 6150 31 0

      duplicate_planet_key_rows
0 0
1 0

```

```

[20]: canonical_audit = pd.DataFrame({
      "variable": [spec.canonical_name for spec in SCHEMA_A.columns],
      "missing_fraction_a": [float(canon_a[spec.canonical_name].isna().mean())
        ↳for spec in SCHEMA_A.columns],

```

```

    "missing_fraction_b": [float(canon_b[spec.canonical_name].isna().mean())
    ↪for spec in SCHEMA_B.columns],
    "invalid_count_a": [int(canon_a[f"{spec.canonical_name}_invalid"].sum())
    ↪for spec in SCHEMA_A.columns],
    "invalid_count_b": [int(canon_b[f"{spec.canonical_name}_invalid"].sum())
    ↪for spec in SCHEMA_B.columns],
})
if canon_a["planet_key"].dropna().str.contains(r"\s{2,}", regex=True).any() or
    ↪canon_b["planet_key"].dropna().str.contains(r"\s{2,}", regex=True).any():
    raise RuntimeError("Identifier normalization left repeated whitespace.")
display(canonical_audit)

```

|   | variable             | missing_fraction_a | missing_fraction_b | invalid_count_a | invalid_count_b |
|---|----------------------|--------------------|--------------------|-----------------|-----------------|
| 0 | planet_radius_rearth | 0.242765           | 0.008130           | 0               | 0               |
| 1 | planet_mass_mearth   | 0.575027           | 0.005041           | 0               | 0               |
| 2 | period_days          | 0.036710           | 0.054309           | 0               | 0               |
| 3 | semi_major_axis_au   | 0.395498           | 0.051382           | 0               | 0               |
| 4 | star_temperature_k   | 0.063505           | 0.047642           | 8               | 3               |
| 5 | star_radius_rsun     | 0.115220           | 0.051057           | 0               | 0               |
| 6 | star_mass_msun       | 0.101286           | 0.001301           | 0               | 0               |
| 7 | distance_pc          | 0.307610           | 0.004390           | 0               | 0               |
| 8 | visual_magnitude_mag | 0.592176           | 0.047967           | 0               | 0               |
| 9 | eccentricity         | 0.650589           | 0.152520           | 1               | 0               |

```

[21]: detection_mapping_audit = pd.concat([
    canon_a.groupby(["discovery_method_source", "discovery_method_class",
    ↪"transit_evidence"], dropna=False).size().rename("n").reset_index().
    ↪assign(source="a"),
    canon_b.groupby(["discovery_method_source", "discovery_method_class",
    ↪"transit_evidence"], dropna=False).size().rename("n").reset_index().
    ↪assign(source="b"),
], ignore_index=True)
detection_mapping_audit = stable_sort(detection_mapping_audit, ["source",
    ↪"transit_evidence", "discovery_method_class", "discovery_method_source"])

```

```

detection_mapping_audit["normalized_label"] =
    ↪detection_mapping_audit["discovery_method_source"].map(normalize_identifier)
detection_mapping_audit["rule_justification"] =
    ↪detection_mapping_audit["discovery_method_source"].map(lambda value:
    ↪detection_rule(value).justification)
if detection_mapping_audit[["discovery_method_class", "transit_evidence"]].
    ↪isna().any().any():
    raise RuntimeError("Detection-method mapping produced a missing controlled_
    ↪classification.")
if detection_mapping_audit.loc[detection_mapping_audit["transit_evidence"].
    ↪eq("confirmed"), "discovery_method_class"].eq("other").any():
    raise RuntimeError("Confirmed transit evidence cannot use the generic_
    ↪discovery family.")
display(detection_mapping_audit)

```

|         |   | discovery_method_source | discovery_method_class | transit_evidence | n    |
|---------|---|-------------------------|------------------------|------------------|------|
| ↪source |   | normalized_label \      |                        |                  |      |
| 0       |   | Astrometry              | astrometry             | absent           | 1    |
| ↪       | a | astrometry              |                        |                  |      |
| 1       |   | Imaging                 | imaging                | absent           | 93   |
| ↪       | a | imaging                 |                        |                  |      |
| 2       |   | Microlensing            | microlensing           | absent           | 65   |
| ↪       | a | microlensing            |                        |                  |      |
| 3       |   | Radial Velocity         | radial_velocity        | absent           | 745  |
| ↪       | a | radial velocity         |                        |                  |      |
| 4       |   | Pulsar                  | timing                 | absent           | 29   |
| ↪       | a | pulsar                  |                        |                  |      |
| 5       |   | Primary Transit         | transit                | confirmed        | 2778 |
| ↪       | a | primary transit         |                        |                  |      |
| 6       |   | Primary Transit, TTV    | transit                | confirmed        | 2    |
| ↪       | a | primary transit, ttv    |                        |                  |      |
| 7       |   | TTV                     | timing                 | supported        | 5    |
| ↪       | a | ttv                     |                        |                  |      |
| 8       |   | Other                   | other                  | unresolved       | 14   |
| ↪       | a | other                   |                        |                  |      |
| 9       |   | Astrometry              | astrometry             | absent           | 6    |
| ↪       | b | astrometry              |                        |                  |      |
| 10      |   | Imaging                 | imaging                | absent           | 94   |
| ↪       | b | imaging                 |                        |                  |      |
| 11      |   | Microlensing            | microlensing           | absent           | 275  |
| ↪       | b | microlensing            |                        |                  |      |
| 12      |   | Disk Kinematics         | other                  | absent           | 1    |
| ↪       | b | disk kinematics         |                        |                  |      |
| 13      |   | Radial Velocity         | radial_velocity        | absent           | 1182 |
| ↪       | b | radial velocity         |                        |                  |      |

|    |                                 |         |            |      |   |
|----|---------------------------------|---------|------------|------|---|
| 14 | Pulsar Timing                   | timing  | absent     | 8    | ▢ |
| ↪  | b pulsar timing                 |         |            |      |   |
| 15 | Pulsation Timing Variations     | timing  | absent     | 2    | ▢ |
| ↪  | b pulsation timing variations   |         |            |      |   |
| 16 | Transit Timing Variations       | timing  | confirmed  | 39   | ▢ |
| ↪  | b transit timing variations     |         |            |      |   |
| 17 | Transit                         | transit | confirmed  | 4517 | ▢ |
| ↪  | b transit                       |         |            |      |   |
| 18 | Orbital Brightness Modulation   | other   | unresolved | 9    | ▢ |
| ↪  | b orbital brightness modulation |         |            |      |   |
| 19 | Eclipse Timing Variations       | timing  | unresolved | 17   | ▢ |
| ↪  | b eclipse timing variations     |         |            |      |   |

↪rule\_justification

|                  |                                                                            |   |
|------------------|----------------------------------------------------------------------------|---|
| 0                | Astrometric discovery is not transit                                       | ▢ |
| ↪evidence.       |                                                                            |   |
| 1                | Direct imaging is not transit                                              | ▢ |
| ↪evidence.       |                                                                            |   |
| 2                | Microlensing is not transit                                                | ▢ |
| ↪evidence.       |                                                                            |   |
| 3                | Radial velocity is not transit                                             | ▢ |
| ↪evidence.       |                                                                            |   |
| 4                | Pulsar discovery is timing based, not transit                              | ▢ |
| ↪based.          |                                                                            |   |
| 5                | The source explicitly identifies primary-transit                           | ▢ |
| ↪discovery.      |                                                                            |   |
| 6                | Primary-transit evidence is explicit; TTV is                               | ▢ |
| ↪additional.     |                                                                            |   |
| 7                | The abbreviation alone supports but does not independently certify transit | ▢ |
| ↪status.         |                                                                            |   |
| 8                | The source label has no controlled transit                                 | ▢ |
| ↪interpretation. |                                                                            |   |
| 9                | Astrometric discovery is not transit                                       | ▢ |
| ↪evidence.       |                                                                            |   |
| 10               | Direct imaging is not transit                                              | ▢ |
| ↪evidence.       |                                                                            |   |
| 11               | Microlensing is not transit                                                | ▢ |
| ↪evidence.       |                                                                            |   |
| 12               | Disk-kinematic discovery is not transit                                    | ▢ |
| ↪evidence.       |                                                                            |   |
| 13               | Radial velocity is not transit                                             | ▢ |
| ↪evidence.       |                                                                            |   |
| 14               | Pulsar timing is not transit                                               | ▢ |
| ↪evidence.       |                                                                            |   |



```

15                                Pulsation timing is not transit_
    ↪evidence.
16                                The source label explicitly states transit timing_
    ↪variations.
17                                The source explicitly identifies transit_
    ↪discovery.
18                                Photometric modulation is insufficient for confirmed transit_
    ↪status.
19                                Eclipse timing alone does not certify a planetary_
    ↪transit.

```

### 1.11 Conservative matching

Candidates are generated by normalized planet identifier. Period agreement is displayed for all candidates before tier assignment. Tier 1 requires a unique identifier in each source and period consistency within 0.1%, where the relative period discrepancy is

$$\delta_P = \frac{|P_A - P_B|}{\min(P_A, P_B)}.$$

This is a matching rule, not independent validation evidence.

All tie handling uses minimum-rank (competition) semantics: when multiple rows share an identical score they receive the same rank, and the next distinct score receives a rank equal to the number of rows already assigned. A top- $k$  set defined by minimum rank may therefore contain more than  $k$  members, because all rows tied at the boundary rank are included.

```

[22]: match_candidates = add_matching_diagnostics(generate_name_candidates(canon_a,
    ↪canon_b))
candidate_period_summary =
    ↪paired_log_summary(match_candidates["period_days_a"],
    ↪match_candidates["period_days_b"])
candidate_period_summary["n_name_candidates"] = len(match_candidates)
display(pd.DataFrame([candidate_period_summary]))

      n  median_signed_log_ratio  median_absolute_log_ratio  q05_log_ratio
    ↪q95_log_ratio  n_name_candidates
0  3282                      0.0                      0.0      -0.00046
    ↪0.000795                      3343

```

```

[23]: tiered_candidates = assign_match_tiers(match_candidates, CONFIG)
manual_decisions = load_match_review_decisions(PATHS.match_review_input)
tiered_candidates = apply_manual_decisions(tiered_candidates, manual_decisions)
match_audit, accepted_matches = finalize_matches(tiered_candidates)
tier1_matches = accepted_matches[accepted_matches["match_tier"].eq("tier1")].
    ↪copy().reset_index(drop=True)
if len(tier1_matches) < CONFIG.minimum_tier1_matches:
    raise RuntimeError(f"Tier-1 cohort too small: {len(tier1_matches)}")
tier_counts = match_audit["match_tier"].value_counts(dropna=False)
matching_flow = pd.DataFrame({

```

```

    "stage": [
        "Name candidates", "Candidates with paired periods", "Candidates_
↳missing at least one period",
        "Tier 1", "Tier 2", "Rejected", "Manually reviewed",
    ],
    "n": [
        len(match_candidates), int(match_candidates["period_available_both"].
↳sum()), int((~match_candidates["period_available_both"]).sum()),
        int(tier_counts.get("tier1", 0)), int(tier_counts.get("tier2", 0)),
↳int(tier_counts.get("rejected", 0)),
        int(match_audit["manual_review"].sum()),
    ],
})
display(matching_flow)

```

|   | stage                                  | n    |
|---|----------------------------------------|------|
| 0 | Name candidates                        | 3343 |
| 1 | Candidates with paired periods         | 3282 |
| 2 | Candidates missing at least one period | 61   |
| 3 | Tier 1                                 | 2981 |
| 4 | Tier 2                                 | 198  |
| 5 | Rejected                               | 164  |
| 6 | Manually reviewed                      | 0    |

```

[24]: availability_records: list[dict[str, Any]] = []
for observable in OBSERVABLES.values():
    eligible_a = semantic_eligibility(tier1_matches, observable, "_a")
    eligible_b = semantic_eligibility(tier1_matches, observable, "_b")
    states = np.select([eligible_a & eligible_b, eligible_a, eligible_b],
↳["both", "a_only", "b_only"], default="neither")
    for planet_key, state in zip(tier1_matches["planet_key"], states):
        availability_records.append({"planet_key": planet_key, "observable":
↳observable.name, "availability_state": state})
availability = pd.DataFrame(availability_records)
availability_summary = pd.crosstab(availability["observable"],
↳availability["availability_state"])
display(availability_summary)

```

| availability_state       | a_only | b_only | both | neither |
|--------------------------|--------|--------|------|---------|
| observable               |        |        |      |         |
| density_g_cm3            | 1      | 2381   | 558  | 41      |
| escape_velocity_km_s     | 1      | 2381   | 558  | 41      |
| geometric_area_ratio_ppm | 1      | 361    | 2588 | 31      |
| gravity_gearth           | 1      | 2381   | 558  | 41      |
| insolation_searth        | 3      | 1490   | 1474 | 14      |

```
[25]: primary_observable = OBSERVABLES["geometric_area_ratio_ppm"]
primary_mask = semantic_eligibility(tier1_matches, primary_observable, "_a") &
↳ semantic_eligibility(tier1_matches, primary_observable, "_b")
primary_cohort = tier1_matches.loc[primary_mask].copy().reset_index(drop=True)
operational_mask = primary_mask & tier1_matches["transit_concordance"].
↳ eq("both_transit")
operational_cohort = tier1_matches.loc[operational_mask].copy().
↳ reset_index(drop=True)
transit_sensitivity_mask = primary_mask & tier1_matches["transit_concordance"].
↳ isin(["both_transit", "a_transit_only", "b_transit_only"])
transit_sensitivity_cohort = tier1_matches.loc[transit_sensitivity_mask].copy().
↳ reset_index(drop=True)
if len(primary_cohort) < CONFIG.minimum_primary_cohort:
    raise RuntimeError("Primary cohort below its declared minimum.")
if len(operational_cohort) < CONFIG.minimum_operational_cohort:
    raise RuntimeError("Concordant-transit operational cohort below its
↳ declared minimum.")
cohort_flow = pd.DataFrame({
    "stage": ["Raw A", "Raw B", "Tier 1", "Primary paired radii", "Concordant
↳ transits"],
    "n": [len(raw_a), len(raw_b), len(tier1_matches), len(primary_cohort),
↳ len(operational_cohort)],
})
display(cohort_flow)
```

|   | stage                | n    |
|---|----------------------|------|
| 0 | Raw A                | 3732 |
| 1 | Raw B                | 6150 |
| 2 | Tier 1               | 2981 |
| 3 | Primary paired radii | 2588 |
| 4 | Concordant transits  | 2561 |

```
[26]: operational_exclusion_audit = tier1_matches.loc[primary_mask &
↳ ~tier1_matches["transit_concordance"].eq("both_transit"), [
    "planet_key", "planet_name_source_a", "planet_name_source_b",
↳ "discovery_method_source_a", "discovery_method_source_b",
    "discovery_method_class_a", "discovery_method_class_b",
↳ "transit_evidence_a", "transit_evidence_b",
    "transit_concordance", "period_relative_difference",
]]
generalizability_rows: list[dict[str, Any]] = []
excluded_primary = tier1_matches.loc[~primary_mask].copy()
log_scale_variables = {"period_days", "semi_major_axis_au", "star_mass_msun",
↳ "distance_pc"}
for source in ("a", "b"):
```

```

    for variable in ("period_days", "semi_major_axis_au", "star_temperature_k",
↳"star_mass_msun", "distance_pc", "visual_magnitude_mag"):
        included_values = primary_cohort[f"{variable}_{source}"].dropna().
↳astype(float)
        excluded_values = excluded_primary[f"{variable}_{source}"].dropna().
↳astype(float)
        scale = "log" if variable in log_scale_variables else "raw"
        included_analysis = np.log(included_values[included_values.gt(0)]) if
↳scale == "log" else included_values
        excluded_analysis = np.log(excluded_values[excluded_values.gt(0)]) if
↳scale == "log" else excluded_values
        pooled_variance = 0.5 * (included_analysis.var(ddof=1) +
↳excluded_analysis.var(ddof=1)) if len(included_analysis) > 1 and
↳len(excluded_analysis) > 1 else np.nan
        standardized_difference = (included_analysis.mean() - excluded_analysis.
↳mean()) / math.sqrt(pooled_variance) if np.isfinite(pooled_variance) and
↳pooled_variance > 0 else np.nan
        generalizability_rows.append({
            "source": source, "variable": variable, "analysis_scale": scale,
            "n_included": len(included_values), "n_excluded":
↳len(excluded_values),
            "median_included_original_scale": float(included_values.median())
↳if len(included_values) else np.nan,
            "median_excluded_original_scale": float(excluded_values.median())
↳if len(excluded_values) else np.nan,
            "standardized_mean_difference": float(standardized_difference) if
↳np.isfinite(standardized_difference) else np.nan,
        })
generalizability_summary = pd.DataFrame(generalizability_rows)
display(operational_exclusion_audit)
display(generalizability_summary)

```

|    | planet_key                | planet_name_source_a | planet_name_source_b      |         |  |
|----|---------------------------|----------------------|---------------------------|---------|--|
|    | discovery_method_source_a |                      | discovery_method_source_b |         |  |
|    | discovery_method_class_a  |                      | discovery_method_class_b  | \       |  |
| 6  | 51 peg b                  | 51 Peg b             | 51 Peg b                  | Radial  |  |
|    | ↳Velocity                 | Radial Velocity      | radial_velocity           |         |  |
|    | ↳radial_velocity          |                      |                           |         |  |
| 9  | 55 cnc e                  | 55 Cnc e             | 55 Cnc e                  | Primary |  |
|    | ↳Transit                  | Radial Velocity      | transit                   |         |  |
|    | ↳radial_velocity          |                      |                           |         |  |
| 30 | bd-10 3166 b              | BD-10 3166 b         | BD-10 3166 b              | Radial  |  |
|    | ↳Velocity                 | Radial Velocity      | radial_velocity           |         |  |
|    | ↳radial_velocity          |                      |                           |         |  |
| 88 | gj 3470 b                 | GJ 3470 b            | GJ 3470 b                 | Primary |  |
|    | ↳Transit                  | Radial Velocity      | transit                   |         |  |
|    | ↳radial_velocity          |                      |                           |         |  |

|     |                  |                 |                 |         |
|-----|------------------|-----------------|-----------------|---------|
| 94  | gj 436 b         | GJ 436 b        | GJ 436 b        | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 290 | hd 149026 b      | HD 149026 b     | HD 149026 b     | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 318 | hd 17156 b       | HD 17156 b      | HD 17156 b      | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 326 | hd 179949 b      | HD 179949 b     | HD 179949 b     | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 333 | hd 189733 b      | HD 189733 b     | HD 189733 b     | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 349 | hd 209458 b      | HD 209458 b     | HD 209458 b     | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 363 | hd 219134 b      | HD 219134 b     | HD 219134 b     | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 427 | hd 46375 b       | HD 46375 b      | HD 46375 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 430 | hd 49674 b       | HD 49674 b      | HD 49674 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 455 | hd 75289 b       | HD 75289 b      | HD 75289 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 456 | hd 76700 b       | HD 76700 b      | HD 76700 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 460 | hd 80606 b       | HD 80606 b      | HD 80606 b      | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |
| 464 | hd 83443 b       | HD 83443 b      | HD 83443 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 467 | hd 88133 b       | HD 88133 b      | HD 88133 b      | Radial  |
|     | ↳Velocity        | Radial Velocity | radial_velocity |         |
|     | ↳radial_velocity |                 |                 |         |
| 478 | hd 97658 b       | HD 97658 b      | HD 97658 b      | Primary |
|     | ↳Transit         | Radial Velocity | transit         |         |
|     | ↳radial_velocity |                 |                 |         |

|      |                                        |                    |                           |         |   |
|------|----------------------------------------|--------------------|---------------------------|---------|---|
| 567  | kelt-4 a b                             | KELT-4 A b         | KELT-4 A b                | Radial  |   |
|      | ↳Velocity                              | Transit            | radial_velocity           |         | ↳ |
|      | ↳ transit                              |                    |                           |         |   |
| 1074 | kepler-138 d                           | Kepler-138 d       | Kepler-138 d              | Radial  |   |
|      | ↳Velocity                              | Transit            | radial_velocity           |         | ↳ |
|      | ↳ transit                              |                    |                           |         |   |
| 1911 | kepler-338 e                           | Kepler-338 e       | Kepler-338 e              |         | ↳ |
|      | ↳ TTV Transit Timing Variations        |                    | timing                    |         | ↳ |
|      | ↳ timing                               |                    |                           |         |   |
| 2508 | kepler-76 b                            | Kepler-76 b        | Kepler-76 b               | Primary | ↳ |
|      | ↳Transit Orbital Brightness Modulation |                    | transit                   |         | ↳ |
|      | ↳ other                                |                    |                           |         |   |
| 2829 | tau boo b                              | tau Boo b          | tau Boo b                 | Radial  | ↳ |
|      | ↳Velocity                              | Radial Velocity    | radial_velocity           |         | ↳ |
|      | ↳radial_velocity                       |                    |                           |         |   |
| 2886 | wasp-157 b                             | WASP-157 b         | WASP-157 b                | Radial  | ↳ |
|      | ↳Velocity                              | Transit            | radial_velocity           |         | ↳ |
|      | ↳ transit                              |                    |                           |         |   |
| 2928 | wasp-53 b                              | WASP-53 b          | WASP-53 b                 | Radial  | ↳ |
|      | ↳Velocity                              | Transit            | radial_velocity           |         | ↳ |
|      | ↳ transit                              |                    |                           |         |   |
| 2954 | wasp-81 b                              | WASP-81 b          | WASP-81 b                 | Radial  | ↳ |
|      | ↳Velocity                              | Transit            | radial_velocity           |         | ↳ |
|      | ↳ transit                              |                    |                           |         |   |
|      | transit_evidence_a                     | transit_evidence_b | transit_concordance       |         | ↳ |
|      | ↳period_relative_difference            |                    |                           |         |   |
| 6    | absent                                 | absent             | neither_confirmed_transit |         | ↳ |
|      | ↳ 3.545441e-06                         |                    |                           |         |   |
| 9    | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 5.430743e-07                         |                    |                           |         |   |
| 30   | absent                                 | absent             | neither_confirmed_transit |         | ↳ |
|      | ↳ 8.601485e-06                         |                    |                           |         |   |
| 88   | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 1.810199e-05                         |                    |                           |         |   |
| 94   | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 2.151381e-05                         |                    |                           |         |   |
| 290  | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 5.563495e-07                         |                    |                           |         |   |
| 318  | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 1.484701e-06                         |                    |                           |         |   |
| 326  | absent                                 | absent             | neither_confirmed_transit |         | ↳ |
|      | ↳ 4.527082e-06                         |                    |                           |         |   |
| 333  | confirmed                              | absent             | a_transit_only            |         | ↳ |
|      | ↳ 1.149387e-06                         |                    |                           |         |   |

|      |              |            |                           |   |
|------|--------------|------------|---------------------------|---|
| 349  | confirmed    | absent     | a_transit_only            | ␣ |
| ↪    | 8.111283e-06 |            |                           |   |
| 363  | confirmed    | absent     | a_transit_only            | ␣ |
| ↪    | 5.625741e-05 |            |                           |   |
| 427  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 2.315142e-06 |            |                           |   |
| 430  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 4.042552e-06 |            |                           |   |
| 455  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 2.849595e-06 |            |                           |   |
| 456  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 1.007311e-05 |            |                           |   |
| 460  | confirmed    | absent     | a_transit_only            | ␣ |
| ↪    | 3.544624e-06 |            |                           |   |
| 464  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 3.081429e-05 |            |                           |   |
| 467  | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 2.272405e-04 |            |                           |   |
| 478  | confirmed    | absent     | a_transit_only            | ␣ |
| ↪    | 6.200399e-05 |            |                           |   |
| 567  | absent       | confirmed  | b_transit_only            | ␣ |
| ↪    | 3.344937e-08 |            |                           |   |
| 1074 | absent       | confirmed  | b_transit_only            | ␣ |
| ↪    | 1.286308e-04 |            |                           |   |
| 1911 | supported    | confirmed  | b_transit_only            | ␣ |
| ↪    | 5.245691e-05 |            |                           |   |
| 2508 | confirmed    | unresolved | a_transit_only            | ␣ |
| ↪    | 6.796430e-07 |            |                           |   |
| 2829 | absent       | absent     | neither_confirmed_transit | ␣ |
| ↪    | 1.002277e-05 |            |                           |   |
| 2886 | absent       | confirmed  | b_transit_only            | ␣ |
| ↪    | 6.326518e-07 |            |                           |   |
| 2928 | absent       | confirmed  | b_transit_only            | ␣ |
| ↪    | 0.000000e+00 |            |                           |   |
| 2954 | absent       | confirmed  | b_transit_only            | ␣ |
| ↪    | 0.000000e+00 |            |                           |   |

|   | source | variable           | analysis_scale | n_included | n_excluded |     |
|---|--------|--------------------|----------------|------------|------------|-----|
| 0 | a      | period_days        | log            | 2588       | 393        | ␣   |
| ↪ |        | 9.488815           |                | 137.48000  |            | -1. |
| ↪ | 281508 |                    |                |            |            |     |
| 1 | a      | semi_major_axis_au | log            | 1227       | 383        | ␣   |
| ↪ |        | 0.077700           |                | 0.53364    |            | -1. |
| ↪ | 332576 |                    |                |            |            |     |

|    |   |                      |     |            |     |     |
|----|---|----------------------|-----|------------|-----|-----|
| 2  | a | star_temperature_k   | raw | 2585       | 362 | 0.  |
|    |   | 5665.000000          |     | 5344.00000 |     |     |
|    |   | 197183               |     |            |     |     |
| 3  | a | star_mass_msun       | log | 2310       | 368 | 0.  |
|    |   | 0.969500             |     | 0.99000    |     |     |
|    |   | 025983               |     |            |     |     |
| 4  | a | distance_pc          | log | 1624       | 355 | 2.  |
|    |   | 674.000000           |     | 41.30000   |     |     |
|    |   | 768456               |     |            |     |     |
| 5  | a | visual_magnitude_mag | raw | 574        | 380 | 2.  |
|    |   | 12.797000            |     | 8.04000    |     |     |
|    |   | 010076               |     |            |     |     |
| 6  | b | period_days          | log | 2588       | 393 | -1. |
|    |   | 9.488815             |     | 137.48000  |     |     |
|    |   | 281512               |     |            |     |     |
| 7  | b | semi_major_axis_au   | log | 2583       | 387 | -1. |
|    |   | 0.084800             |     | 0.53000    |     |     |
|    |   | 259502               |     |            |     |     |
| 8  | b | star_temperature_k   | raw | 2588       | 392 | 0.  |
|    |   | 5663.000000          |     | 5313.50000 |     |     |
|    |   | 188494               |     |            |     |     |
| 9  | b | star_mass_msun       | log | 2588       | 393 | 0.  |
|    |   | 0.960000             |     | 0.99000    |     |     |
|    |   | 016023               |     |            |     |     |
| 10 | b | distance_pc          | log | 2588       | 393 | 2.  |
|    |   | 692.149000           |     | 43.61970   |     |     |
|    |   | 453384               |     |            |     |     |
| 11 | b | visual_magnitude_mag | raw | 2588       | 393 | 2.  |
|    |   | 14.560500            |     | 8.07000    |     |     |
|    |   | 826939               |     |            |     |     |

## 1.12 Primary exact certificates

The primary analysis includes every Tier-1 match with positive paired planetary and stellar radii. Complete-row and coordinate-wise factorial widths are reported separately.

```
[27]: primary_certificates = add_source_ranks(build_certificates(primary_cohort,
    ↪primary_observable))
primary_verification = verify_certificates(primary_certificates,
    ↪primary_cohort, primary_observable, CONFIG.floating_tolerance,
    ↪"primary_certificate_verification")
display(report_table(primary_verification))
if not primary_verification.passed:
    raise RuntimeError("Primary certificate verification failed.")
```



|                 | stage                            | status        | computational_pass | major_count |
|-----------------|----------------------------------|---------------|--------------------|-------------|
| ↪blocking_count | issues                           | eligible_rows | enumerated_rows    | failures    |
| 0               | primary_certificate_verification | passed        | True               | 0           |
| ↪               | 0                                | 2588          | 2588               | 0           |

```
[28]: primary_summary = summarize_widths(primary_certificates, CONFIG.
      ↪equivalence_factors)
threshold_sensitivity = threshold_curve(primary_certificates, CONFIG.
      ↪equivalence_factors)
composition_samples = host_composition_sensitivity(
      primary_certificates, "factorial_width", "host_key_a", CONFIG.
      ↪composition_resamples, CONFIG.seed
)
composition_bounds = composition_interval(composition_samples, CONFIG.
      ↪composition_interval_level)
primary_summary["host_composition_interval_lower"] = composition_bounds[0]
primary_summary["host_composition_interval_upper"] = composition_bounds[1]
display(pd.DataFrame([primary_summary]))
display(threshold_sensitivity)
```

|                            | n_primary           | median_factorial_width  | q75_factorial_width | q90_factorial_width |
|----------------------------|---------------------|-------------------------|---------------------|---------------------|
| ↪q95_factorial_width       | q99_factorial_width | maximum_factorial_width |                     |                     |
| ↪median_complete_row_width | \                   |                         |                     |                     |
| 0                          | 2588                | 1.004868                | 1.013016            | 1.20001             |
| ↪                          | 1.462761            | 3.026051                | 13.567213           |                     |
| ↪                          | 1.004149            |                         |                     |                     |

|                                   | median_factorial_inflation  | q90_factorial_inflation | q95_factorial_inflation |
|-----------------------------------|-----------------------------|-------------------------|-------------------------|
| ↪q99_factorial_inflation          | maximum_factorial_inflation |                         |                         |
| ↪count_factorial_at_or_below_1p01 | \                           |                         |                         |
| 0                                 | 1.0                         | 1.024341                | 1.195588                |
| ↪                                 | 2.11868                     | 12.729919               |                         |
| ↪                                 | 1787                        |                         |                         |

|                                      | fraction_factorial_at_or_below_1p01 | count_factorial_above_1p01 |
|--------------------------------------|-------------------------------------|----------------------------|
| ↪fraction_factorial_above_1p01       | count_factorial_at_or_below_1p05    |                            |
| ↪fraction_factorial_at_or_below_1p05 | \                                   |                            |
| 0                                    | 0.690495                            | 801                        |
| ↪                                    | 0.309505                            | 2130                       |
| ↪0.823029                            |                                     |                            |

|                                  | count_factorial_above_1p05         | fraction_factorial_above_1p05 |
|----------------------------------|------------------------------------|-------------------------------|
| ↪count_factorial_at_or_below_1p1 | fraction_factorial_at_or_below_1p1 |                               |
| ↪count_factorial_above_1p1       | \                                  |                               |
| 0                                | 458                                | 0.176971                      |
| ↪                                | 2203                               | 0.851236                      |
|                                  |                                    | 385                           |

```

fraction_factorial_above_1p1 count_factorial_at_or_below_1p25
↪fraction_factorial_at_or_below_1p25 count_factorial_above_1p25
↪fraction_factorial_above_1p25 \
0          0.148764          2367          0.
↪          0.914606          221          0.
↪085394

count_factorial_at_or_below_1p5 fraction_factorial_at_or_below_1p5
↪count_factorial_above_1p5 fraction_factorial_above_1p5
↪count_factorial_at_or_below_2p0 \
0          2461          0.950927          0.
↪          127          0.049073          2535

fraction_factorial_at_or_below_2p0 count_factorial_above_2p0
↪fraction_factorial_above_2p0 count_factorial_at_or_below_5p0
↪fraction_factorial_at_or_below_5p0 \
0          0.979521          53          0.
↪          0.020479          2579          0.
↪996522

count_factorial_above_5p0 fraction_factorial_above_5p0
↪count_factorial_at_or_below_10p0 fraction_factorial_at_or_below_10p0
↪count_factorial_above_10p0 \
0          9          0.003478          0.
↪          2585          0.998841          3

fraction_factorial_above_10p0 fraction_inflation_above_1p01
↪fraction_inflation_above_1p05 fraction_inflation_above_1p25
↪fraction_inflation_above_2p0 \
0          0.001159          0.120943          0.
↪          0.08694          0.042117          0.011592

host_composition_interval_lower host_composition_interval_upper
0          1.004559          1.005201

factor count_at_or_below fraction_at_or_below count_above fraction_above
↪n_primary
0 1.01          1787          0.690495          801          0.309505
↪          2588
1 1.05          2130          0.823029          458          0.176971
↪          2588
2 1.10          2203          0.851236          385          0.148764
↪          2588
3 1.25          2367          0.914606          221          0.085394
↪          2588
4 1.50          2461          0.950927          127          0.049073
↪          2588

```

|   |       |      |          |    |          |
|---|-------|------|----------|----|----------|
| 5 | 2.00  | 2535 | 0.979521 | 53 | 0.020479 |
| ↪ | 2588  |      |          |    |          |
| 6 | 5.00  | 2579 | 0.996522 | 9  | 0.003478 |
| ↪ | 2588  |      |          |    |          |
| 7 | 10.00 | 2585 | 0.998841 | 3  | 0.001159 |
| ↪ | 2588  |      |          |    |          |

### 1.13 Transit-only operational ranking

Ranking is restricted to the conservative cohort classified as transit-related in both source artifacts. The score remains a geometric area ratio, not a complete mission-priority function.

Ranking uses minimum-rank (competition) semantics. Boundary ties at the top-25 cut are resolved by including all tied members; the guaranteed pool therefore may exceed 25 systems.

```
[29]: operational_certificates = □
      ↪ add_source_ranks(build_certificates(operational_cohort, primary_observable))
operational_verification = verify_certificates(operational_certificates, □
      ↪ operational_cohort, primary_observable, CONFIG.floating_tolerance, □
      ↪ "operational_certificate_verification")
if not operational_verification.passed:
    raise RuntimeError("Operational certificate verification failed.")
operational_ranking = robust_top_k(operational_certificates, CONFIG.primary_k)
ranking_verification = verify_ranking(operational_ranking, CONFIG.primary_k, □
      ↪ len(operational_certificates))
if not ranking_verification.passed:
    raise RuntimeError("Robust-ranking verification failed.")
operational_certificates = operational_certificates.merge(
    operational_ranking[["planet_key", "best_rank", "worst_rank", □
      ↪ "top_k_status"]],
    on="planet_key", how="left", validate="one_to_one",
)
ranking_summary = ranking_agreement(operational_certificates, CONFIG.primary_k)
ranking_summary.update({
    "guaranteed_top25": int(operational_ranking["top_k_status"].
      ↪ eq("guaranteed").sum()),
    "possible_top25": int(operational_ranking["top_k_status"].eq("possible").
      ↪ sum()),
    "excluded_top25": int(operational_ranking["top_k_status"].eq("excluded").
      ↪ sum()),
    "robust_candidate_pool": int(operational_ranking["top_k_status"].
      ↪ ne("excluded").sum()),
})
k_sensitivity = k_curve(operational_certificates, CONFIG.k_grid)
display(pd.DataFrame([ranking_summary]))
```

```

    n_ranking spearman_rho kendall_tau top_k_intersection top_k_union
↳top_k_jaccard median_rank_displacement p95_rank_displacement
↳p99_rank_displacement \
0      2561      0.996729      0.976452      21      29      0.
↳724138      5.0      77.0      250.4

    maximum_rank_displacement guaranteed_top25 possible_top25 excluded_top25
↳robust_candidate_pool
0      1115      13      31      2517
↳      44

```

```

[30]: secondary_certificate_frames: list[pd.DataFrame] = []
secondary_summary_rows: list[dict[str, Any]] = []
secondary_reports: list[ValidationReport] = []
for observable_name in ("density_g_cm3", "gravity_gearth",
↳"escape_velocity_km_s", "insolation_searth"):
    observable = OBSERVABLES[observable_name]
    base = operational_cohort if observable_name in {"density_g_cm3",
↳"gravity_gearth", "escape_velocity_km_s"} else tier1_matches
    mask = semantic_eligibility(base, observable, "_a") &
↳semantic_eligibility(base, observable, "_b")
    cohort = base.loc[mask].copy().reset_index(drop=True)
    certificates = build_certificates(cohort, observable)
    report = verify_certificates(certificates, cohort, observable, CONFIG.
↳floating_tolerance, f"{observable_name}_certificate_verification")
    secondary_reports.append(report)
    if not report.passed:
        raise RuntimeError(f"Secondary certificate verification failed:
↳{observable_name}")
    summary = paired_log_summary(certificates["source_value_a"],
↳certificates["source_value_b"])
    summary.pop("n", None)
    summary.update({
        "observable": observable_name, "n_secondary": len(certificates),
        "median_complete_row_width": float(certificates["complete_row_width"].
↳median()),
        "median_factorial_width": float(certificates["factorial_width"].
↳median()),
        "q95_factorial_width": float(certificates["factorial_width"].quantile(0.
↳95)),
        "q99_factorial_width": float(certificates["factorial_width"].quantile(0.
↳99)),
        "maximum_factorial_width": float(certificates["factorial_width"].max()),
        "count_factorial_above_2": int(certificates["factorial_width"].gt(2).
↳sum()),

```

```

        "count_factorial_above_10": int(certificates["factorial_width"].gt(10).
↳sum()),
    })
    secondary_summary_rows.append(summary)
    secondary_certificate_frames.append(certificates)
secondary_certificates = pd.concat(secondary_certificate_frames,
↳ignore_index=True, sort=False)
secondary_summary = pd.DataFrame(secondary_summary_rows)
display(secondary_summary)

```

```

    median_signed_log_ratio  median_absolute_log_ratio  q05_log_ratio
↳q95_log_ratio              observable  n_secondary  median_complete_row_width
↳median_factorial_width  \
0          -0.000006          0.053442          -0.611566          1.
↳776839      density_g_cm3          513          1.054895
↳          1.134855
1          -0.000006          0.027441          -0.441070          1.
↳871348      gravity_gearth          513          1.027821
↳          1.103149
2          -0.000003          0.018910          -0.178712          0.
↳941153  escape_velocity_km_s          513          1.019090
↳          1.032889
3          0.000000          0.007609          -0.263850          0.
↳175810      insolation_searth          1474          1.007638
↳          1.007910

    q95_factorial_width  q99_factorial_width  maximum_factorial_width
↳count_factorial_above_2  count_factorial_above_10
0          8.818423          140.593643          1807.293690
↳          94          25
1          8.310094          139.071000          987.066291
↳          75          25
2          2.846316          11.723792          23.218382
↳          41          7
3          1.689725          3.099854          7.703079
↳          49          0

```

```

[31]: forensic_cases = select_forensic_cases(primary_certificates,
↳operational_ranking, secondary_certificates, CONFIG)
forensic_review_decisions = load_forensic_review_decisions(PATHS.
↳forensic_review_input)
forensic_cases = apply_forensic_review_decisions(forensic_cases,
↳forensic_review_decisions)
forensic_columns = [
    "selection_reasons", "review_priority", "review_classification",
↳"review_finding", "evidence_reference",

```

```

    "reviewer", "review_date", "second_reviewer", "adjudication",
    "selected_primary_extreme", "selected_rank_displacement",
    ↪ "selected_operational_candidate",
    "selected_secondary_extreme", "selected_stable_reference",
    "planet_key", "planet_name_source_a", "planet_name_source_b",
    "observable", "observable_unit",
    "source_row_a", "source_row_b", "discovery_method_class_a",
    ↪ "discovery_method_class_b",
    "transit_evidence_a", "transit_evidence_b", "transit_concordance",
    "planet_radius_rearth_a", "planet_radius_rearth_b", "star_radius_rsun_a",
    ↪ "star_radius_rsun_b",
    "planet_mass_mearth_a", "planet_mass_mearth_b", "star_temperature_k_a",
    ↪ "star_temperature_k_b",
    "semi_major_axis_au_a", "semi_major_axis_au_b",
    "source_value_a", "source_value_b", "complete_row_width",
    ↪ "factorial_lower", "factorial_upper",
    "factorial_width", "factorial_inflation", "rank_a", "rank_b",
    ↪ "rank_displacement",
    "best_rank", "worst_rank", "top_k_status",
    "contribution_planet_radius_rearth", "contribution_star_radius_rsun",
]
forensic_table = forensic_cases.reindex(columns=forensic_columns)
forensic_report = ValidationReport("forensic_completeness")
if forensic_table.duplicated(["planet_key", "observable"]).any():
    forensic_report.add("blocking", "duplicate_forensic_case", "A
    ↪ planet-observable forensic case appears more than once.")
if forensic_table["selection_reasons"].isna().any() or
    ↪ forensic_table["selection_reasons"].astype("string").str.strip().eq("").
    ↪ any():
    forensic_report.add("blocking", "missing_selection_reason", "A selected
    ↪ forensic case has no rule-recorded reason.")
expected_primary_extremes = set(primary_certificates.
    ↪ loc[primary_certificates["factorial_width"].ge(CONFIG.
    ↪ extreme_width_threshold), "planet_key"])
observed_primary_extremes = set(forensic_table.loc[forensic_table["observable"].
    ↪ eq(primary_observable.name), "planet_key"])
if not expected_primary_extremes.issubset(observed_primary_extremes):
    forensic_report.add("blocking", "missing_primary_extreme", "Not every
    ↪ primary extreme is present in the forensic audit.")
expected_operational = set(operational_ranking.
    ↪ loc[operational_ranking["top_k_status"].ne("excluded"), "planet_key"])
observed_operational = set(forensic_table.
    ↪ loc[forensic_table["selected_operational_candidate"].fillna(False),
    ↪ "planet_key"])
if expected_operational != observed_operational:

```

```

        forensic_report.add("blocking", "operational_candidate_loss", "Operational_
        ↪candidate flags are incomplete after forensic aggregation.")
forensic_report.metrics["rows"] = len(forensic_table)
if not forensic_report.passed:
    raise RuntimeError("Forensic completeness verification failed.")

priority_zero = forensic_table["review_priority"].eq(0)
required_review_fields = [
    "review_classification", "review_finding", "evidence_reference", "reviewer",
    "review_date", "second_reviewer", "adjudication",
]
completed_fields = forensic_table[required_review_fields].astype("string").
    ↪apply(lambda column: column.str.strip().ne(""))
completed_fields["review_classification"] &=
    ↪~forensic_table["review_classification"].isin(["pending", "unresolved"])
reviewer_one = forensic_table["reviewer"].astype("string").str.strip().str.
    ↪casefold()
reviewer_two = forensic_table["second_reviewer"].astype("string").str.strip().
    ↪str.casefold()
if CONFIG.is_solo_author:
    lead_matches = reviewer_one.str.contains(CONFIG.lead_author.casefold(),
    ↪regex=False)
    sentinel_matches = reviewer_two.str.contains(SOLO_AUTHOR_SENTINEL.
    ↪casefold(), regex=False)
    independent_reviewers = lead_matches & sentinel_matches
else:
    independent_reviewers = reviewer_one.ne(reviewer_two)
parsed_review_dates = pd.to_datetime(forensic_table["review_date"],
    ↪errors="coerce", utc=True)
nonfuture_review_dates = parsed_review_dates.le(pd.Timestamp.now(tz="UTC"))
completed_rows = completed_fields.all(axis=1) & independent_reviewers &
    ↪parsed_review_dates.notna() & nonfuture_review_dates
priority_zero_review_complete = bool((~priority_zero | completed_rows).all())
acceptable_findings = forensic_table["review_classification"].
    ↪isin(["confirmed_source_difference", "confirmed_agreement",
    ↪"corrected_source_error", "source_semantic_mismatch"])
if not priority_zero_review_complete:
    priority_zero_findings_status: ReviewFindingStatus = "not_assessed"
elif bool((~priority_zero | acceptable_findings).all()):
    priority_zero_findings_status = "acceptable"
else:
    priority_zero_findings_status = "correction_required"
forensic_review_report = ValidationReport("priority_zero_forensic_review")
reviewer_placeholder_matches = forensic_table[["reviewer", "second_reviewer"]].
    ↪map(contains_placeholder).any(axis=1)
if reviewer_placeholder_matches.any():

```

```

        forensic_review_report.add(
            "major", "reviewer_placeholder",
            "A reviewer identity contains a placeholder term.",
            int(reviewer_placeholder_matches.sum()),
        )
    if not priority_zero_review_complete:
        forensic_review_report.add(
            "major", "human_review_pending",
            "Priority-0 cases require documented self-audit review (solo-author_
↪mode) before submission-ready status.",
            int((priority_zero & ~completed_rows).sum()),
        )
    elif priority_zero_findings_status == "correction_required":
        forensic_review_report.add(
            "major", "correction_required",
            "At least one completed priority-0 review identified a matching, unit,
↪or data error requiring correction and rerun.",
            int((priority_zero & ~acceptable_findings).sum()),
        )
    forensic_review_report.metrics.update(
        priority_zero_cases=int(priority_zero.sum()),
        completed_priority_zero=int((priority_zero & completed_rows).sum()),
        acceptable_priority_zero_findings=int((priority_zero & acceptable_findings).
↪sum()),
        findings_status=priority_zero_findings_status,
        review_file_present=PATHS.forensic_review_input.is_file(),
        review_mode=author_mode_label(),
    )
display(forensic_table.head(30))
display(report_table(forensic_review_report))

```

|                                          | selection_reasons | review_priority |   |
|------------------------------------------|-------------------|-----------------|---|
| ↪review_classification \                 |                   |                 |   |
| 0 operational_candidate;stable_reference |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 1 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 2 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 3 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 4 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 5 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |
| 6 operational_candidate                  |                   | 0               | ↪ |
| ↪confirmed_agreement                     |                   |                 |   |



|    |                              |   |   |
|----|------------------------------|---|---|
| 7  | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_agreement         |   |   |
| 8  | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 9  | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 10 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 11 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 12 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 13 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 14 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 15 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 16 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 17 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 18 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 19 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 20 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 21 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 22 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 23 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 24 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 25 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 26 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 27 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 28 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |
| 29 | operational_candidate        | 0 | ␣ |
|    | ↳confirmed_source_difference |   |   |

□

```

↪          review_finding \
0   Source A value 24664.9 vs Source B value 24664.9 (ppm); factorial width 1.
    ↪0000. Planet radius 10.5813 vs 10.5813 R_e...
1   Source A value 20158.3 vs Source B value 20158.3 (ppm); factorial width 1.
    ↪0000. Planet radius 10.6822 vs 10.6822 R_e...
2   Source A value 25356.8 vs Source B value 25356.8 (ppm); factorial width 1.
    ↪0000. Planet radius 20.8487 vs 20.8487 R_e...
3   Source A value 22251.5 vs Source B value 22251.5 (ppm); factorial width 1.
    ↪0000. Planet radius 13.2266 vs 13.2266 R_e...
4   Source A value 32301.1 vs Source B value 32301.1 (ppm); factorial width 1.
    ↪0000. Planet radius 11.1866 vs 11.1866 R_e...
5   Source A value 56767.6 vs Source B value 56767.6 (ppm); factorial width 1.
    ↪0000. Planet radius 14.9080 vs 14.9080 R_e...
6   Source A value 22797.2 vs Source B value 22797.2 (ppm); factorial width 1.
    ↪0000. Planet radius 11.9040 vs 11.9040 R_e...
7   Source A value 22435 vs Source B value 22435 (ppm); factorial width 1.0000.□
    ↪Planet radius 16.0289 vs 16.0289 R_earth...
8   Source A value 32101.4 vs Source B value 32097.6 (ppm); factorial width 1.
    ↪0001. dominant contributor(s): planet-radi...
9   Source A value 27230.8 vs Source B value 27248.3 (ppm); factorial width 1.
    ↪0006. dominant contributor(s): planet-radi...
10  Source A value 28222.9 vs Source B value 28201.9 (ppm); factorial width 1.
    ↪0007. dominant contributor(s): planet-radi...
11  Source A value 27795.1 vs Source B value 27586.5 (ppm); factorial width 1.
    ↪0102. dominant contributor(s): planet-radi...
12  Source A value 21373.4 vs Source B value 20740.2 (ppm); factorial width 1.
    ↪0305. dominant contributor(s): star-radius...
13  Source A value 20341.7 vs Source B value 20846 (ppm); factorial width 1.0330.
    ↪dominant contributor(s): planet-radius...
14  Source A value 27039.6 vs Source B value 26005.7 (ppm); factorial width 1.
    ↪0398. dominant contributor(s): planet-radi...
15  Source A value 22183.5 vs Source B value 21010.4 (ppm); factorial width 1.
    ↪0558. dominant contributor(s): planet-radi...
16  Source A value 18666.3 vs Source B value 20158 (ppm); factorial width 1.0799.
    ↪dominant contributor(s): planet-radius...
17  Source A value 22726.7 vs Source B value 22105.5 (ppm); factorial width 1.
    ↪0896. dominant contributor(s): planet-radi...
18  Source A value 21216.2 vs Source B value 19548.6 (ppm); factorial width 1.
    ↪1164. dominant contributor(s): planet-radi...
19  Source A value 21660.6 vs Source B value 21348.5 (ppm); factorial width 1.
    ↪1195. dominant contributor(s): planet-radi...
20  Source A value 30748.5 vs Source B value 34614.9 (ppm); factorial width 1.
    ↪1257. dominant contributor(s): planet-radi...

```

21 Source A value 20806.4 vs Source B value 20061.3 (ppm); factorial width 1.  
↳1802. dominant contributor(s): planet-radi...

22 Source A value 19081.2 vs Source B value 19123.9 (ppm); factorial width 1.  
↳1931. dominant contributor(s): planet-radi...

23 Source A value 18381 vs Source B value 18594.8 (ppm); factorial width 1.2047.  
↳ dominant contributor(s): planet-radius...

24 Source A value 18535.3 vs Source B value 20465.5 (ppm); factorial width 1.  
↳2080. dominant contributor(s): planet-radi...

25 Source A value 26019.3 vs Source B value 25114.4 (ppm); factorial width 1.  
↳2120. dominant contributor(s): planet-radi...

26 Source A value 20046.1 vs Source B value 25226.9 (ppm); factorial width 1.  
↳2584. dominant contributor(s): planet-radi...

27 Source A value 24060.2 vs Source B value 30622.6 (ppm); factorial width 1.  
↳2727. dominant contributor(s): planet-radi...

28 Source A value 28866.4 vs Source B value 22453 (ppm); factorial width 1.2856.  
↳ dominant contributor(s): planet-radius...

29 Source A value 21932.6 vs Source B value 16596.5 (ppm); factorial width 1.  
↳3215. dominant contributor(s): planet-radi...

↳ evidence\_reference

↳ reviewer \

0 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=307,  
↳source\_row\_b=347; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

1 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=332,  
↳source\_row\_b=1217; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

2 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=333,  
↳source\_row\_b=1550; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

3 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=348,  
↳source\_row\_b=915; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

4 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=357,  
↳source\_row\_b=297; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

5 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3399,  
↳source\_row\_b=1556; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

6 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3482,  
↳source\_row\_b=2088; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

7 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=345,  
↳source\_row\_b=1607; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

8 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=2639,  
↳source\_row\_b=94; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

9 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3613,  
↳source\_row\_b=414; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

10 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3082,  
↳source\_row\_b=863; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

11 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=96,  
↳source\_row\_b=4230; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

12 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3533,  
↳source\_row\_b=835; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

13 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3576,  
↳source\_row\_b=4395; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

14 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=2635,  
↳source\_row\_b=336; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

15 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3450,  
↳source\_row\_b=6118; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

16 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3612,  
↳source\_row\_b=4242; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

17 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=281,  
↳source\_row\_b=1029; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

18 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3596,  
↳source\_row\_b=1670; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

19 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3477,  
↳source\_row\_b=1849; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

20 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3676,  
↳source\_row\_b=173; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

21 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3537,  
↳source\_row\_b=4326; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

22 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3593,  
↳source\_row\_b=1671; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

23 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=265,  
↳source\_row\_b=844; frozen inputs kepler.cs... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

24 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3583,  
↳source\_row\_b=1281; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

25 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3665,  
↳source\_row\_b=905; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

26 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3525,  
↳source\_row\_b=1028; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

27 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3645,  
↳source\_row\_b=1245; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

28 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3564,  
↳source\_row\_b=1909; frozen inputs kepler... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

29 forensic\_cases.csv (protocol 2.4.0 corrected-input run), source\_row\_a=3574,  
↳source\_row\_b=345; frozen inputs kepler.c... Khaled Tarchi (Lead Author -  
↳Solo-Author Self-Audit)

|   | review_date | second_reviewer           |                                                                                                                                 |
|---|-------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|   |             |                           | ↳                                                                                                                               |
|   |             |                           | ↳adjudication \                                                                                                                 |
| 0 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 1 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 2 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 3 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 4 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 5 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |
| 6 | 2026-08-14  | N/A (Solo-Author Project) | Exact agreement between the two<br>↳frozen source snapshots for this system/observable (factorial width 1.0000).<br>↳Reclass... |

- 7 2026-08-14 N/A (Solo-Author Project) Exact agreement between the two  
↳ frozen source snapshots for this system/observable (factorial width 1.0000).  
↳ Reclass...
- 8 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 9 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 10 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 11 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 12 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 13 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 14 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 15 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 16 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 17 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 18 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 19 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 20 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...
- 21 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳ the two frozen source snapshots for this system/observable; no pipeline  
↳ matchin...

22 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

23 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

24 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

25 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

26 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

27 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

28 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

29 2026-08-14 N/A (Solo-Author Project) Confirmed genuine difference between  
↳the two frozen source snapshots for this system/observable; no pipeline  
↳matchin...

|        | selected_primary_extreme | selected_rank_displacement | selected_operational_candidate | selected_secondary_extreme | selected_stable_reference | planet_key \ |
|--------|--------------------------|----------------------------|--------------------------------|----------------------------|---------------------------|--------------|
| 0      | False                    | False                      | False                          | True                       | hat-p-54 b                | ↳            |
| ↳ True |                          | False                      |                                | False                      | hats-22 b                 | ↳            |
| 1      | False                    | False                      | False                          | False                      | hats-23 b                 | ↳            |
| ↳ True |                          | False                      |                                | False                      | hats-43 b                 | ↳            |
| 2      | False                    | False                      | False                          | False                      | hats-6 b                  | ↳            |
| ↳ True |                          | False                      |                                | False                      | ngts-1 b                  | ↳            |
| 3      | False                    | False                      | False                          | False                      | qatar-6 b                 | ↳            |
| ↳ True |                          | False                      |                                | False                      | hats-34 b                 | ↳            |
| 4      | False                    | False                      | False                          | False                      | kepler-45 b               | ↳            |
| ↳ True |                          |                            |                                |                            |                           |              |

|    |      |       |       |       |       |               |   |
|----|------|-------|-------|-------|-------|---------------|---|
| 9  |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-52 b      | ␣ |
| 10 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | kepler-785 b  | ␣ |
| 11 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | corot-2 b     | ␣ |
| 12 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-107 b     | ␣ |
| 13 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-19 b      | ␣ |
| 14 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | kepler-447 b  | ␣ |
| 15 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | ogle-tr-113 b | ␣ |
| 16 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-50 b      | ␣ |
| 17 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | hat-p-32 b    | ␣ |
| 18 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-39 b      | ␣ |
| 19 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | qatar-1 b     | ␣ |
| 20 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | wt-2 b        | ␣ |
| 21 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-110 b     | ␣ |
| 22 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-36 b      | ␣ |
| 23 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | hat-p-18 b    | ␣ |
| 24 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-25 b      | ␣ |
| 25 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-98 b      | ␣ |
| 26 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-10 b      | ␣ |
| 27 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-80 b      | ␣ |
| 28 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-140 b     | ␣ |
| 29 |      | False |       | False |       |               |   |
| ↪  | True |       | False |       | False | was-17 b      | ␣ |

```

planet_name_source_a planet_name_source_b observable␣
↪observable_unit source_row_a source_row_b discovery_method_class_a␣
↪discovery_method_class_b \

```



|          |              |              |                          |   |
|----------|--------------|--------------|--------------------------|---|
| 0        | HAT-P-54 b   | HAT-P-54 b   | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 307          | 347          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 1        | HATS-22 b    | HATS-22 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 332          | 1217         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 2        | HATS-23 b    | HATS-23 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 333          | 1550         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 3        | HATS-43 b    | HATS-43 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 348          | 915          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 4        | HATS-6 b     | HATS-6 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 357          | 297          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 5        | NGTS-1 b     | NGTS-1 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3399         | 1556         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 6        | Qatar-6 b    | Qatar-6 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3482         | 2088         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 7        | HATS-34 b    | HATS-34 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 345          | 1607         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 8        | Kepler-45 b  | Kepler-45 b  | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 2639         | 94           | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 9        | WASP-52 b    | WASP-52 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3613         | 414          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 10       | Kepler-785 b | Kepler-785 b | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3082         | 863          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 11       | CoRoT-2 b    | CoRoT-2 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 96           | 4230         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 12       | WASP-107 b   | WASP-107 b   | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3533         | 835          | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 13       | WASP-19 b    | WASP-19 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3576         | 4395         | transit                  | ┐ |
| ↪transit |              |              |                          |   |
| 14       | Kepler-447 b | Kepler-447 b | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 2635         | 336          | transit                  | ┐ |
| ↪transit |              |              |                          |   |

|          |               |               |                          |   |
|----------|---------------|---------------|--------------------------|---|
| 15       | OGLE-TR-113 b | OGLE-TR-113 b | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3450          | 6118          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 16       | WASP-50 b     | WASP-50 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3612          | 4242          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 17       | HAT-P-32 b    | HAT-P-32 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 281           | 1029          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 18       | WASP-39 b     | WASP-39 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3596          | 1670          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 19       | Qatar-1 b     | Qatar-1 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3477          | 1849          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 20       | WTS-2 b       | WTS-2 b       | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3676          | 173           | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 21       | WASP-110 b    | WASP-110 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3537          | 4326          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 22       | WASP-36 b     | WASP-36 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3593          | 1671          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 23       | HAT-P-18 b    | HAT-P-18 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 265           | 844           | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 24       | WASP-25 b     | WASP-25 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3583          | 1281          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 25       | WASP-98 b     | WASP-98 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3665          | 905           | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 26       | WASP-10 b     | WASP-10 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3525          | 1028          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 27       | WASP-80 b     | WASP-80 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3645          | 1245          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 28       | WASP-140 b    | WASP-140 b    | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3564          | 1909          | transit                  | ┐ |
| ↪transit |               |               |                          |   |
| 29       | WASP-17 b     | WASP-17 b     | geometric_area_ratio_ppm | ┐ |
| ↪ ppm    | 3574          | 345           | transit                  | ┐ |
| ↪transit |               |               |                          |   |

|         | transit_evidence_a | transit_evidence_b | transit_concordance | planet_radius_rearth_a | planet_radius_rearth_b | star_radius_rsun_a | star_radius_rsun_b | planet_mass_mearth_a | \   |
|---------|--------------------|--------------------|---------------------|------------------------|------------------------|--------------------|--------------------|----------------------|-----|
| 0       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 10. |
| ↳581296 |                    | 10.581296          | 0.617               | 0.617                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 1       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 10. |
| ↳682177 |                    | 10.682177          | 0.689               | 0.689                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 2       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 20. |
| ↳848740 |                    | 20.848740          | 1.199               | 1.199                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 3       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 13. |
| ↳226620 |                    | 13.226620          | 0.812               | 0.812                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 4       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 11. |
| ↳186582 |                    | 11.186582          | 0.570               | 0.570                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 5       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 14. |
| ↳907970 |                    | 14.907970          | 0.573               | 0.573                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 6       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 11. |
| ↳903958 |                    | 11.903958          | 0.722               | 0.722                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 7       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 16. |
| ↳028870 |                    | 16.028870          | 0.980               | 0.980                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 8       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 10. |
| ↳760640 |                    | 10.760000          | 0.550               | 0.550                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 9       | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 14. |
| ↳235430 |                    | 14.240000          | 0.790               | 0.790                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 10      | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 13. |
| ↳024858 |                    | 13.020000          | 0.710               | 0.710                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 11      | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 16. |
| ↳421185 |                    | 16.432000          | 0.902               | 0.906                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 12      | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 10. |
| ↳536460 |                    | 10.536460          | 0.660               | 0.670                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |
| 13      | confirmed          | confirmed          | both_transit        |                        |                        |                    |                    |                      | 15. |
| ↳636555 |                    | 15.860708          | 1.004               | 1.006                  |                        |                    |                    |                      | ↳   |
| ↳       | NaN                |                    |                     |                        |                        |                    |                    |                      |     |

|         |           |           |              |       |   |
|---------|-----------|-----------|--------------|-------|---|
| 14      | confirmed | confirmed | both_transit | 18.   |   |
| ↪494850 |           | 18.490000 | 1.030        | 1.050 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 15      | confirmed | confirmed | both_transit | 12.   |   |
| ↪441990 |           | 12.251000 | 0.765        | 0.774 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 16      | confirmed | confirmed | both_transit | 12.   |   |
| ↪755842 |           | 13.069694 | 0.855        | 0.843 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 17      | confirmed | confirmed | both_transit | 22.   |   |
| ↪832733 |           | 22.193820 | 1.387        | 1.367 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 18      | confirmed | confirmed | both_transit | 14.   |   |
| ↪235430 |           | 14.336311 | 0.895        | 0.939 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 19      | confirmed | confirmed | both_transit | 13.   |   |
| ↪226620 |           | 12.811887 | 0.823        | 0.803 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 20      | confirmed | confirmed | both_transit | 14.   |   |
| ↪571700 |           | 15.277867 | 0.761        | 0.752 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 21      | confirmed | confirmed | both_transit | 13.   |   |
| ↪876742 |           | 13.192970 | 0.881        | 0.853 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 22      | confirmed | confirmed | both_transit | 14.   |   |
| ↪224221 |           | 14.874343 | 0.943        | 0.985 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 23      | confirmed | confirmed | both_transit | 10.   |   |
| ↪614923 |           | 11.153000 | 0.717        | 0.749 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 24      | confirmed | confirmed | both_transit | 14.   |   |
| ↪123340 |           | 13.809464 | 0.950        | 0.884 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 25      | confirmed | confirmed | both_transit | 12.   |   |
| ↪329900 |           | 12.823096 | 0.700        | 0.741 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 26      | confirmed | confirmed | both_transit | 12.   |   |
| ↪105720 |           | 12.106000 | 0.783        | 0.698 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 27      | confirmed | confirmed | both_transit | 10.   |   |
| ↪670968 |           | 11.197791 | 0.630        | 0.586 | ┐ |
| ↪       | NaN       |           |              |       |   |
| 28      | confirmed | confirmed | both_transit | 16.   |   |
| ↪140960 |           | 14.235405 | 0.870        | 0.870 | ┐ |
| ↪       | NaN       |           |              |       |   |

|         |           |           |              |       |
|---------|-----------|-----------|--------------|-------|
| 29      | confirmed | confirmed | both_transit | 22.   |
| ↵317119 |           | 20.960830 | 1.380        | 1.490 |
| ↵       | NaN       |           |              |       |

  

|                       |                      |                      |                |
|-----------------------|----------------------|----------------------|----------------|
| planet_mass_mearth_b  | star_temperature_k_a | star_temperature_k_b |                |
| ↵semi_major_axis_au_a | semi_major_axis_au_b | source_value_a       | source_value_b |
| ↵complete_row_width   | \                    |                      |                |

  

|         |     |     |              |              |    |
|---------|-----|-----|--------------|--------------|----|
| 0       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 24664.949087 | 24664.949087 | 1. |
| ↵000000 |     |     |              |              |    |
| 1       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 20158.300747 | 20158.300747 | 1. |
| ↵000000 |     |     |              |              |    |
| 2       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 25356.778972 | 25356.778972 | 1. |
| ↵000000 |     |     |              |              |    |
| 3       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 22251.460360 | 22251.460360 | 1. |
| ↵000000 |     |     |              |              |    |
| 4       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 32301.146242 | 32301.146242 | 1. |
| ↵000000 |     |     |              |              |    |
| 5       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 56767.608774 | 56767.608774 | 1. |
| ↵000000 |     |     |              |              |    |
| 6       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 22797.183824 | 22797.183824 | 1. |
| ↵000000 |     |     |              |              |    |
| 7       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 22435.021308 | 22435.021308 | 1. |
| ↵000000 |     |     |              |              |    |
| 8       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 32101.376103 | 32097.557693 | 1. |
| ↵000119 |     |     |              |              |    |
| 9       | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 27230.763739 | 27248.250329 | 1. |
| ↵000642 |     |     |              |              |    |
| 10      | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 28222.931486 | 28201.882284 | 1. |
| ↵000746 |     |     |              |              |    |
| 11      | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 27795.131759 | 27586.543078 | 1. |
| ↵007561 |     |     |              |              |    |
| 12      | NaN | NaN | NaN          | NaN          |    |
| ↵       | NaN | NaN | 21373.438649 | 20740.186847 | 1. |
| ↵030533 |     |     |              |              |    |

|         |     |     |              |              |    |
|---------|-----|-----|--------------|--------------|----|
| 13      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 20341.714396 | 20845.964099 |    |
| ↪024789 |     |     |              |              |    |
| 14      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 27039.583754 | 26005.670084 |    |
| ↪039757 |     |     |              |              |    |
| 15      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 22183.478652 | 21010.383656 |    |
| ↪055834 |     |     |              |              |    |
| 16      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 18666.326478 | 20158.049828 |    |
| ↪079915 |     |     |              |              |    |
| 17      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 22726.664389 | 22105.478195 |    |
| ↪028101 |     |     |              |              |    |
| 18      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 21216.216285 | 19548.636324 |    |
| ↪085304 |     |     |              |              |    |
| 19      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 21660.621174 | 21348.527883 |    |
| ↪014619 |     |     |              |              |    |
| 20      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 30748.501372 | 34614.861103 |    |
| ↪125741 |     |     |              |              |    |
| 21      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 20806.356492 | 20061.339626 |    |
| ↪037137 |     |     |              |              |    |
| 22      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 19081.232420 | 19123.877254 |    |
| ↪002235 |     |     |              |              |    |
| 23      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 18380.960826 | 18594.846322 |    |
| ↪011636 |     |     |              |              |    |
| 24      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 18535.335118 | 20465.480830 |    |
| ↪104133 |     |     |              |              |    |
| 25      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 26019.314653 | 25114.364651 |    |
| ↪036033 |     |     |              |              |    |
| 26      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 20046.130333 | 25226.867049 |    |
| ↪258441 |     |     |              |              |    |
| 27      | NaN | NaN | NaN          | NaN          | 1. |
| ↪ NaN   |     | NaN | 24060.207068 | 30622.617323 |    |
| ↪272750 |     |     |              |              |    |

|    |        |     |              |              |    |
|----|--------|-----|--------------|--------------|----|
| 28 | NaN    | NaN | NaN          | NaN          | 1. |
| ↪  | NaN    | NaN | 28866.427679 | 22452.980747 |    |
| ↪  | 285639 |     |              |              |    |

|    |        |     |              |              |    |
|----|--------|-----|--------------|--------------|----|
| 29 | NaN    | NaN | NaN          | NaN          | 1. |
| ↪  | NaN    | NaN | 21932.639991 | 16596.527850 |    |
| ↪  | 321520 |     |              |              |    |

|         | factorial_lower | factorial_upper   | factorial_width | factorial_inflation |                |
|---------|-----------------|-------------------|-----------------|---------------------|----------------|
| ↪rank_a | rank_b          | rank_displacement | best_rank       | worst_rank          | top_k_status \ |
| 0       | 24664.949087    | 24664.949087      | 1.000000        | 1.000000            |                |
| ↪15     | 15              | 0                 | 10.0            | 20.0                | guaranteed     |
| 1       | 20158.300747    | 20158.300747      | 1.000000        | 1.000000            |                |
| ↪36     | 29              | 7                 | 24.0            | 43.0                | possible       |
| 2       | 25356.778972    | 25356.778972      | 1.000000        | 1.000000            |                |
| ↪14     | 11              | 3                 | 9.0             | 17.0                | guaranteed     |
| 3       | 22251.460360    | 22251.460360      | 1.000000        | 1.000000            |                |
| ↪23     | 21              | 2                 | 17.0            | 30.0                | possible       |
| 4       | 32301.146242    | 32301.146242      | 1.000000        | 1.000000            |                |
| ↪2      | 3               | 1                 | 2.0             | 5.0                 | guaranteed     |
| 5       | 56767.608774    | 56767.608774      | 1.000000        | 1.000000            |                |
| ↪1      | 1               | 0                 | 1.0             | 1.0                 | guaranteed     |
| 6       | 22797.183824    | 22797.183824      | 1.000000        | 1.000000            |                |
| ↪20     | 18              | 2                 | 14.0            | 26.0                | possible       |
| 7       | 22435.021308    | 22435.021308      | 1.000000        | 1.000000            |                |
| ↪22     | 20              | 2                 | 16.0            | 29.0                | possible       |
| 8       | 32097.557693    | 32101.376103      | 1.000119        | 1.000000            |                |
| ↪3      | 4               | 1                 | 3.0             | 6.0                 | guaranteed     |
| 9       | 27230.763739    | 27248.250329      | 1.000642        | 1.000000            |                |
| ↪9      | 9               | 0                 | 7.0             | 15.0                | guaranteed     |
| 10      | 28201.882284    | 28222.931486      | 1.000746        | 1.000000            |                |
| ↪6      | 6               | 0                 | 5.0             | 11.0                | guaranteed     |
| 11      | 27550.241924    | 27831.755588      | 1.010218        | 1.002637            |                |
| ↪7      | 7               | 0                 | 6.0             | 13.0                | guaranteed     |
| 12      | 20740.186847    | 21373.438649      | 1.030533        | 1.000000            |                |
| ↪28     | 27              | 1                 | 19.0            | 40.0                | possible       |
| 13      | 20260.913227    | 20929.098470      | 1.032979        | 1.007992            |                |
| ↪35     | 26              | 9                 | 20.0            | 41.0                | possible       |
| 14      | 26005.670084    | 27039.583754      | 1.039757        | 1.000000            |                |
| ↪11     | 10              | 1                 | 8.0             | 16.0                | guaranteed     |
| 15      | 21010.383656    | 22183.478652      | 1.055834        | 1.000000            |                |
| ↪24     | 25              | 1                 | 18.0            | 38.0                | possible       |
| 16      | 18666.326478    | 20158.049828      | 1.079915        | 1.000000            |                |
| ↪49     | 30              | 19                | 25.0            | 57.0                | possible       |
| 17      | 21472.569702    | 23396.537586      | 1.089601        | 1.059819            |                |
| ↪21     | 22              | 1                 | 12.0            | 34.0                | possible       |

|     |              |              |      |          |            |   |
|-----|--------------|--------------|------|----------|------------|---|
| 18  | 19274.486657 | 21517.984043 |      | 1.116397 | 1.028649   | ┘ |
| ↪29 | 37           | 8            | 18.0 | 50.0     | possible   |   |
| 19  | 20323.539840 | 22753.042962 |      | 1.119541 | 1.103411   | ┘ |
| ↪26 | 23           | 3            | 15.0 | 40.0     | possible   |   |
| 20  | 30748.501372 | 34614.861103 |      | 1.125741 | 1.000000   | ┘ |
| ↪4  | 2            | 2            | 2.0  | 8.0      | guaranteed |   |
| 21  | 18806.421946 | 22194.726098 |      | 1.180167 | 1.137909   | ┘ |
| ↪33 | 32           | 1            | 18.0 | 55.0     | possible   |   |
| 22  | 17488.692672 | 20865.318723 |      | 1.193075 | 1.190414   | ┘ |
| ↪46 | 40           | 6            | 20.0 | 76.0     | possible   |   |
| 23  | 16843.908959 | 20291.675919 |      | 1.204689 | 1.190832   | ┘ |
| ↪52 | 47           | 5            | 23.0 | 80.0     | possible   |   |
| 24  | 17720.634668 | 21406.374695 |      | 1.207991 | 1.094063   | ┘ |
| ↪50 | 28           | 22           | 19.0 | 69.0     | possible   |   |
| 25  | 23219.641874 | 28142.490728 |      | 1.212012 | 1.169859   | ┘ |
| ↪12 | 14           | 2            | 6.0  | 24.0     | guaranteed |   |
| 26  | 20046.130333 | 25226.867049 |      | 1.258441 | 1.000000   | ┘ |
| ↪38 | 13           | 25           | 10.0 | 44.0     | possible   |   |
| 27  | 24060.207068 | 30622.617323 |      | 1.272750 | 1.000000   | ┘ |
| ↪16 | 5            | 11           | 5.0  | 22.0     | guaranteed |   |
| 28  | 22452.980747 | 28866.427679 |      | 1.285639 | 1.000000   | ┘ |
| ↪5  | 19           | 14           | 5.0  | 28.0     | possible   |   |
| 29  | 16596.527850 | 21932.639991 |      | 1.321520 | 1.000000   | ┘ |
| ↪25 | 64           | 39           | 18.0 | 84.0     | possible   |   |

|    | contribution_planet_radius_rearth | contribution_star_radius_rsun |
|----|-----------------------------------|-------------------------------|
| 0  | 8.881784e-16                      | 0.000000                      |
| 1  | 0.000000e+00                      | 0.000000                      |
| 2  | 0.000000e+00                      | 0.000000                      |
| 3  | 0.000000e+00                      | 0.000000                      |
| 4  | 0.000000e+00                      | 0.000000                      |
| 5  | 0.000000e+00                      | 0.000000                      |
| 6  | 0.000000e+00                      | 0.000000                      |
| 7  | 8.881784e-16                      | 0.000000                      |
| 8  | 1.189556e-04                      | 0.000000                      |
| 9  | 6.419569e-04                      | 0.000000                      |
| 10 | 7.460974e-04                      | 0.000000                      |
| 11 | 1.316767e-03                      | 0.008850                      |
| 12 | 0.000000e+00                      | 0.030076                      |
| 13 | 2.846679e-02                      | 0.003980                      |
| 14 | 5.245391e-04                      | 0.038463                      |
| 15 | 3.093895e-02                      | 0.023392                      |
| 16 | 4.861350e-02                      | 0.028269                      |
| 17 | 5.676259e-02                      | 0.029049                      |
| 18 | 1.412324e-02                      | 0.095984                      |
| 19 | 6.371611e-02                      | 0.049203                      |
| 20 | 9.464778e-02                      | 0.023794                      |



```

21          1.010601e-01          0.064596
22          8.938313e-02          0.087151
23          9.889536e-02          0.087326
24          4.494916e-02          0.144010
25          7.844143e-02          0.113841
26          4.625859e-05          0.229827
27          9.637949e-02          0.144800
28          2.512559e-01          0.000000
29          1.253972e-01          0.153385

          stage status computational_pass major_count
blocking_count issues priority_zero_cases completed_priority_zero
acceptable_priority_zero_findings \
0 priority_zero_forensic_review passed True 0
  0 113 113
  113

findings_status review_file_present review_mode
0 acceptable True solo_author_self_audit

```

## 1.14 Publication figures

Figures are generated only from canonical result objects. Main panels emphasize exact widths, availability, agreement, decision stability, and rule-selected extreme cases.

```

[32]: FIGURE_PATHS: list[Path] = []
fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
sns.barplot(data=cohort_flow, x="n", y="stage", color=sns.color_palette()[0],
ax=axes[0])
for index, value in enumerate(cohort_flow["n"]):
    axes[0].text(value, index, f" {value:,"}, va="center")
axes[0].set(title="Study flow", xlabel="Rows or matched planets", ylabel="")
heat = availability_summary.reindex(columns=["a_only", "b_only", "both",
neither"], fill_value=0)
sns.heatmap(heat, annot=True, fmt="g", cmap="Blues", cbar=False, ax=axes[1])
axes[1].set(title="Observable availability", xlabel="Availability state",
ylabel="")
FIGURE_PATHS.append(save_figure(fig, "figure_01_flow_and_availability"))

```

```

[33]: fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
period_values = match_candidates["period_absolute_log_ratio"].dropna()
sns.ecdfplot(period_values, ax=axes[0])
axes[0].axvline(math.log1p(CONFIG.tier1_period_relative_tolerance),
color="red", linestyle="--", label="Tier-1 tolerance")
axes[0].set(xscale="log", xlabel="Absolute period log ratio", ylabel="Empirical
cumulative fraction", title="Name-candidate period consistency")
axes[0].legend()
parameter_rows = []

```

```

for variable in ("planet_radius_rearth", "star_radius_rsun",
    ↪ "planet_mass_mearth", "star_temperature_k"):
    a = tier1_matches[f"{variable}_a"]
    b = tier1_matches[f"{variable}_b"]
    valid = a.notna() & b.notna() & a.gt(0) & b.gt(0)
    parameter_rows.extend({"variable": variable, "log_ratio": value} for value
    ↪ in np.log(a[valid] / b[valid]))
sns.boxplot(data=pd.DataFrame(parameter_rows), x="variable", y="log_ratio",
    ↪ showfliers=False, ax=axes[1])
axes[1].axhline(0, color="black", linewidth=1)
axes[1].tick_params(axis="x", rotation=20)
axes[1].set(title="Base-parameter source differences", xlabel="", ylabel="log(A/
    ↪ B); outliers omitted from panel only")
FIGURE_PATHS.append(save_figure(fig, "figure_02_matching_and_parameters"))

```

```

[34]: fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
plot_widths = primary_certificates[["complete_row_width", "factorial_width"]].
    ↪ melt(var_name="width_type", value_name="width")
sns.ecdfplot(data=plot_widths, x="width", hue="width_type", ax=axes[0])
axes[0].set(xscale="log", xlabel="Multiplicative width", ylabel="Empirical
    ↪ cumulative fraction", title="Complete-row and factorial stability")
axes[0].axvline(1.25, color="red", linestyle="--", linewidth=1)
sns.scatterplot(data=primary_certificates, x="complete_row_width",
    ↪ y="factorial_width", alpha=0.35, s=18, ax=axes[1])
maximum = float(primary_certificates[["complete_row_width", "factorial_width"]].
    ↪ max().max())
axes[1].plot([1, maximum], [1, maximum], color="black", linestyle="--",
    ↪ linewidth=1)
axes[1].set(xscale="log", yscale="log", xlabel="Complete-row width",
    ↪ ylabel="Factorial width", title="Additional hybrid-coordinate stress")
FIGURE_PATHS.append(save_figure(fig, "figure_03_width_distributions"))

```

```

[35]: fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
axes[0].scatter(operational_certificates["source_value_a"],
    ↪ operational_certificates["source_value_b"], alpha=0.35, s=16)
finite = operational_certificates[["source_value_a", "source_value_b"]].
    ↪ to_numpy(float)
lo, hi = np.nanmin(finite), np.nanmax(finite)
axes[0].plot([lo, hi], [lo, hi], color="black", linestyle="--")
axes[0].set(xscale="log", yscale="log", xlabel="Source A area ratio (ppm)",
    ↪ ylabel="Source B area ratio (ppm)", title="Transit-only source agreement")
mean_log = 0.5 * (np.log(operational_certificates["source_value_a"]) + np.
    ↪ log(operational_certificates["source_value_b"]))
axes[1].scatter(mean_log, operational_certificates["signed_log_ratio"], alpha=0.
    ↪ 35, s=16)
axes[1].axhline(0, color="black", linewidth=1)

```

```
axes[1].set(xlabel="Mean log area ratio", ylabel="log(A/B)", title="Log-scale_
↳agreement by magnitude")
FIGURE_PATHS.append(save_figure(fig, "figure_04_agreement_and_ranking"))
```

```
[36]: top_plot = operational_ranking[operational_ranking["top_k_status"].
↳ne("excluded")].copy()
top_plot = top_plot.sort_values("factorial_lower").tail(min(50, len(top_plot)))
fig, ax = plt.subplots(figsize=(8, max(5, 0.22 * len(top_plot))))
y = np.arange(len(top_plot))
colors = top_plot["top_k_status"].map({"guaranteed": sns.color_palette()[2],
↳"possible": sns.color_palette()[1]})
ax.hlines(y, top_plot["factorial_lower"], top_plot["factorial_upper"],
↳color=colors, linewidth=1.5)
ax.scatter(top_plot["factorial_lower"], y, color=colors, s=18)
ax.scatter(top_plot["factorial_upper"], y, color=colors, marker="|", s=60)
ax.set_yticks(y, top_plot["planet_key"], fontsize=7)
ax.set_xscale("log")
ax.set(xlabel="Geometric area-ratio interval (ppm)", ylabel="",
↳title="Concordant-transit robust top-25 candidates")
FIGURE_PATHS.append(save_figure(fig, "figure_05_robust_transit_top25"))
```

```
[37]: fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
axes[0].plot(threshold_sensitivity["factor"],
↳threshold_sensitivity["fraction_at_or_below"], marker="o")
axes[0].set(xscale="log", ylim=(0, 1.02), xlabel="Equivalence factor",
↳ylabel="Fraction at or below factor", title="Equivalence-threshold_
↳sensitivity")
axes[1].plot(k_sensitivity["k"], k_sensitivity["guaranteed"], marker="o",
↳label="guaranteed")
axes[1].plot(k_sensitivity["k"], k_sensitivity["possible"], marker="o",
↳label="possible")
axes[1].set(xlabel="k", ylabel="Planets", title="Transit-only robust top-k_
↳sensitivity")
axes[1].legend()
FIGURE_PATHS.append(save_figure(fig, "figure_06_sensitivity"))
```

```
[38]: extreme_plot = forensic_table[
    forensic_table["selected_primary_extreme"].fillna(False)
    | forensic_table["selected_secondary_extreme"].fillna(False)
    | forensic_table["selected_rank_displacement"].fillna(False)
].nlargest(25, "factorial_width").sort_values("factorial_width")
fig, ax = plt.subplots(figsize=(8, max(5, 0.28 * len(extreme_plot))))
ax.barh(extreme_plot["planet_key"], extreme_plot["factorial_width"], color=sns.
↳color_palette()[3])
ax.set_xscale("log")
```

```
ax.set(xlabel="Factorial multiplicative width", ylabel="", title="Rule-selected_
↳extreme diagnostic cases")
FIGURE_PATHS.append(save_figure(fig, "figure_07_forensic_cases"))
```

## 1.15 Canonical publication tables

Sample sizes use distinct names so values from different cohorts cannot overwrite one another. All manuscript numbers must be drawn from these tables or `results_summary.json`.

```
[39]: sample_rows = pd.concat([
    cohort_flow.assign(section="cohort"),
    matching_flow.assign(section="matching"),
], ignore_index=True)
sample_rows["denominator"] = np.nan
sample_rows["fraction"] = np.nan
availability_rows = availability_summary.reset_index().
    ↳melt(id_vars="observable", var_name="stage", value_name="n")
availability_rows["stage"] = availability_rows["observable"] + ":" +
    ↳availability_rows["stage"]
availability_rows["section"] = "availability"
availability_rows["denominator"] = len(tier1_matches)
availability_rows["fraction"] = availability_rows["n"] / len(tier1_matches)
table_01_sample_and_availability = pd.concat([
    sample_rows[["section", "stage", "n", "denominator", "fraction"]],
    availability_rows[["section", "stage", "n", "denominator", "fraction"]],
], ignore_index=True)
table_02_primary_stability = pd.DataFrame([primary_summary])
table_03_operational_top25 =
    ↳operational_ranking[operational_ranking["top_k_status"].ne("excluded")][[
        "planet_key", "planet_name_source_a", "planet_name_source_b",
    ↳"transit_concordance",
        "source_value_a", "source_value_b", "factorial_lower", "factorial_upper",
    ↳"factorial_width",
        "rank_a", "rank_b", "rank_displacement", "best_rank", "worst_rank",
    ↳"top_k_status", "period_relative_difference",
    ]].copy()
table_04_secondary_results = secondary_summary.copy()
if int(table_02_primary_stability.loc[0, "n_primary"]) !=
    ↳len(primary_certificates):
    raise RuntimeError("Primary table sample size is inconsistent.")
if len(table_03_operational_top25) !=
    ↳int(ranking_summary["robust_candidate_pool"]):
    raise RuntimeError("Operational table candidate count is inconsistent.")
display(table_01_sample_and_availability)
display(table_02_primary_stability)
display(table_03_operational_top25.head(30))
display(table_04_secondary_results)
```

|           | section      | stage                                  | n    | denominator | ␣  |
|-----------|--------------|----------------------------------------|------|-------------|----|
| ↪fraction |              |                                        |      |             |    |
| 0         | cohort       | Raw A                                  | 3732 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 1         | cohort       | Raw B                                  | 6150 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 2         | cohort       | Tier 1                                 | 2981 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 3         | cohort       | Primary paired radii                   | 2588 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 4         | cohort       | Concordant transits                    | 2561 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 5         | matching     | Name candidates                        | 3343 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 6         | matching     | Candidates with paired periods         | 3282 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 7         | matching     | Candidates missing at least one period | 61   | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 8         | matching     | Tier 1                                 | 2981 | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 9         | matching     | Tier 2                                 | 198  | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 10        | matching     | Rejected                               | 164  | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 11        | matching     | Manually reviewed                      | 0    | NaN         | ␣  |
| ↪ NaN     |              |                                        |      |             |    |
| 12        | availability | density_g_cm3:a_only                   | 1    | 2981.0      | 0. |
| ↪000335   |              |                                        |      |             |    |
| 13        | availability | escape_velocity_km_s:a_only            | 1    | 2981.0      | 0. |
| ↪000335   |              |                                        |      |             |    |
| 14        | availability | geometric_area_ratio_ppm:a_only        | 1    | 2981.0      | 0. |
| ↪000335   |              |                                        |      |             |    |
| 15        | availability | gravity_gearth:a_only                  | 1    | 2981.0      | 0. |
| ↪000335   |              |                                        |      |             |    |
| 16        | availability | insolation_searth:a_only               | 3    | 2981.0      | 0. |
| ↪001006   |              |                                        |      |             |    |
| 17        | availability | density_g_cm3:b_only                   | 2381 | 2981.0      | 0. |
| ↪798725   |              |                                        |      |             |    |
| 18        | availability | escape_velocity_km_s:b_only            | 2381 | 2981.0      | 0. |
| ↪798725   |              |                                        |      |             |    |
| 19        | availability | geometric_area_ratio_ppm:b_only        | 361  | 2981.0      | 0. |
| ↪121100   |              |                                        |      |             |    |
| 20        | availability | gravity_gearth:b_only                  | 2381 | 2981.0      | 0. |
| ↪798725   |              |                                        |      |             |    |
| 21        | availability | insolation_searth:b_only               | 1490 | 2981.0      | 0. |
| ↪499832   |              |                                        |      |             |    |

|    |                                      |                                    |                         |                     |    |
|----|--------------------------------------|------------------------------------|-------------------------|---------------------|----|
| 22 | availability                         | density_g_cm3:both                 | 558                     | 2981.0              | 0. |
|    | ↪187186                              |                                    |                         |                     |    |
| 23 | availability                         | escape_velocity_km_s:both          | 558                     | 2981.0              | 0. |
|    | ↪187186                              |                                    |                         |                     |    |
| 24 | availability                         | geometric_area_ratio_ppm:both      | 2588                    | 2981.0              | 0. |
|    | ↪868165                              |                                    |                         |                     |    |
| 25 | availability                         | gravity_gearth:both                | 558                     | 2981.0              | 0. |
|    | ↪187186                              |                                    |                         |                     |    |
| 26 | availability                         | insolation_searth:both             | 1474                    | 2981.0              | 0. |
|    | ↪494465                              |                                    |                         |                     |    |
| 27 | availability                         | density_g_cm3:neither              | 41                      | 2981.0              | 0. |
|    | ↪013754                              |                                    |                         |                     |    |
| 28 | availability                         | escape_velocity_km_s:neither       | 41                      | 2981.0              | 0. |
|    | ↪013754                              |                                    |                         |                     |    |
| 29 | availability                         | geometric_area_ratio_ppm:neither   | 31                      | 2981.0              | 0. |
|    | ↪010399                              |                                    |                         |                     |    |
| 30 | availability                         | gravity_gearth:neither             | 41                      | 2981.0              | 0. |
|    | ↪013754                              |                                    |                         |                     |    |
| 31 | availability                         | insolation_searth:neither          | 14                      | 2981.0              | 0. |
|    | ↪004696                              |                                    |                         |                     |    |
|    | n_primary                            | median_factorial_width             | q75_factorial_width     | q90_factorial_width | ↪  |
|    | ↪q95_factorial_width                 | q99_factorial_width                | maximum_factorial_width | ↪                   |    |
|    | ↪median_complete_row_width           | \                                  |                         |                     |    |
| 0  | 2588                                 | 1.004868                           | 1.013016                | 1.20001             | ↪  |
|    | ↪1.462761                            | 3.026051                           | 13.567213               |                     | ↪  |
|    | ↪1.004149                            |                                    |                         |                     |    |
|    | median_factorial_inflation           | q90_factorial_inflation            | q95_factorial_inflation | ↪                   |    |
|    | ↪q99_factorial_inflation             | maximum_factorial_inflation        | ↪                       |                     |    |
|    | ↪count_factorial_at_or_below_1p01    | \                                  |                         |                     |    |
| 0  |                                      | 1.0                                | 1.024341                | 1.195588            | ↪  |
|    | ↪2.11868                             |                                    | 12.729919               |                     | ↪  |
|    | ↪1787                                |                                    |                         |                     |    |
|    | fraction_factorial_at_or_below_1p01  | count_factorial_above_1p01         | ↪                       |                     |    |
|    | ↪fraction_factorial_above_1p01       | count_factorial_at_or_below_1p05   | ↪                       |                     |    |
|    | ↪fraction_factorial_at_or_below_1p05 | \                                  |                         |                     |    |
| 0  |                                      | 0.690495                           | 801                     |                     | ↪  |
|    | ↪0.309505                            | 2130                               |                         |                     | ↪  |
|    | ↪0.823029                            |                                    |                         |                     |    |
|    | count_factorial_above_1p05           | fraction_factorial_above_1p05      | ↪                       |                     |    |
|    | ↪count_factorial_at_or_below_1p1     | fraction_factorial_at_or_below_1p1 | ↪                       |                     |    |
|    | ↪count_factorial_above_1p1           | \                                  |                         |                     |    |
| 0  |                                      | 458                                | 0.176971                |                     | ↪  |
|    | ↪2203                                | 0.851236                           | 385                     |                     |    |

```

fraction_factorial_above_1p1 count_factorial_at_or_below_1p25
↵fraction_factorial_at_or_below_1p25 count_factorial_above_1p25
↵fraction_factorial_above_1p25 \
0          0.148764          2367          0.
↵          0.914606          221          0.
↵085394

count_factorial_at_or_below_1p5 fraction_factorial_at_or_below_1p5
↵count_factorial_above_1p5 fraction_factorial_above_1p5
↵count_factorial_at_or_below_2p0 \
0          2461          0.950927          2535
↵          127          0.049073

fraction_factorial_at_or_below_2p0 count_factorial_above_2p0
↵fraction_factorial_above_2p0 count_factorial_at_or_below_5p0
↵fraction_factorial_at_or_below_5p0 \
0          0.979521          53          0.
↵          0.020479          2579          0.
↵996522

count_factorial_above_5p0 fraction_factorial_above_5p0
↵count_factorial_at_or_below_10p0 fraction_factorial_at_or_below_10p0
↵count_factorial_above_10p0 \
0          9          0.003478          3
↵          2585          0.998841

fraction_factorial_above_10p0 fraction_inflation_above_1p01
↵fraction_inflation_above_1p05 fraction_inflation_above_1p25
↵fraction_inflation_above_2p0 \
0          0.001159          0.120943          0.011592
↵          0.08694          0.042117

host_composition_interval_lower host_composition_interval_upper
0          1.004559          1.005201

planet_key planet_name_source_a planet_name_source_b transit_concordance
↵source_value_a source_value_b factorial_lower factorial_upper
↵factorial_width rank_a \
0          ngts-1 b          NGTS-1 b          NGTS-1 b          both_transit
↵          56767.608774          56767.608774          56767.608774          1.
↵000000          1
1          hats-6 b          HATS-6 b          HATS-6 b          both_transit
↵          32301.146242          32301.146242          32301.146242          1.
↵000000          2

```

|    |              |              |              |              |    |
|----|--------------|--------------|--------------|--------------|----|
| 2  | kepler-45 b  | Kepler-45 b  | Kepler-45 b  | both_transit | ␣  |
| ↪  | 32101.376103 | 32097.557693 | 32097.557693 | 32101.376103 | 1. |
| ↪  | 000119       | 3            |              |              |    |
| 3  | wts-2 b      | WTS-2 b      | WTS-2 b      | both_transit | ␣  |
| ↪  | 30748.501372 | 34614.861103 | 30748.501372 | 34614.861103 | 1. |
| ↪  | 125741       | 4            |              |              |    |
| 4  | kepler-785 b | Kepler-785 b | Kepler-785 b | both_transit | ␣  |
| ↪  | 28222.931486 | 28201.882284 | 28201.882284 | 28222.931486 | 1. |
| ↪  | 000746       | 6            |              |              |    |
| 5  | corot-2 b    | CoRoT-2 b    | CoRoT-2 b    | both_transit | ␣  |
| ↪  | 27795.131759 | 27586.543078 | 27550.241924 | 27831.755588 | 1. |
| ↪  | 010218       | 7            |              |              |    |
| 6  | wasp-52 b    | WASP-52 b    | WASP-52 b    | both_transit | ␣  |
| ↪  | 27230.763739 | 27248.250329 | 27230.763739 | 27248.250329 | 1. |
| ↪  | 000642       | 9            |              |              |    |
| 7  | kepler-447 b | Kepler-447 b | Kepler-447 b | both_transit | ␣  |
| ↪  | 27039.583754 | 26005.670084 | 26005.670084 | 27039.583754 | 1. |
| ↪  | 039757       | 11           |              |              |    |
| 8  | hats-23 b    | HATS-23 b    | HATS-23 b    | both_transit | ␣  |
| ↪  | 25356.778972 | 25356.778972 | 25356.778972 | 25356.778972 | 1. |
| ↪  | 000000       | 14           |              |              |    |
| 9  | hat-p-54 b   | HAT-P-54 b   | HAT-P-54 b   | both_transit | ␣  |
| ↪  | 24664.949087 | 24664.949087 | 24664.949087 | 24664.949087 | 1. |
| ↪  | 000000       | 15           |              |              |    |
| 10 | wasp-80 b    | WASP-80 b    | WASP-80 b    | both_transit | ␣  |
| ↪  | 24060.207068 | 30622.617323 | 24060.207068 | 30622.617323 | 1. |
| ↪  | 272750       | 16           |              |              |    |
| 11 | wasp-98 b    | WASP-98 b    | WASP-98 b    | both_transit | ␣  |
| ↪  | 26019.314653 | 25114.364651 | 23219.641874 | 28142.490728 | 1. |
| ↪  | 212012       | 12           |              |              |    |
| 12 | qatar-2 b    | Qatar-2 b    | Qatar-2 b    | both_transit | ␣  |
| ↪  | 27125.612413 | 27515.212988 | 22899.983488 | 32592.469046 | 1. |
| ↪  | 423253       | 10           |              |              |    |
| 13 | qatar-6 b    | Qatar-6 b    | Qatar-6 b    | both_transit | ␣  |
| ↪  | 22797.183824 | 22797.183824 | 22797.183824 | 22797.183824 | 1. |
| ↪  | 000000       | 19           |              |              |    |
| 14 | wasp-140 b   | WASP-140 b   | WASP-140 b   | both_transit | ␣  |
| ↪  | 28866.427679 | 22452.980747 | 22452.980747 | 28866.427679 | 1. |
| ↪  | 285639       | 5            |              |              |    |
| 15 | hats-34 b    | HATS-34 b    | HATS-34 b    | both_transit | ␣  |
| ↪  | 22435.021308 | 22435.021308 | 22435.021308 | 22435.021308 | 1. |
| ↪  | 000000       | 21           |              |              |    |
| 16 | hats-43 b    | HATS-43 b    | HATS-43 b    | both_transit | ␣  |
| ↪  | 22251.460360 | 22251.460360 | 22251.460360 | 22251.460360 | 1. |
| ↪  | 000000       | 22           |              |              |    |



|    |               |               |               |              |    |
|----|---------------|---------------|---------------|--------------|----|
| 17 | hat-p-32 b    | HAT-P-32 b    | HAT-P-32 b    | both_transit | ┐  |
| ↳  | 22726.664389  | 22105.478195  | 21472.569702  | 23396.537586 | 1. |
| ↳  | 089601 20     |               |               |              |    |
| 18 | ogle-tr-113 b | OGLE-TR-113 b | OGLE-TR-113 b | both_transit | ┐  |
| ↳  | 22183.478652  | 21010.383656  | 21010.383656  | 22183.478652 | 1. |
| ↳  | 055834 23     |               |               |              |    |
| 19 | wasp-43 b     | WASP-43 b     | WASP-43 b     | both_transit | ┐  |
| ↳  | 25419.933248  | 25312.738299  | 20482.830071  | 31414.024119 | 1. |
| ↳  | 533676 13     |               |               |              |    |
| 20 | qatar-1 b     | Qatar-1 b     | Qatar-1 b     | both_transit | ┐  |
| ↳  | 21660.621174  | 21348.527883  | 20323.539840  | 22753.042962 | 1. |
| ↳  | 119541 25     |               |               |              |    |
| 21 | wasp-107 b    | WASP-107 b    | WASP-107 b    | both_transit | ┐  |
| ↳  | 21373.438649  | 20740.186847  | 20740.186847  | 21373.438649 | 1. |
| ↳  | 030533 27     |               |               |              |    |
| 22 | wasp-19 b     | WASP-19 b     | WASP-19 b     | both_transit | ┐  |
| ↳  | 20341.714396  | 20845.964099  | 20260.913227  | 20929.098470 | 1. |
| ↳  | 032979 33     |               |               |              |    |
| 23 | hats-22 b     | HATS-22 b     | HATS-22 b     | both_transit | ┐  |
| ↳  | 20158.300747  | 20158.300747  | 20158.300747  | 20158.300747 | 1. |
| ↳  | 000000 34     |               |               |              |    |
| 24 | wasp-10 b     | WASP-10 b     | WASP-10 b     | both_transit | ┐  |
| ↳  | 20046.130333  | 25226.867049  | 20046.130333  | 25226.867049 | 1. |
| ↳  | 258441 36     |               |               |              |    |
| 29 | wasp-39 b     | WASP-39 b     | WASP-39 b     | both_transit | ┐  |
| ↳  | 21216.216285  | 19548.636324  | 19274.486657  | 21517.984043 | 1. |
| ↳  | 116397 28     |               |               |              |    |
| 33 | wasp-110 b    | WASP-110 b    | WASP-110 b    | both_transit | ┐  |
| ↳  | 20806.356492  | 20061.339626  | 18806.421946  | 22194.726098 | 1. |
| ↳  | 180167 31     |               |               |              |    |
| 34 | wasp-50 b     | WASP-50 b     | WASP-50 b     | both_transit | ┐  |
| ↳  | 18666.326478  | 20158.049828  | 18666.326478  | 20158.049828 | 1. |
| ↳  | 079915 46     |               |               |              |    |
| 37 | wasp-67 b     | WASP-67 b     | WASP-67 b     | both_transit | ┐  |
| ↳  | 27285.010731  | 17994.380483  | 17994.380483  | 27285.010731 | 1. |
| ↳  | 516307 8      |               |               |              |    |
| 39 | wasp-25 b     | WASP-25 b     | WASP-25 b     | both_transit | ┐  |
| ↳  | 18535.335118  | 20465.480830  | 17720.634668  | 21406.374695 | 1. |
| ↳  | 207991 47     |               |               |              |    |

|   | rank_b                     | rank_displacement | best_rank | worst_rank | top_k_status | ┐ |
|---|----------------------------|-------------------|-----------|------------|--------------|---|
| ↳ | period_relative_difference |                   |           |            |              |   |
| 0 | 1                          | 0                 | 1         | 1          | guaranteed   | ┐ |
| ↳ | 0.000000e+00               |                   |           |            |              |   |
| 1 | 3                          | 1                 | 2         | 5          | guaranteed   | ┐ |
| ↳ | 0.000000e+00               |                   |           |            |              |   |

|               |    |    |    |    |            |   |
|---------------|----|----|----|----|------------|---|
| 2             | 4  | 1  | 3  | 6  | guaranteed | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 3             | 2  | 2  | 2  | 8  | guaranteed | ␣ |
| ↪5.889820e-07 |    |    |    |    |            |   |
| 4             | 6  | 0  | 5  | 11 | guaranteed | ␣ |
| ↪2.026588e-09 |    |    |    |    |            |   |
| 5             | 7  | 0  | 6  | 13 | guaranteed | ␣ |
| ↪1.663804e-06 |    |    |    |    |            |   |
| 6             | 9  | 0  | 7  | 15 | guaranteed | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 7             | 10 | 1  | 8  | 16 | guaranteed | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 8             | 11 | 3  | 9  | 17 | guaranteed | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 9             | 15 | 0  | 10 | 20 | guaranteed | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 10            | 5  | 11 | 5  | 22 | guaranteed | ␣ |
| ↪6.323646e-07 |    |    |    |    |            |   |
| 11            | 14 | 2  | 6  | 24 | guaranteed | ␣ |
| ↪1.215132e-07 |    |    |    |    |            |   |
| 12            | 8  | 2  | 2  | 25 | guaranteed | ␣ |
| ↪1.293829e-06 |    |    |    |    |            |   |
| 13            | 17 | 2  | 14 | 26 | possible   | ␣ |
| ↪1.711260e-06 |    |    |    |    |            |   |
| 14            | 18 | 13 | 5  | 28 | possible   | ␣ |
| ↪4.382859e-07 |    |    |    |    |            |   |
| 15            | 19 | 2  | 16 | 29 | possible   | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 16            | 20 | 2  | 17 | 30 | possible   | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 17            | 21 | 1  | 12 | 34 | possible   | ␣ |
| ↪3.720916e-07 |    |    |    |    |            |   |
| 18            | 24 | 1  | 18 | 38 | possible   | ␣ |
| ↪1.396185e-06 |    |    |    |    |            |   |
| 19            | 12 | 1  | 4  | 40 | possible   | ␣ |
| ↪3.110114e-06 |    |    |    |    |            |   |
| 20            | 22 | 3  | 15 | 40 | possible   | ␣ |
| ↪5.915392e-07 |    |    |    |    |            |   |
| 21            | 26 | 1  | 19 | 40 | possible   | ␣ |
| ↪2.761526e-06 |    |    |    |    |            |   |
| 22            | 25 | 8  | 20 | 41 | possible   | ␣ |
| ↪1.876176e-06 |    |    |    |    |            |   |
| 23            | 28 | 6  | 24 | 43 | possible   | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |
| 24            | 13 | 23 | 10 | 44 | possible   | ␣ |
| ↪0.000000e+00 |    |    |    |    |            |   |

|                                                                                                                                                                     |           |            |          |             |          |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|----------|-------------|----------|---|
| 29                                                                                                                                                                  | 35        | 7          | 18       | 50          | possible | ␣ |
| ↪8.655427e-06                                                                                                                                                       |           |            |          |             |          |   |
| 33                                                                                                                                                                  | 30        | 1          | 18       | 55          | possible | ␣ |
| ↪1.190981e-06                                                                                                                                                       |           |            |          |             |          |   |
| 34                                                                                                                                                                  | 29        | 17         | 25       | 57          | possible | ␣ |
| ↪2.097084e-06                                                                                                                                                       |           |            |          |             |          |   |
| 37                                                                                                                                                                  | 48        | 40         | 7        | 64          | possible | ␣ |
| ↪0.000000e+00                                                                                                                                                       |           |            |          |             |          |   |
| 39                                                                                                                                                                  | 27        | 20         | 19       | 69          | possible | ␣ |
| ↪9.827801e-07                                                                                                                                                       |           |            |          |             |          |   |
| median_signed_log_ratio median_absolute_log_ratio q05_log_ratio ␣<br>↪q95_log_ratio observable n_secondary median_complete_row_width ␣<br>↪median_factorial_width \ |           |            |          |             |          |   |
| 0                                                                                                                                                                   | -0.000006 |            | 0.053442 | -0.611566   | 1.       |   |
| ↪776839 density_g_cm3 513 1.054895 ␣                                                                                                                                |           |            |          |             |          |   |
| ↪ 1.134855                                                                                                                                                          |           |            |          |             |          |   |
| 1                                                                                                                                                                   | -0.000006 |            | 0.027441 | -0.441070   | 1.       |   |
| ↪871348 gravity_gearth 513 1.027821 ␣                                                                                                                               |           |            |          |             |          |   |
| ↪ 1.103149                                                                                                                                                          |           |            |          |             |          |   |
| 2                                                                                                                                                                   | -0.000003 |            | 0.018910 | -0.178712   | 0.       |   |
| ↪941153 escape_velocity_km_s 513 1.019090 ␣                                                                                                                         |           |            |          |             |          |   |
| ↪ 1.032889                                                                                                                                                          |           |            |          |             |          |   |
| 3                                                                                                                                                                   | 0.000000  |            | 0.007609 | -0.263850   | 0.       |   |
| ↪175810 insolation_searth 1474 1.007638 ␣                                                                                                                           |           |            |          |             |          |   |
| ↪ 1.007910                                                                                                                                                          |           |            |          |             |          |   |
| q95_factorial_width q99_factorial_width maximum_factorial_width ␣<br>↪count_factorial_above_2 count_factorial_above_10                                              |           |            |          |             |          |   |
| 0                                                                                                                                                                   | 8.818423  | 140.593643 |          | 1807.293690 |          | ␣ |
| ↪ 94 25                                                                                                                                                             |           |            |          |             |          |   |
| 1                                                                                                                                                                   | 8.310094  | 139.071000 |          | 987.066291  |          | ␣ |
| ↪ 75 25                                                                                                                                                             |           |            |          |             |          |   |
| 2                                                                                                                                                                   | 2.846316  | 11.723792  |          | 23.218382   |          | ␣ |
| ↪ 41 7                                                                                                                                                              |           |            |          |             |          |   |
| 3                                                                                                                                                                   | 1.689725  | 3.099854   |          | 7.703079    |          | ␣ |
| ↪ 49 0                                                                                                                                                              |           |            |          |             |          |   |

```
[40]: all_reports = [
    physics_reference_report, ranking_reference_report,␣
    ↪contract_reference_report,
    bound_property_report, ranking_property_report, provenance_report,␣
    ↪reproduction_report,
    raw_report_a, raw_report_b, canonical_report_a, canonical_report_b,
    primary_verification, operational_verification, ranking_verification,
    *secondary_reports, forensic_report, forensic_review_report,
]
```

```

validation_issues = validation_issues_frame(all_reports)
quality_gates = {
    "generator_script_available": input_manifest["code_assets"]["generator"] is_
↳not None,
    "raw_schema_a": raw_report_a.passed,
    "raw_schema_b": raw_report_b.passed,
    "canonical_a": canonical_report_a.passed,
    "canonical_b": canonical_report_b.passed,
    "tier1_minimum": len(tier1_matches) >= CONFIG.minimum_tier1_matches,
    "primary_minimum": len(primary_cohort) >= CONFIG.minimum_primary_cohort,
    "operational_minimum": len(operational_cohort) >= CONFIG.
↳minimum_operational_cohort,
    "operational_all_concordant_transits":_
↳operational_cohort["transit_concordance"].eq("both_transit").all(),
    "primary_certificates_fully_verified": primary_verification.passed and_
↳primary_verification.metrics["eligible_rows"] == len(primary_certificates),
    "operational_certificates_fully_verified": operational_verification.passed,
    "robust_ranking_verified": ranking_verification.passed,
    "primary_table_count_consistent": int(table_02_primary_stability.loc[0,_
↳"n_primary"]) == len(primary_certificates),
    "operational_table_count_consistent": len(table_03_operational_top25) ==_
↳int(ranking_summary["robust_candidate_pool"]),
    "physics_reference_tests": physics_reference_report.passed,
    "ranking_reference_tests": ranking_reference_report.passed,
    "data_contract_reference_tests": contract_reference_report.passed,
    "randomized_bound_property_tests": bound_property_report.passed,
    "randomized_ranking_property_tests": ranking_property_report.passed,
    "secondary_certificates_fully_verified": all(report.passed for report in_
↳secondary_reports),
    "threshold_partitions_consistent": bool(np.
↳allclose(threshold_sensitivity["fraction_at_or_below"] +_
↳threshold_sensitivity["fraction_above"], 1.0)),
    "threshold_counts_consistent":_
↳bool((threshold_sensitivity["count_at_or_below"] +_
↳threshold_sensitivity["count_above"]).eq(len(primary_certificates)).all()),
    "forensic_case_selection_complete": forensic_report.passed,
    "detection_mapping_complete": not_
↳detection_mapping_audit[["discovery_method_class", "transit_evidence"]].
↳isna().any().any(),
    "operational_exclusion_count_consistent": len(operational_exclusion_audit)_
↳== len(primary_cohort) - len(operational_cohort),
    "no_blocking_validation_issues": not any(report.blocking_count for report_
↳in all_reports),
}
computationally_definitive = all(quality_gates.values())

```

```

solo_author_forensic_self_audit_complete = priority_zero_review_complete and
    ↪priority_zero_findings_status == "acceptable"
external_reproduction_complete = False
final_package_audit_complete = False
computationally_ready = (
    computationally_definitive and solo_author_forensic_self_audit_complete
    and provenance_documented and clean_environment_self_reproduction_passed
)
submission_ready = computationally_ready and external_reproduction_complete and
    ↪final_package_audit_complete
results_summary = {
    "protocol_version": CONFIG.protocol_version,
    "publication_contract": PUBLICATION_CONTRACT,
    "sample_flow": dict(zip(cohort_flow["stage"], cohort_flow["n"])),
    "matching_flow": matching_flow.to_dict(orient="records"),
    "candidate_period_matching_diagnostic": candidate_period_summary,
    "missingness": canonical_audit.to_dict(orient="records"),
    "detection_method_mapping": detection_mapping_audit.
    ↪to_dict(orient="records"),
    "operational_exclusions": {"n": len(operational_exclusion_audit)},
    "generalizability": generalizability_summary.to_dict(orient="records"),
    "primary": primary_summary,
    "operational_ranking": ranking_summary,
    "secondary": secondary_summary.to_dict(orient="records"),
    "threshold_sensitivity": threshold_sensitivity.to_dict(orient="records"),
    "k_sensitivity": k_sensitivity.to_dict(orient="records"),
    "quality_gates": quality_gates,
    "status": {
        "computationally_definitive": computationally_definitive,
        "review_mode": author_mode_label(),
        "lead_author": CONFIG.lead_author,
        "forensic_case_selection_complete": forensic_report.passed,
        "priority_zero_review_complete": priority_zero_review_complete,
        "priority_zero_findings_status": priority_zero_findings_status,
        "solo_author_forensic_self_audit_complete":
    ↪solo_author_forensic_self_audit_complete,
        "provenance_documented": provenance_documented,
        "clean_environment_self_reproduction_passed":
    ↪clean_environment_self_reproduction_passed,
        "external_reproduction_complete": external_reproduction_complete,
        "final_package_audit_complete": final_package_audit_complete,
        "computationally_ready": computationally_ready,
        "submission_ready": submission_ready,
    },
    "limitations": [
        "Two derivative artifacts do not span all measurement or epistemic
    ↪uncertainty.",

```

```

        "Coordinate-wise hybrid vertices may not correspond to a jointly_
        ↪published parameter set.",
        "Mass-dependent observables have source-semantic limitations.",
        "The geometric area ratio is not an observed transit depth or_
        ↪mission-priority score.",
    ],
}
display(pd.DataFrame({"quality_gate": quality_gates.keys(), "passed":_
        ↪quality_gates.values()}))

```

|    | quality_gate                            | passed |
|----|-----------------------------------------|--------|
| 0  | generator_script_available              | True   |
| 1  | raw_schema_a                            | True   |
| 2  | raw_schema_b                            | True   |
| 3  | canonical_a                             | True   |
| 4  | canonical_b                             | True   |
| 5  | tier1_minimum                           | True   |
| 6  | primary_minimum                         | True   |
| 7  | operational_minimum                     | True   |
| 8  | operational_all_concordant_transits     | True   |
| 9  | primary_certificates_fully_verified     | True   |
| 10 | operational_certificates_fully_verified | True   |
| 11 | robust_ranking_verified                 | True   |
| 12 | primary_table_count_consistent          | True   |
| 13 | operational_table_count_consistent      | True   |
| 14 | physics_reference_tests                 | True   |
| 15 | ranking_reference_tests                 | True   |
| 16 | data_contract_reference_tests           | True   |
| 17 | randomized_bound_property_tests         | True   |
| 18 | randomized_ranking_property_tests       | True   |
| 19 | secondary_certificates_fully_verified   | True   |
| 20 | threshold_partitions_consistent         | True   |
| 21 | threshold_counts_consistent             | True   |
| 22 | forensic_case_selection_complete        | True   |
| 23 | detection_mapping_complete              | True   |
| 24 | operational_exclusion_count_consistent  | True   |
| 25 | no_blocking_validation_issues           | True   |

```

[41]: TABLE_PATHS: list[Path] = []
      for frame, filename in (
          (table_01_sample_and_availability, "table_01_sample_and_availability.csv"),
          (table_02_primary_stability, "table_02_primary_stability.csv"),
          (table_03_operational_top25, "table_03_operational_top25.csv"),
          (table_04_secondary_results, "table_04_secondary_results.csv"),
      ):
          path = PATHS.table_dir / filename
          atomic_csv(frame, path)

```

```

TABLE_PATHS.append(path)

atomic_csv(match_audit, PATHS.artifact_dir / "match_audit.csv.gz", gzip=True)
atomic_csv(primary_certificates, PATHS.artifact_dir / "primary_certificates.csv.
↳gz", gzip=True)
atomic_csv(secondary_certificates, PATHS.artifact_dir / "secondary_certificates.
↳csv.gz", gzip=True)
atomic_csv(forensic_table, PATHS.artifact_dir / "forensic_cases.csv")
forensic_review_record = forensic_table[[
    "planet_key", "observable", "review_priority", "selection_reasons",
↳"review_classification",
    "review_finding", "evidence_reference", "reviewer", "review_date",
↳"second_reviewer", "adjudication",
]].copy()
atomic_csv(forensic_review_record, PATHS.artifact_dir / "forensic_review_record.
↳csv")
atomic_csv(detection_mapping_audit, PATHS.artifact_dir /
↳"detection_method_mapping.csv")
atomic_csv(operational_exclusion_audit, PATHS.artifact_dir /
↳"operational_exclusion_audit.csv")
atomic_csv(generalizability_summary, PATHS.artifact_dir /
↳"generalizability_summary.csv")
atomic_json(results_summary, PATHS.artifact_dir / "results_summary.json")
atomic_json({"issues": validation_issues.to_dict(orient="records")}, PATHS.
↳artifact_dir / "validation_report.json")
environment_specification = {
    "protocol_version": CONFIG.protocol_version,
    "python_supported": f"{sys.version_info.major}.{sys.version_info.minor}",
    "python_implementation": platform.python_implementation(),
    "direct_dependencies": SOFTWARE_VERSIONS,
    "exact_direct_dependency_constraints": {
        name: f"{name}=={version}" for name, version in SOFTWARE_VERSIONS.
↳items() if name != "python"
    },
    "execution_platform": platform.platform(),
    "generator": input_manifest["code_assets"]["generator"],
}
atomic_json(environment_specification, PATHS.artifact_dir /
↳"environment_specification.json")

exported_provenance = copy.deepcopy(provenance_payload) if provenance_payload
↳is not None else {}
exported_provenance["authorship"] = {
    "lead_author": CONFIG.lead_author,
    "review_mode": author_mode_label(),
    "self_audit_note": (

```

```

        "Single-author project: the lead author performed the priority-0
↪forensic review as a "
        "documented self-audit. No second reviewer identity exists."
    ),
}
atomic_json(exported_provenance, PATHS.artifact_dir / "source_provenance.json")

exported_reproduction = copy.deepcopy(reproduction_payload) if
↪reproduction_payload is not None else {}
exported_reproduction["authorship"] = {
    "lead_author": CONFIG.lead_author,
    "review_mode": author_mode_label(),
    "self_audit_note": (
        "Clean-environment reproduction executed and certified by the lead
↪author on a fresh "
        "environment as a solo-author self-audit."
    ),
}
atomic_json(exported_reproduction, PATHS.artifact_dir /
↪"clean_environment_self_reproduction.json")

```

```

[42]: output_paths = [
    PATHS.artifact_dir / "match_audit.csv.gz",
    PATHS.artifact_dir / "primary_certificates.csv.gz",
    PATHS.artifact_dir / "secondary_certificates.csv.gz",
    PATHS.artifact_dir / "forensic_cases.csv",
    PATHS.artifact_dir / "forensic_review_record.csv",
    PATHS.artifact_dir / "detection_method_mapping.csv",
    PATHS.artifact_dir / "operational_exclusion_audit.csv",
    PATHS.artifact_dir / "generalizability_summary.csv",
    PATHS.artifact_dir / "results_summary.json",
    PATHS.artifact_dir / "validation_report.json",
    PATHS.artifact_dir / "environment_specification.json",
    PATHS.artifact_dir / "source_provenance.json",
    PATHS.artifact_dir / "clean_environment_self_reproduction.json",
    *TABLE_PATHS, *FIGURE_PATHS,
]
manifest = {
    **input_manifest,
    "outputs": {path.as_posix(): {"sha256": sha256_file(path), "bytes": path.
↪stat().st_size} for path in output_paths},
    "quality_gates": quality_gates,
    "computationally_definitive": computationally_definitive,
    "priority_zero_review_complete": priority_zero_review_complete,
    "priority_zero_findings_status": priority_zero_findings_status,
    "solo_author_forensic_self_audit_complete":
↪solo_author_forensic_self_audit_complete,

```



```

        "provenance_documented": provenance_documented,
        "clean_environment_self_reproduction_passed":␣
↪clean_environment_self_reproduction_passed,
        "external_reproduction_complete": external_reproduction_complete,
        "final_package_audit_complete": final_package_audit_complete,
        "computationally_ready": computationally_ready,
        "submission_ready": submission_ready,
        "interpretation_note": "Computational certification is distinct from human␣
↪forensic review and publication readiness.",
    }
}
atomic_json(manifest, PATHS.artifact_dir / "analysis_manifest.json")

audit_errors: list[str] = []
reloaded_manifest = json.loads((PATHS.artifact_dir / "analysis_manifest.json").
↪read_text(encoding="utf-8"))
expected_relative_outputs = {
    "artifacts/match_audit.csv.gz", "artifacts/primary_certificates.csv.gz",␣
↪"artifacts/secondary_certificates.csv.gz",
    "artifacts/forensic_cases.csv", "artifacts/forensic_review_record.csv",␣
↪"artifacts/detection_method_mapping.csv", "artifacts/
↪operational_exclusion_audit.csv",
    "artifacts/generalizability_summary.csv", "artifacts/results_summary.json",␣
↪"artifacts/validation_report.json", "artifacts/environment_specification.
↪json",
    "artifacts/source_provenance.json", "artifacts/
↪clean_environment_self_reproduction.json",
    "tables/table_01_sample_and_availability.csv", "tables/
↪table_02_primary_stability.csv",
    "tables/table_03_operational_top25.csv", "tables/table_04_secondary_results.
↪csv",
    "figures/figure_01_flow_and_availability.pdf", "figures/
↪figure_02_matching_and_parameters.pdf",
    "figures/figure_03_width_distributions.pdf", "figures/
↪figure_04_agreement_and_ranking.pdf",
    "figures/figure_05_robust_transit_top25.pdf", "figures/
↪figure_06_sensitivity.pdf",
    "figures/figure_07_forensic_cases.pdf",
}
observed_relative_outputs = {Path(path).relative_to(PATHS.output_root).
↪as_posix() for path in reloaded_manifest["outputs"]}
if observed_relative_outputs != expected_relative_outputs:
    audit_errors.append(
        f"Artifact contract mismatch;␣
↪missing={sorted(expected_relative_outputs-observed_relative_outputs)}, "
        ␣
↪f"unexpected={sorted(observed_relative_outputs-expected_relative_outputs)}"

```

```

    )
for path_text, metadata in reloaded_manifest["outputs"].items():
    path = Path(path_text)
    if not path.is_file() or path.stat().st_size == 0:
        audit_errors.append(f"Missing or empty artifact: {path}")
    elif sha256_file(path) != metadata["sha256"]:
        audit_errors.append(f"Hash mismatch: {path}")
reloaded_primary = pd.read_csv(PATHS.artifact_dir / "primary_certificates.csv.
↪gz")
reloaded_table_02 = pd.read_csv(PATHS.table_dir / "table_02_primary_stability.
↪csv")
reloaded_table_03 = pd.read_csv(PATHS.table_dir / "table_03_operational_top25.
↪csv")
reloaded_secondary = pd.read_csv(PATHS.artifact_dir / "secondary_certificates.
↪csv.gz")
reloaded_forensic = pd.read_csv(PATHS.artifact_dir / "forensic_cases.csv")
reloaded_review_record = pd.read_csv(PATHS.artifact_dir / ↵
↪"forensic_review_record.csv")
if len(reloaded_primary) != len(primary_certificates):
    audit_errors.append("Primary certificate count changed after export.")
if int(reloaded_table_02.loc[0, "n_primary"]) != len(primary_certificates):
    audit_errors.append("Reloaded primary table count is inconsistent.")
if not reloaded_table_03["transit_concordance"].eq("both_transit").all():
    audit_errors.append("Reloaded operational table contains a non-concordant ↵
↪transit.")
if len(reloaded_secondary) != len(secondary_certificates):
    audit_errors.append("Secondary certificate count changed after export.")
if len(reloaded_forensic) != len(forensic_table):
    audit_errors.append("Forensic audit count changed after export.")
if len(reloaded_review_record) != len(forensic_table):
    audit_errors.append("Forensic review record count changed after export.")
if sha256_file(PATHS.input_a) != input_manifest["inputs"]["a"]["sha256"] or ↵
↪sha256_file(PATHS.input_b) != input_manifest["inputs"]["b"]["sha256"]:
    audit_errors.append("An input file changed during execution.")
generator_record = input_manifest["code_assets"]["generator"]
if generator_record is None:
    audit_errors.append("The notebook generator was not available during ↵
↪execution.")
elif not GENERATOR_PATH.is_file() or sha256_file(GENERATOR_PATH) != ↵
↪generator_record["sha256"]:
    audit_errors.append("The notebook generator changed during execution.")
optional_path_lookup = {
    "forensic_review": PATHS.forensic_review_input,
    "match_review": PATHS.match_review_input,
    "source_provenance": PATHS.provenance_input,
    "clean_environment_self_reproduction": PATHS.reproduction_input,

```

```

}
for key, record in input_manifest["optional_inputs"].items():
    if record is not None:
        path = optional_path_lookup[key]
        if not path.is_file() or sha256_file(path) != record["sha256"]:
            audit_errors.append(f"Optional input changed during execution:␣
↪{key}.".")
if audit_errors:
    raise RuntimeError("Definitive artifact audit failed:\n- " + "\n- ".
↪join(audit_errors))
reloaded_manifest["artifact_hash_audit"] = True
reloaded_manifest["publication_package_complete"] = True
reloaded_manifest["computationally_ready"] = bool(computationally_ready)
reloaded_manifest["submission_ready"] = bool(submission_ready)
atomic_json(reloaded_manifest, PATHS.artifact_dir / "analysis_manifest.json")
print(f"Verified {len(output_paths)} publication artifacts.")

```

Verified 24 publication artifacts.

```

[43]: ## Final package audit: corrections ledger, raw copies, supplement exports, and␣
↪table S1

FINAL_PACKAGE_AUDIT_ERRORS: list[str] = []

# 1) Corrections ledger (K2-136 d documented source-error correction)
CORRECTIONS_LEDGER = [
    {
        "planet_key": "k2-136 d",
        "observable": "geometric_area_ratio_ppm",
        "column": "pl_rade",
        "issue_type": "unit_error",
        "issue_severity": "corrected_source_error",
        "recorded_value": "1.450 R_Jup",
        "corrected_value": "0.1293603354 R_Jup",
        "resolution": "Source A radius re-interpreted as an Earth-normalized␣
↪entry and converted to canonical units (1.450 R_earth = 1.450 / 11.209␣
↪R_Jup) before the geometric-area-ratio pipeline.",
        "impact": "After correction K2-136 d has factorial width 1.2257 and␣
↪rank displacement 85 (rank_a 1465 to rank_b 1380); it is not an extreme case␣
↪and is not forced into the forensic case set.",
        "documentation": "source_provenance.json source_a.derivative_status and␣
↪source_a.preprocessing_history",
    },
]
corrections_dir = Path("data/corrections")
corrections_dir.mkdir(parents=True, exist_ok=True)

```

```

atomic_csv(pd.DataFrame(CORRECTIONS_LEDGER), corrections_dir / "corrections.
↳csv")

# 2) Raw frozen-input copies for journal deposit
raw_dir = Path("data/raw")
raw_dir.mkdir(parents=True, exist_ok=True)
raw_ledger_rows: list[dict[str, Any]] = []
for input_key, source_path in (("a", PATHS.input_a), ("b", PATHS.input_b)):
    target = raw_dir / source_path.name
    shutil.copy2(source_path, target)
    if sha256_file(target) != input_manifest["inputs"][input_key]["sha256"]:
        FINAL_PACKAGE_AUDIT_ERRORS.append(f"Raw copy hash mismatch: {target}")
    raw_ledger_rows.append({
        "key": input_key,
        "file": source_path.name,
        "sha256": sha256_file(target),
        "bytes": target.stat().st_size,
        "source_path": source_path.as_posix(),
    })
atomic_csv(pd.DataFrame(raw_ledger_rows), raw_dir / "SHA256SUMS.csv")

# 3) Supplement exports: forensic tables and table S1 for the
↳1,115-displacement planet
supplement_dir = Path("data/supplement")
supplement_dir.mkdir(parents=True, exist_ok=True)
for filename in ("forensic_cases.csv", "forensic_review_record.csv"):
    shutil.copy2(PATHS.artifact_dir / filename, supplement_dir / filename)
table_s1 = operational_certificates[[
    "planet_key", "rank_a", "rank_b", "rank_displacement", "source_value_a",
    "source_value_b", "factorial_width", "top_k_status", "best_rank",
↳"worst_rank",
]].copy()
table_s1 = table_s1.sort_values(["rank_displacement", "planet_key"],
↳ascending=[False, True]).reset_index(drop=True)
table_s1_path = supplement_dir / "table_s1_largest_rank_displacements.csv"
atomic_csv(table_s1, table_s1_path)
if table_s1_path.stat().st_size == 0:
    FINAL_PACKAGE_AUDIT_ERRORS.append("Table S1 is empty.")
if table_s1.iloc[0]["planet_key"] != "kepler-458 c" or int(table_s1.
↳iloc[0]["rank_displacement"]) != 1115:
    FINAL_PACKAGE_AUDIT_ERRORS.append("Table S1 does not lead with the
↳1,115-displacement planet (kepler-458 c).")
supplement_ledger_rows: list[dict[str, Any]] = []
for filename in ("forensic_cases.csv", "forensic_review_record.csv",
↳"table_s1_largest_rank_displacements.csv"):
    path = supplement_dir / filename

```

```

        supplement_ledger_rows.append({"file": filename, "sha256":  

↪ sha256_file(path), "bytes": path.stat().st_size})
atomic_csv(pd.DataFrame(supplement_ledger_rows), supplement_dir / "SHA256SUMS.  

↪ csv")

# 4) Full-package hash record appended to the manifest, with authoritative  

↪ status
package_paths = [*output_paths]
package_paths.extend([
    corrections_dir / "corrections.csv",
    raw_dir / "SHA256SUMS.csv",
    raw_dir / PATHS.input_a.name,
    raw_dir / PATHS.input_b.name,
    supplement_dir / "SHA256SUMS.csv",
    *[supplement_dir / name for name in ("forensic_cases.csv",  

↪ "forensic_review_record.csv", "table_s1_largest_rank_displacements.csv")],
])
for path in package_paths:
    if not path.is_file() or path.stat().st_size == 0:
        FINAL_PACKAGE_AUDIT_ERRORS.append(f"Final package path missing or empty:  

↪ {path}")

final_package_record = {
    "date_utc": pd.Timestamp.now(tz="UTC").isoformat(),
    "paths": {path.as_posix(): {"sha256": sha256_file(path), "bytes": path.  

↪ stat().st_size} for path in package_paths},
}
reloaded_manifest["final_package"] = final_package_record
reloaded_manifest["final_package_audit_complete"] = bool(not  

↪ FINAL_PACKAGE_AUDIT_ERRORS)
reloaded_manifest["final_package_audit_errors"] =  

↪ list(FINAL_PACKAGE_AUDIT_ERRORS)
if not FINAL_PACKAGE_AUDIT_ERRORS:
    reloaded_manifest["publication_package_complete"] = True
reloaded_manifest["computationally_ready"] = bool(computationally_ready)
reloaded_manifest["submission_ready"] = bool(submission_ready)
atomic_json(reloaded_manifest, PATHS.artifact_dir / "analysis_manifest.json")

final_package_audit_complete = not bool(FINAL_PACKAGE_AUDIT_ERRORS)
submission_ready = computationally_ready and external_reproduction_complete and  

↪ final_package_audit_complete
reloaded_results = json.loads((PATHS.artifact_dir / "results_summary.json").  

↪ read_text(encoding="utf-8"))
reloaded_results["status"]["final_package_audit_complete"] =  

↪ bool(final_package_audit_complete)
reloaded_results["status"]["submission_ready"] = bool(submission_ready)

```

```

reloaded_results["status"]["computationally_ready"] =
    ↪bool(computationally_ready)
atomic_json(reloaded_results, PATHS.artifact_dir / "results_summary.json")

# Re-hash results_summary.json after status update and patch both hash locations
results_path = PATHS.artifact_dir / "results_summary.json"
reloaded_manifest["outputs"][results_path.as_posix()] = {
    "sha256": sha256_file(results_path),
    "bytes": results_path.stat().st_size,
}
if "final_package" in reloaded_manifest and results_path.as_posix() in
    ↪reloaded_manifest["final_package"].get("paths", {}):
    reloaded_manifest["final_package"]["paths"][results_path.as_posix()] = {
        "sha256": sha256_file(results_path),
        "bytes": results_path.stat().st_size,
    }
atomic_json(reloaded_manifest, PATHS.artifact_dir / "analysis_manifest.json")

if FINAL_PACKAGE_AUDIT_ERRORS:
    raise RuntimeError("Final package audit failed:\n- " + "\n- ").
    ↪join(FINAL_PACKAGE_AUDIT_ERRORS))
print(f"Final package audit complete: {len(package_paths)} hashed paths;
    ↪corrections ledger, raw copies, supplement exports, and table S1 written.")

```

Final package audit complete: 32 hashed paths; corrections ledger, raw copies, supplement exports, and table S1 written.

```

[44]: artifact_hash_audit = not audit_errors
computationally_definitive = all(quality_gates.values()) and artifact_hash_audit
publication_package_complete = artifact_hash_audit
solo_author_forensic_self_audit_complete = priority_zero_review_complete and
    ↪priority_zero_findings_status == "acceptable"
clean_environment_self_reproduction_passed = reproduction_payload is not None
    ↪and reproduction_report.major_count == 0 and reproduction_report.passed
external_reproduction_complete = False
computationally_ready = (
    computationally_definitive and solo_author_forensic_self_audit_complete and
    ↪publication_package_complete
    and provenance_documented and clean_environment_self_reproduction_passed
)
submission_ready = computationally_ready and external_reproduction_complete and
    ↪final_package_audit_complete
final_status = pd.DataFrame({
    "item": [
        *quality_gates.keys(), "artifact_hash_audit",
        ↪"computationally_definitive",
        "forensic_case_selection_complete", "priority_zero_review_complete",
    ]
})

```

```

        "solo_author_forensic_self_audit_complete", "provenance_documented",
        "clean_environment_self_reproduction_passed",
    ↪ "publication_package_complete",
        "review_mode", "external_reproduction_complete",
    ↪ "final_package_audit_complete",
        "computationally_ready", "submission_ready",
    ],
    "passed": [
        *quality_gates.values(), artifact_hash_audit,
    ↪ computationally_definitive,
        forensic_report.passed, priority_zero_review_complete,
        solo_author_forensic_self_audit_complete, provenance_documented,
        clean_environment_self_reproduction_passed,
    ↪ publication_package_complete,
        CONFIG.is_solo_author, external_reproduction_complete,
    ↪ final_package_audit_complete,
        computationally_ready, submission_ready,
    ],
})
display(final_status)
display(pd.DataFrame({"item": ["priority_zero_findings_status"], "value":
    ↪ [priority_zero_findings_status]}))
print(f"COMPUTATIONAL CERTIFICATION: {'PASS' if computationally_definitive else
    ↪ 'FAIL'}")
publication_blockers = []
if not computationally_definitive:
    publication_blockers.append("automated computational certification failed")
if not priority_zero_review_complete:
    publication_blockers.append("priority-0 forensic reviews are incomplete")
if priority_zero_findings_status == "correction_required":
    publication_blockers.append("a priority-0 review identified an error
    ↪ requiring correction and rerun")
if not provenance_documented:
    publication_blockers.append("source provenance is incomplete")
if not clean_environment_self_reproduction_passed:
    publication_blockers.append("clean-environment self-reproduction is
    ↪ missing")
if not publication_package_complete:
    publication_blockers.append("artifact package audit failed")
if not external_reproduction_complete:
    publication_blockers.append("external independent reproduction is pending")
if not final_package_audit_complete:
    publication_blockers.append("final package audit is incomplete")
if computationally_ready:
    print("READINESS STATUS: COMPUTATIONALLY READY; SOLO-AUTHOR SELF-AUDIT
    ↪ COMPLETE; FINAL PACKAGE AUDIT COMPLETE; EXTERNAL REPRODUCTION PENDING")

```



```

else:
    print("READINESS STATUS: NOT COMPUTATIONALLY READY - " + "; ".
        ↪join(publication_blockers))

```

|    | item                                       | passed |
|----|--------------------------------------------|--------|
| 0  | generator_script_available                 | True   |
| 1  | raw_schema_a                               | True   |
| 2  | raw_schema_b                               | True   |
| 3  | canonical_a                                | True   |
| 4  | canonical_b                                | True   |
| 5  | tier1_minimum                              | True   |
| 6  | primary_minimum                            | True   |
| 7  | operational_minimum                        | True   |
| 8  | operational_all_concordant_transits        | True   |
| 9  | primary_certificates_fully_verified        | True   |
| 10 | operational_certificates_fully_verified    | True   |
| 11 | robust_ranking_verified                    | True   |
| 12 | primary_table_count_consistent             | True   |
| 13 | operational_table_count_consistent         | True   |
| 14 | physics_reference_tests                    | True   |
| 15 | ranking_reference_tests                    | True   |
| 16 | data_contract_reference_tests              | True   |
| 17 | randomized_bound_property_tests            | True   |
| 18 | randomized_ranking_property_tests          | True   |
| 19 | secondary_certificates_fully_verified      | True   |
| 20 | threshold_partitions_consistent            | True   |
| 21 | threshold_counts_consistent                | True   |
| 22 | forensic_case_selection_complete           | True   |
| 23 | detection_mapping_complete                 | True   |
| 24 | operational_exclusion_count_consistent     | True   |
| 25 | no_blocking_validation_issues              | True   |
| 26 | artifact_hash_audit                        | True   |
| 27 | computationally_definitive                 | True   |
| 28 | forensic_case_selection_complete           | True   |
| 29 | priority_zero_review_complete              | True   |
| 30 | solo_author_forensic_self_audit_complete   | True   |
| 31 | provenance_documented                      | True   |
| 32 | clean_environment_self_reproduction_passed | True   |
| 33 | publication_package_complete               | True   |
| 34 | review_mode                                | True   |
| 35 | external_reproduction_complete             | False  |
| 36 | final_package_audit_complete               | True   |
| 37 | computationally_ready                      | True   |
| 38 | submission_ready                           | False  |

|   | item                          | value      |
|---|-------------------------------|------------|
| 0 | priority_zero_findings_status | acceptable |

COMPUTATIONAL CERTIFICATION: PASS



READINESS STATUS: COMPUTATIONALLY READY; SOLO-AUTHOR SELF-AUDIT COMPLETE; FINAL PACKAGE AUDIT COMPLETE; EXTERNAL REPRODUCTION PENDING

### 1.16 Limitations and prohibited interpretations

1. Two derivative source artifacts do not span all epistemic or measurement uncertainty.
2. Coordinate-wise hybrid vertices need not correspond to a published complete parameter set.
3. Shared literature means the sources are not independent replications.
4. Complete-case cohorts are selected toward better-characterized systems.
5. The geometric area ratio is not an observed transit depth or full target-priority metric.
6. Mass-dependent results are secondary because mass semantics may differ.
7. Host resampling measures composition sensitivity, not uncertainty about the frozen estimand.
8. Extreme cases require source-provenance interpretation and are not automatically errors.
9. The analysis measures reproducibility under substitution, not truth, causality, or habitability.
10. Observed discrepancies confound catalog curation decisions with the temporal information-epoch difference between sources (Source A extends to ~February 2018; Source B is frozen in March 2026); disentangling these two effects requires access to same-epoch catalog versions.
11. Of the 444 selected forensic cases, only all 113 priority-0 cases received documented self-audit review; the remaining 331 lower-priority cases have not been manually reviewed and their classification is provisional.

### 1.17 Reproducible conclusion contract

All manuscript numbers must be copied from `results_summary.json` or the four canonical tables, with the same denominators, population labels, and rounding policy. A definitive execution requires a clean kernel and all cells run once in notebook order.

COMPUTATIONAL CERTIFICATION: PASS certifies the implemented mathematics, controlled cohort construction, exhaustive finite-vertex verification, randomized property tests, robust-ranking logic, deterministic exports, generator availability, and artifact hashes. READINESS STATUS: COMPUTATIONALLY READY; SOLO-AUTHOR SELF-AUDIT COMPLETE; FINAL PACKAGE AUDIT COMPLETE; EXTERNAL REPRODUCTION PENDING reflects complete source provenance, documented self-audit review of every priority-0 forensic case with no unresolved correction-required finding, a clean-environment reproduction using the same protocol and frozen inputs, certified by the lead author as a solo-author self-audit, and a completed final package audit covering the raw-input copies, corrections ledger, supplement exports, and table S1. External independent reproduction remains pending.

These statuses do not establish that either source is correct, do not quantify total astrophysical uncertainty, do not guarantee peer-review acceptance, and do not replace scientific judgment about provenance, population selection, or interpretation.

The clean-environment self-reproduction was performed by the lead author (Khaled Tarchi) using a separate Python environment with identical frozen inputs. This constitutes self-reproduction, not independent reproduction; independent reproduction requires verification by an unaffiliated party.